

“the plaintiffs”). The plaintiff, Christine Campbell, has dismissed her case against the defendants.

[2] The action arises out of the care and treatment received by Mr. Campbell from certain defendants between March 17 and April 16, 2009 (“the relevant period”). The defendants who remain in the action are Dr. David Kenneth Roberts (“Dr. Roberts”), Dr. Murray David Weingarten (“Dr. Weingarten”), Dr. Erik Olav Silmberg (“Dr. Silmberg”) and Dr. Pdraig Richard Dominic Warde (“Dr. Warde”) (collectively “the defendant physicians”). The plaintiffs are dismissing the action against the other named defendants.

[3] During the relevant period Dr. Weingarten was an internist at Southlake Regional Health Centre (“Southlake”), Dr. Roberts was a community respirologist at Southlake, Dr. Silmberg was a community radiologist at Southlake, and Dr. Warde was a radiation oncologist at Princess Margaret Hospital. They all treated Mr. Campbell at various times during the relevant period.

[4] The case was heard over 29 days. Most of the case was heard in May and June 2013 but, due to scheduling and other issues, the case could not be completed until late March 2014. The parties consented to calling more than three experts each given the issues at stake and the different areas of practice of the defendant physicians.

[5] This is a difficult and tragic case in which Mr. Campbell has been left with permanent cognitive impairments resulting from two brain abscesses. Mr. Campbell claims that had his condition been properly diagnosed from an early stage, his condition was entirely treatable and full recovery likely. The defendants submit that all possible steps were taken to diagnose and treat Mr. Campbell. While his condition was misdiagnosed, the standard of care did not fall below that of a reasonable physician. The condition was somewhat rare and the symptoms replicated those of cancer. According to the defendants the treatment given to Mr. Campbell was the correct treatment, given the information available. According to the defendants the result is very unfortunate, but does not come close to negligence in all of the circumstances.

Background Facts

[6] Prior to the events which gave rise to this action Mr. Campbell was a healthy and active 58 year old man. He had a loving marriage with his wife, Carol, and a close relationship with his children, Riki and Noel. He had been working full-time as a consultant for the Magna group of companies. His work was very specialized and involved problem solving issues for which he created imaginative and cutting edge solutions. Mr. Campbell was a smoker. He had smoked a package of cigarettes a day for 40 years. Whether from the negligence of the defendants or an unfortunate series of unforeseeable events, Mr. Campbell has been left with permanent cognitive defects which affect his memory, concentration, language, executive function, informational processing and vision. He will never be able to work again. He will have significant care needs, although the parties have different views on the cost of that care.

[7] The parties filed an agreed statement of facts in this proceeding. The factual background set out herein contains many of those facts. Where there are facts in dispute, they have been

noted. It is also important for certain medical concepts to be explained, as they will be referred to throughout this judgment.

[8] Mr. Campbell suffers from Hereditary Hemorrhagic Telangiectasia (“HHT”). He was diagnosed with this condition in 1990. HHT is a rare genetic disorder that leads to abnormal blood vessel formation in various organs. HHT patients can develop arteriovenous malformations (“AVMs”). AVMs are abnormal connections between veins and arteries which create an alternate blood flow and may prevent the body’s normal filtering system from stopping the spread of infections. AVMs may develop in the lungs and these are referred to as pulmonary AVMs (“PAVMs”).

[9] In 2008 Mr. Campbell attended the HHT clinic at St. Michael’s Hospital under the care of an HHT specialist, namely, Dr. Marie Faughnan. At that time he underwent investigations to determine if his HHT had caused any abnormalities, including the possible presence of AVMs.

[10] On June 19, 2008, Mr. Campbell underwent a contrast echocardiogram (also known as a “bubble study”), which showed a positive finding. This meant that the bubbles injected into Mr. Campbell’s system returned to the right side of the heart more quickly than they should; this can be indicative of the presence of AVMs. Dr. Cox opined at trial that a patient who coughed up blood and had a positive bubble study (such as Mr. Campbell) could have pulmonary AVMs that could not be detected by a CT scan.

[11] On July 1, 2008, Dr. Faughnan ordered both a CT scan of Mr. Campbell’s chest and an MRI of his head to look for AVMs. That testing took place on November 27, 2008. On November 28, 2008, Dr. Faughnan noted that the chest CT scan “did not show any significant pulmonary AVMs” and that the head MRI “did not show any cerebral AVMs.” She also noted that there was no history of infection or abscess. Dr. Faughnan did not testify at trial. Dr. Roberts testified at trial that he was aware of the possibility of pulmonary AVMs and noted this in his discharge summary of April 16, 2009.

[12] Mr. Campbell’s evidence was that, after undergoing the tests in 2008, Dr. Faughnan gave him a “clean bill of health” and had no particular concerns.

[13] The plaintiffs submit that while Mr. Campbell’s history of HHT may have increased his risk of developing brain abscesses, that possibility is not critical to the liability analysis. Brain abscess should have been considered a first priority with respect to the differential diagnosis based on Mr. Campbell’s symptoms, as a lung infection can spread to the brain without pulmonary AVMs being present. As well, the plaintiffs submit that the defendants did not adequately consider the possibility that Mr. Campbell had microscopic pulmonary AVMs, which would tend to increase the risk of abscess formation.

[14] There were many references in the evidence by both the experts and the defendant physicians to the term “differential diagnosis.” This concept is part of every doctor’s medical training and involves the physician considering a patient’s complaints, their symptoms and history, and the relevant findings in order to create a list of likely explanations for the patient’s presentation. A differential diagnosis is updated constantly as investigations of the patient’s

condition proceeds. Prioritizing a differential diagnosis involves considering both the treatability and the seriousness of a condition. Curable and more serious conditions should rank ahead of other less serious conditions when considering a differential diagnosis.

[15] Brain abscesses were also discussed at length during the trial given Mr. Campbell's diagnosis. An abscess is an area of localized infection. When present in the brain it can be particularly serious because the growth of an abscess may cause the death of brain cells, and pus can cause the swelling of brain tissue. Dr. Weingarten and Dr. Roberts, as well as the experts, agreed that a brain abscess is a serious and urgent medical condition. Left untreated, bacteria will increase swelling which can cause permanent neurological disability or death.

[16] At trial, reference was made to a medical text entitled *Principles and Practice of Infectious Diseases* (7th ed.)¹ (hereinafter the "Mandell text"). That text makes reference to the classical symptoms of brain abscess as being fever, headache and neurological impairment (which in Mr. Campbell's case presented as visual impairment). Not all patients present with a fever. Depending on the size of the abscess, either surgery or antibiotic treatment is mandated. Antibiotic treatment is the suggested treatment for multiple brain abscesses which measure less than 3 centimetres and optimally, less than 2.5 centimetres. Abscesses larger than three centimetres require surgical excision. According to *Harrison's Principles of Internal Medicine*² "bacterial abscess can be successfully treated in the majority of patients."

[17] It should be noted that, according to the Mandell text, brain abscess can be caused by the spread of infection from other parts of the body (hematogenous dissemination) with the most common source of infection being the lungs. Where brain abscess is caused by hematogenous dissemination it usually results in multiple brain abscesses.

[18] The Mandell text indicates that the risk factor for brain abscess for patients with HHT is 1,000 times greater than the general population.

[19] There was also much discussion of lung infections at trial. Lung infections are a common medical condition, especially among smokers. Lung infections are more common than lung cancer. According to the experts, a lung infection is usually treated with antibiotics, which will shrink any lung lesions and dissipate pus material. Testing, such as bronchoscopy or biopsies, is not indicated for lung infections, as the use of antibiotics would mean that procedures would not produce any positive culture for those tests. Spiculated borders found on lung lesions are often indicative of cancerous lesions but may also be found in lung infections.

[20] Mr. Campbell had no health related complaints after the HHT testing until he went on a vacation to Louisiana with his wife in late February and early March 2009. His evidence was that, even before the vacation, he experienced severe headaches which prompted him to take Tylenol on the vacation. He had also experienced a cough and fatigue before the trip. During

¹ Gerald L. Mandell, John E. Bennett & Raphael Dolin, *Principles and Practice of Infectious Diseases*, 7th ed., Vol. 1 (Philadelphia: Elsevier Inc., 2010) at 1269-1275.

² Eugene Braunwalk et al., *Harrison's Principles of Internal Medicine*, 15th ed. (New York: McGraw-Hill, 2001), at 2469.

the trip, the headache and fatigue worsened. Of note was the fact that Mr. Campbell was an hour late for his plane when he left for his trip and took his golf clubs on the trip but never golfed. These things were unusual for him. He complained to both his daughter and his son about his headaches and fatigue while away. His son testified that his father had never shared health concerns with him before and his daughter testified that her father had never complained about headaches before.

[21] Upon his return from vacation, the headaches worsened and he had trouble concentrating. When he coughed, the pain in his head was severe. Despite these symptoms, Mr. Campbell returned to work immediately following his vacation.

[22] Mr. Campbell's wife Carol has a B.Sc. in nursing and has been working as a public health nurse since 2001. Her evidence was that she had no concerns about her husband's health prior to March 2009.

[23] The plaintiffs take the position that there is no evidence that Mr. Campbell was suffering any cognitive impairment while on vacation and his cough and headache during the vacation are irrelevant to the considerations in this case.

[24] On March 14, 2009, Mr. Campbell went to Southlake. He complained he had been feeling unwell for the last six days. This included a productive cough, head and body aches and a low grade fever at times but with no chills. He told the emergency room physician about his HHT condition. The defense noted that Mr. Campbell did not tell the emergency physician, Dr. Kouchakan, about the symptoms he had experienced before and during his vacation.

[25] Dr. Mohsen Kouchakan, a community internist, initially examined Mr. Campbell. Dr. Kouchakan ordered lab tests, a chest x-ray and a head CT scan. The chest x-ray showed a 3.4 centimetre mass in the lung which the radiologist concluded could be early pneumonia. The head CT scan was done "without contrast". A without contrast CT scan has less sensitivity in terms of being able to detect abnormalities than a "with contrast" CT scan. The CT scan was interpreted by a radiologist and showed no bleeding or swelling in the brain, but did show decreased white matter density in the right occipital lobe suggestive of an underlying mass such as a metastatic deposit or early infarction. A further CT scan was recommended.

[26] Dr. Kouchakan's opinion was that Mr. Campbell had a "likely diagnosis of community acquired pneumonia." While he noted that there were some findings on the CT scan of the head, Mr. Campbell had had a recent comprehensive MRI of the head for the purpose of the HHT investigations. Dr. Kouchakan offered to admit Mr. Campbell for further investigation, including a CT scan of the chest to rule out lung malignancy and potentially a blood transfusion to treat anemia, but Mr. Campbell did not want to stay in the hospital. Mr. Campbell agreed at trial that Dr. Kouchakan wanted him to stay longer to understand what was wrong with him.

[27] Mr. Campbell was sent home with a ten day course of antibiotics and told to return to the hospital if he had any concerns and otherwise follow-up with his family doctor in a week

[28] Mr. Campbell took the antibiotics and pain medication prescribed by Dr. Kouchakan but on March 16, 2009, while driving to work, he lost his left field of vision in both eyes. He knew

that something “major” was wrong and drove straight to the office of his family doctor who advised him to return to Southlake. Mr. Campbell did not go to Southlake immediately. He returned home, ate dinner with his wife and slept at home that night.

[29] The period between March 17 and March 26, 2009, shall be referred to in this judgment as Mr. Campbell’s “first admission to Southlake.”

[30] On March 17, 2009, Mr. Campbell returned to Southlake complaining of a headache, left side visual impairments and severe fatigue. He reported feeling tired and achy over the previous several weeks and had begun to feel signs of the left side visual impairments the week before. His headaches worsened considerably when he coughed.

[31] Dr. Weingarten examined Mr. Campbell and noted that his white blood cell count and platelets were normal. He ordered a chest CT, head CT and head MRI. He documented a plan to consult a neurologist, Dr. Sheldon Baryshnik, and an otolaryngologist, Dr. D.M. Finklestein. He then admitted Mr. Campbell for further evaluation. Mr. Campbell told Dr. Weingarten and the ER physician that he had HHT. His HHT records were ordered from St. Michael’s Hospital. His antibiotics were continued and a course of steroids ordered. Dr. Weingarten reviewed all available records for Mr. Campbell from the March 14 visit, and was aware he had been given antibiotics for possible pneumonia and that he had left the hospital against medical advice. He was also aware that Mr. Campbell had been a pack a day smoker for decades. Dr. Weingarten ordered a blood transfusion, iron supplements and ordered that Mr. Campbell be admitted. Mr. Campbell did not inform Dr. Weingarten, or any other physicians at Southlake throughout his treatment there, about the symptoms he had experienced before and during his vacation to Louisiana.

[32] Mr. Campbell had a “with contrast” chest and head CT on March 17, 2009. Dr. Lisa Thain, a community radiologist, interpreted both CTs. She noted three lung lesions on the chest CT which she said may have been “primary bronchogenic carcinomas.” She also noted 2 lesions in the brain; one measuring 1.6 centimetres in diameter and the second measuring 2.2 centimetres in diameter. She suggested a biopsy of the cranial mass in the right upper lobe and concluded that there was evidence of two brain metastases although no evidence of “regional metastatic disease.”

[33] Dr. Weingarten reviewed the results of the “with contrast” head and chest CT and listed tumour, infections and AVMs as the differential diagnosis for the lung. His evidence was that finding spiculated borders on the lung lesions was a “hallmark for malignant tumours.” The spiculated borders did not look like “an average pneumonia or an abscess or anything infective.” While Dr. Weingarten testified that infection could not be completely ruled out where spiculated borders are found, the usual chest infections do not have them whereas cancerous lesions very frequently do.

[34] The differential diagnosis for the brain lesion was either an area of infarction (stroke) or a metastasis (cancer) with surrounding edema. He ordered a needle lung biopsy, which took place on March 24, and ordered steroids to assist in reducing the swelling in Mr. Campbell’s head. Dr. Weingarten’s evidence was that he ordered a needle lung biopsy because he assumed that the

disease process in Mr. Campbell's lungs and brain were linked. That is, if the lung biopsy showed cancer, then the lesions in Mr. Campbell's brain were also cancerous and a brain biopsy would not have been necessary.

[35] Dr. Weingarten's evidence was that he did not know what the diagnosis was as of March 17, 2009. Even if he had thought a brain abscess was likely (which he did not) he would have ordered the same series of tests he ordered on March 17, 2009 – a CT and MRI of the head.

[36] The plaintiffs note that Dr. Weingarten did not list brain abscess as part of his differential diagnosis for the lesions in the head, even though he did list infection as a possibility for the lung lesions. According to the plaintiffs, this illustrates Dr. Weingarten's failure to consider that the lung and brain lesions were related and that the lung infection had potentially spread to the brain.

[37] On March 18, 2009, Dr. Weingarten met with Mr. Campbell in his room at Southlake. The evidence of Mr. Campbell and Dr. Weingarten does not align with respect to what happened at that meeting. According to Mr. Campbell, Dr. Weingarten told him in a cursory fashion that he had lung cancer, two inoperable brain tumours and his condition was terminal. After this very short conversation he left the room. Mr. Campbell also testified that Nurse Norris was in the room when this conversation took place.

[38] Dr. Weingarten did not recall this interaction and denied that he would have told Mr. Campbell his condition was terminal, as he never told any patient their condition was terminal. His intention on that date was to discuss the results of the March 17 tests with Mr. Campbell. He has established a usual practice for discussing results where imaging is suggestive of, but not confirmed to be, cancer. He denied that the conversation was cursory and agreed he may well have drawn a diagram of the lesions, as suggested by Mrs. Campbell.

[39] Although Dr. Weingarten could not specifically recall the details of the conversation, he was confident he would have informed Mr. Campbell that he had metastatic disease in his brain and the lesions in his lungs were likely cancerous, but a tissue diagnosis and further MRI of the head was needed. He would not have discussed a prognosis with Mr. Campbell, as there was no diagnosis at that point. His contemporaneous note indicated that CT studies to that point suggested metastatic disease to the brain and lung, that he was awaiting results of a head MRI, intended to order a lung biopsy, and that he discussed all of this with Mr. Campbell.

[40] Mrs. Campbell's evidence was that her husband did not tell her he had cancer at that point. The information she had was from later in the first admission, when Dr. Roberts told her that the test results did not indicate cancer. Riki Campbell's evidence was that her father told her he had spots on his lungs but "they weren't sure what they were." Noel Campbell testified that he did not learn of the cancer diagnosis until the second admission to Southlake.

[41] Nurse Kate Norris testified for the plaintiffs. She has 20 years of nursing experience and was working a shift at Southlake on March 18, 2009. She was aware of Mr. Campbell's circumstances and understood that he was to be given test results and "bad news" that day. Nurse Norris did not recall being in the room when Dr. Weingarten met with Mr. Campbell. Her notes indicated that the meeting lasted 45 minutes.

[42] She further testified that Mr. Campbell was upset after his meeting with Dr. Weingarten and left the hospital for about an hour, telling her he needed to take care of a few things. Nurse Norris allowed him to leave without signing the prescribed form given the circumstances. On March 24 Mr. Campbell left a thank you card for Nurse Norris in the emergency department. It was produced at trial and said “thank you for giving me some time to accept some difficult news, you are very special.”

[43] Mr. Campbell’s evidence was that as a result of this discussion with Dr. Weingarten he called a family friend, Patsy Russell, to send flowers to his wife to help her deal with the news. Ms. Russell’s evidence at trial was that she recalled speaking to Mr. Campbell about arranging for the flowers and that he told her he had tumours in his lung and brain.

[44] Also as a result of Dr. Weingarten’s news, Mr. Campbell sent an email to Dr. Faughnan. Unfortunately, the email was sent to the wrong address but was produced at trial as Exhibit 5. The email stated “they have discovered some spots on my lungs as well as two tumours on my brain. The brain tumours have one on the side and one on the back of my head. They will not be able to operate.” Mrs. Campbell visited her husband in the hospital around this time and testified that he told her the doctor had informed him that he had two inoperable brain tumours. Her evidence was that throughout the first admission she could not recall discussions with her husband’s physicians about any possibility other than cancer.

[45] The plaintiffs’ position on the interaction with Dr. Weingarten on March 18 is that he had a “tunnel vision” type view of Mr. Campbell’s situation in which either he told Mr. Campbell his condition was terminal or was so negative that Mr. Campbell drew this conclusion himself. Dr. Weingarten had failed to list abscess in his differential diagnosis to that point, and failed to take steps to rule it in or out.

[46] Mr. Campbell received a blood transfusion and steroids on March 18. He testified that he felt greatly improved after receiving those treatments.

[47] On March 20 Dr. Weingarten ordered an MRI of Mr. Campbell’s head. The plaintiffs noted that Dr. Weingarten made no mention of the possibility of infection or Mr. Campbell’s recent fever in the MRI requisition form. The form only indicated “lesion R occipital lobe.” While he testified that he ordered the MRI to help determine if the brain lesions were cancer or something else, according to the plaintiffs, this is not borne out on the requisition form. Dr. Weingarten’s evidence was that a requisition form for an MRI is not meant to lead a radiologist to one conclusion or another; it is to receive an explanation of what is shown on the MRI.

[48] Dr. Silmberg interpreted the March 20 MRI of Mr. Campbell’s head. He noted that there were two ring-enhancing lesions. The one in the left basal ganglia measured 1.5 centimetres in diameter and the one in the right occipital lobe measured 2.4 centimetres in diameter. He gave a differential diagnosis of ring-enhancing lesions but, given the history of the lung lesions, his view was that the head lesions were most likely metastases. His evidence at trial was that, if Dr. Weingarten had noted the recent fever on the requisition form, he would have listed abscess on his report. He further testified that mentioning that there was a differential diagnosis was his “quiet way” of raising the possibility of infection.

[49] Dr. Weingarten's evidence was that he knew that Dr. Silmberg's differential diagnosis for ring-enhancing lesions included abscess. However, in his view, tumour was most likely and abscess, although not impossible, was quite unlikely given the clinical situation.

[50] By March 23 Mr. Campbell's symptoms had improved because of the antibiotics and steroids. The 10 day course of antibiotics ended on March 23 and was not renewed. Dr. Weingarten noted that Mr. Campbell's visual field deficit was improved "presumably secondary to steroids." He saw no reason to put Mr. Campbell on another course of antibiotics.

[51] Dr. James Doran, a community radiologist, was to administer the needle lung biopsy on March 24. However, the procedure was deferred when Dr. Doran noted that Mr. Campbell's lung lesions had shrunk by 50 per cent compared to their size on March 17. He felt this new piece of information should be considered by Mr. Campbell's physicians before taking further steps. Mr. Campbell's evidence was that Dr. Doran told him he had only seen such shrinkage once before in his life.

[52] The plaintiffs called Dr. Doran as a witness. He is a radiologist at Southlake and was formerly a defendant in this action. His evidence was that he understood that the lung biopsy was to test for cancer. As was his normal practice he did a chest CT scan before doing the biopsy. He noted shrinkage of the lesions from the CT scan done seven days before. He was surprised by the change, and although he has seen cancerous lesions shrink in the past, they normally do not change much in size over time and this change raised doubt as to whether Mr. Campbell had cancer. He wrote a report in which he noted shrinkage of Mr. Campbell's lesions to be about 50 per cent. He explained that his reference to a "vascular component" related to the appearance of the lesions being possibly AVMs with infection. Dr. Doran's evidence was that the shrinkage did not exclude cancer. It could have been a combination of cancer and infection, or a "collapse of surrounding lung because of compression of the tumour." He confirmed that cancer patients can present with infectious elements.

[53] Dr. Doran did not know that Mr. Campbell had just completed a 10 day course of antibiotics. His evidence was that if he had known this, he would likely have considered infection to be more likely.

[54] By March 24 Mr. Campbell was feeling much better. His headaches were gone and his vision was back to normal. He was regularly leaving the hospital at night.

[55] Dr. Weingarten was puzzled by the shrinkage of the lesions and questioned whether the lesions were inflammatory as opposed to malignant or even steroid sensitive (lymphoma). Dr. Weingarten testified that at this point he began to doubt whether Mr. Campbell had cancer. Infection was becoming a higher possibility in his mind, but still not more likely than cancer. The diagnosis was not clear to Dr. Weingarten at this point. For example, the shrinkage of the lung lesions was unusual for malignancy but Mr. Campbell's lack of cognitive impairment was unusual for brain abscess. The plaintiffs submit that, notwithstanding Dr. Weingarten's change in view, he failed to take steps to rule infection in or out.

[56] Given this change in events, Dr. Weingarten consulted Dr. Roberts on March 24 and Dr. Weingarten was only minimally involved in Mr. Campbell's care after that point. Dr. Roberts saw Mr. Campbell on March 25, reviewed his medical records from St. Michael's Hospital and noted that, based on the shrinkage, the lung lesions could be small cell carcinoma or another differential diagnosis which included infection, inflammation or structural disturbances in the lung. During his meeting with Mr. Campbell, Dr. Roberts did a comprehensive examination and gathered a detailed history including the fact that he was a long time smoker and had lived in Mississippi. The smoking habit was germane because of it being a risk factor for lung disease and other lung infections. The residence in Mississippi was of note as residence in a river valley can result in histoplasmosis, which can cause brain masses and pneumonias. He also spoke with Mr. Campbell about the risks and benefits of a bronchoscopy.

[57] Mr. Campbell's evidence was that during the meeting on March 25, Dr. Roberts told him he would find the cancer by doing a bronchoscopy and that he was so confident it was cancer he was going to order chemotherapy. Mr. Campbell was not cross-examined about this conversation. However, Mrs. Campbell's evidence was that Dr. Roberts told her that the test results were negative for cancer.

[58] On March 26 Dr. Roberts did a bronchoscopy on Mr. Campbell in order to collect specimens from inside his lungs and evaluate the nature of the lesions. His view was that a bronchoscopy would give a "window on a fairly wide range of options." This included evidence of infection or abnormal cells. Dr. Roberts also ordered a bone scan, to test for hematogenous spread of infection to other organs, as well as liver function tests.

[59] After the bronchoscopy was completed Dr. Roberts discharged Mr. Campbell to await the results. Mr. Campbell continued to take steroids on discharge. Dr. Roberts' evidence was that on discharge Mr. Campbell was told not to drive and he gave him his personal pager number. Mr. Campbell disputed that Dr. Roberts gave him any discharge instructions at all. However, his discharge note makes reference to no driving and further indicated, "he understands that." Mr. Campbell was also encouraged by Dr. Roberts to return to the ER if there were problems.

[60] In his discharge note Dr. Roberts noted that Mr. Campbell had shown no signs of infection during his stay and his platelet count was normal. He recited the fact that a head MRI conducted on November 27, 2008, an echocardiogram done in 2008 and a chest x-ray done on June 29, 2008, were all normal. Dr. Roberts' view at that point was that Mr. Campbell probably had "primary lung malignancy" and that he would consider a needle lung biopsy if the bronchoscopy results were inconclusive. He further noted that a tissue diagnosis of cancer would be needed before Mr. Campbell could be referred to an oncologist. Copies of Dr. Roberts' notes were sent to Dr. Faughnan to keep her apprised and to allow her to make comment if she wished. Dr. Faughnan did not make any comments and made no recommendations about AVM precautions after the definitive chest CT on November 27, 2009.

[61] Dr. Roberts' discharge note indicates a lung tumour NYD (not yet diagnosed). He goes on to say that it is "likely cancer" which he describes as a "probability consideration." His evidence was that the case was incomplete as there were still unanswered questions. His reason for noting lung cancer as a probability was based on Mr. Campbell's smoking history, the

radiology reports and, finally, the lack of suggestion of infection, due to the absence of fever or elevated white blood cell counts.

[62] Mr. Campbell testified that he felt fine on discharge and went to work on March 27. He felt fine on March 28 and planned to return to work on March 30. He went to work on March 30, but did not feel well according to Mrs. Campbell.

[63] On March 30, 2009, Mr. Campbell returned to Southlake by ambulance. He had been found by his wife disoriented as to place, person and time. He presented in this manner at Southlake and also had a low-grade fever, lethargy and visual agnosia (a symptom which affected his ability to interpret visual cues). Mr. Campbell was started on IV antibiotics due to the fever. The period of March 30, 2009 to April 16, 2009, shall be referred to in this judgment as the “second admission to Southlake.”

[64] A chest x-ray done on March 31 showed that the lung lesions had shrunk to the point of being “barely perceptible.” A head CT scan done that day noted that there was some progression of the swelling around the brain lesions. The bronchoscopy results were available on April 1 and noted that there were no malignant cells. However, expert evidence at trial made it clear that a biopsy result which shows no malignant cells does not mean that cancer is not present. It simply means that cancerous cells were not present in the tissue samples taken. The defendants submit that the lack of cancerous cells from the bronchoscopy was therefore not conclusive that cancer was not present.

[65] On March 31 Dr. Roberts resumed care of Mr. Campbell and made an order to consult radiation oncology. He did a thorough physical examination of Mr. Campbell and was concerned about the fact that he had been disoriented for more than 20 hours. These were new findings from the first admission and Dr. Roberts wrote that Mr. Campbell showed inflammatory reaction with fever. He also considered whether Mr. Campbell’s course of steroids (Decadron) was related to the precipitation of an infection. He wrote, “It appears we have a new set of rules from last week.” That change led him to arrange a consultation with neurologist Dr. Baryshnik.

[66] On April 2 Dr. Roberts consulted with Dr. Baryshnik regarding Mr. Campbell. His hope was that Dr. Baryshnik would provide a different perspective on the range of diagnostic considerations. Dr. Baryshnik concluded from the head MRI that there were “enhancing ring lesions + edema – extensive” and “looks like 2 mets.” He listed three possibilities for Mr. Campbell’s delirium; paraneoplastic, edema or steroid. Dr. Roberts commented on Dr. Baryshnik’s note as not suggesting any new or re-ordering of priorities. Dr. Baryshnik did not suggest surgery.

[67] Mrs. Campbell testified that she went to the hospital to visit her husband every day at this time. His condition worsened each day and the invasive testing was taking a toll on him. Dr. Roberts spoke with her on several occasions, but never mentioned any other diagnosis than cancer. Ms. Russell’s evidence was similar to that of Mrs. Campbell’s with respect to her description of Mr. Campbell’s condition at this point. He was confused, in restraints, and sometimes did not recognize his wife. He was visually impaired as well.

[68] On April 3 Mr. Campbell underwent a needle lung biopsy as ordered by Dr. Roberts being “the next gentlest procedure.” Those results were reported on April 9 and were negative for malignancy. The results also indicated that the changes were “non-specifically reactive/inflammatory in nature. They would be compatible with either a primary inflammatory process or a secondary phenomenon related to adjacent (but not biopsied) neoplasia.” Dr. Roberts’ evidence was that the biopsy gave no definitive answer. Dr. Roberts did not communicate these results to Dr. Warde even though Dr. Warde’s consult note indicated that he was awaiting the biopsy results “with interest.” Dr. Roberts’ evidence was that he did not think that the results of the needle biopsy would have changed Dr. Warde’s appraisal. Dr. Warde agreed with this.

[69] During the needle biopsy Mr. Campbell suffered a pneumothorax. This is caused when air gets between the lung and chest wall causing a lung collapse. As a result, daily chest x-rays were ordered by Dr. Roberts.

[70] Mr. Campbell’s condition and in particular his cognition continued to deteriorate. However, he did not exhibit a fever or elevation of his white blood cell count. As a result of the decline in cognition, Dr. Roberts sought a consult on April 4 with a senior radiation oncologist, Dr. Warde. His purpose in doing so was to obtain an independent opinion from a specialist who had experience with both cancerous and non-cancerous brain lesions. The oncology referral form listed a diagnosis of metastatic brain tumour plus confusion and agitation, and indicated the patient had been advised of this. Dr. Roberts indicated that a reference to some form of cancer was necessary on the form or it may have been “blocked” administratively. During a telephone discussion with Dr. Warde, Dr. Roberts made it clear that there was no firm diagnosis of cancer. Dr. Warde confirmed that he was aware that cancer was not a firm diagnosis at that time and he referred to that in his dictated note.

[71] On April 6 Dr. Warde reviewed the imaging and medical records of Mr. Campbell. His evidence was that he would normally review a patient’s records in chronological order in order to see the progression of any disease. He concluded that the most likely diagnosis was primary lung cancer with two brain metastases. He noted that Mr. Campbell was confused and not oriented to time, place or person. He told Mrs. Campbell there was no definite diagnosis of cancer. However, in his opinion, whole brain radiation was warranted even without a firm tissue diagnosis of cancer. Mrs. Campbell recalled Dr. Warde telling her he was 99.9 per cent sure it was cancer and that a thorough medical workup had been performed on Mr. Campbell. Dr. Warde denied telling Mrs. Campbell this. He testified that he does not normally use percentages but acknowledges he would have told Mrs. Campbell that he had a high level of certainty about cancer or he would not have recommended moving ahead.

[72] Based on her conversations with Dr. Warde along with his recommendations and assurances, as Mr. Campbell’s Power of Attorney for Personal Care at that point, Mrs. Campbell consented to the whole brain radiation therapy.

[73] Between April 7 and 9 Mr. Campbell received three fractions of whole brain radiation at Princess Margaret Hospital. He received a further fraction on April 10. Another fraction of radiation scheduled for April 13 was cancelled by Dr. Warde, as Mr. Campbell’s condition had

not improved. Mrs. Campbell testified that her husband's condition deteriorated during the radiation therapy; he was losing weight and may have been semi-comatose.

[74] On April 11 Mr. Campbell underwent an open lung biopsy to obtain tissue samples. This procedure is more invasive than a bronchoscopy or a needle lung biopsy, but more tissue is collected thus making it more likely to find cancerous cells if they are present. Pending the results of the lung biopsy on April 11, Mr. Campbell had no ongoing fever and the results of his chest x-rays were good.

[75] On April 15 Dr. Roberts received preliminary results from the April 11 lung biopsy which showed no cancerous cells. The final results were available on April 16. The results indicated organizing pneumonia with abscess formation but no malignancy. Dr. Roberts interpreted this as the first conclusive evidence that Mr. Campbell did not have lung cancer.

[76] Given the preliminary results of the lung biopsy, Dr. Roberts ordered a CT scan of Mr. Campbell's head on an expedited basis on April 15. The CT scan showed a progression of edema and increased intracranial pressure. On the same day, Dr. Roberts consulted an infectious disease ("ID") physician, Dr. Michael Lingley. Dr. Lingley noted at least two brain lesions and that the lung lesions did not look like typical AVMs. Although the possibility of malignant disease was high, the lack of a diagnosis to that point required a brain biopsy for histology and all types of infection. Dr. Lingley also noted that "AVM (HHT) is associated with brain abscess." He then recommended a certain course of antibiotics if it was agreed to be an abscess. The plaintiffs note that this is the first mention of abscess in the medical records to that point. Dr. Roberts testified in cross-examination that, throughout his care of Mr. Campbell, he was unable to distinguish the likelihood of brain metastases versus an infectious or inflammatory disorder.

[77] On April 16 Dr. Roberts sought a neurological consultation at St. Michael's Hospital. After speaking with neurosurgeon Dr. Ginsberg, Mr. Campbell was transferred to St. Michael's for neurosurgical assessment.

[78] Mrs. Campbell testified she was not consulted about this transfer. However, Dr. Roberts' note of April 16 indicated that he spoke to Mrs. Campbell. His evidence was that he would have made every effort to contact her to avoid a situation where she came to visit her husband at Southlake and found he was no longer there. Throughout his time at Southlake, Mrs. Campbell understood her husband had cancer and his doctors were trying to figure out the type of cancer he had. No one ever mentioned the possibility of infection or abscess to her or that he would be transferred to St. Michael's Hospital because of the possibility of neurosurgery.

[79] Once at St. Michael's Hospital, Mr. Campbell underwent a head MRI on April 17, 2009. The radiologist reported that the images were "more in keeping with intracranial abscesses given the changes on the diffusion weighted sequences." As the definitive results of the April 11 open lung biopsy were now available, a craniotomy was scheduled for April 17, 2009. The craniotomy was cancelled as Mr. Campbell experienced a significant epistaxis (nosebleed) just before the surgery.

[80] The craniotomy took place on April 18. The results confirmed that the lesions in the brain were abscesses and not cancerous tumours. The abscesses were drained and antibiotic therapy commenced. On May 8, Mr. Campbell underwent a second craniotomy to drain and remove recurrent brain abscesses. He was discharged to Southlake on May 26 where Dr. Roberts resumed his care. Dr. Roberts ordered another imaging study of Mr. Campbell's chest. His evidence was that he was still looking for cancer which the plaintiffs submit is yet more evidence of his tunnel vision and relentless pursuit of a cancer diagnosis despite all of the other evidence available.

[81] Mr. Campbell underwent rehabilitation therapy at Westpark Health Centre between May 29 and July 7.

[82] On July 29 Mr. Campbell was examined by Dr. Faughnan who noted that he had no "detectable pulmonary AVMs" but that if the abscesses were caused by his HHT condition, they could have been caused by "microscopic pulmonary AVMs."

[83] The parties agree that Mr. Campbell has suffered permanent cognitive deficits which affect his memory, concentration, language, executive function, informational processing and vision.

The Legal Issues in this Action

[84] The plaintiffs herein have the burden of proof, on a balance of probabilities, of showing that the defendants were negligent. In proving negligence, the plaintiffs must prove each of the following elements in relation to each defendant physician:

- (a) The defendant physicians owed them a duty of care (this is acknowledged by the physicians and need not be proven);
- (b) The defendant physicians breached that duty of care;
- (c) That the plaintiffs suffered loss or injury (and establish damages based on evidence); and
- (d) That the defendant physicians' conduct was the legal cause of the plaintiffs' injury or loss.

[85] Where liability is established in relation to any defendant physician, the court must determine if Mr. Campbell was contributorily negligent.

Summary of the Parties' Positions

The Plaintiffs

[86] The plaintiffs submit that the defendants did not apply the principles of differential diagnosis. A differential diagnosis is meant to ensure that all possible explanations for a patient's symptoms are considered and that the most serious or life threatening of those

conditions are treated promptly. The essence of a differential diagnosis is the consideration of all of the possible causes for a patient's symptoms and then ruling them in or out starting first with the most serious.

[87] The plaintiffs take the view that the possibility of a brain abscess should have been considered in the face of Mr. Campbell's symptoms of fever, headache and neurologic deficit. However, the defendants did not list brain abscess in their differential diagnosis nor did they take steps to rule it in or out.

[88] The defendants' argument that brain abscess was statistically less probable than cancer does not excuse their failure to consider it. While cancer may be more prevalent than brain abscess, that is irrelevant. The principles of differential diagnosis required the defendants to go beyond the most likely explanation and consider all possible explanations.

[89] Mr. Campbell presented with symptoms of a lung infection. He reacted to the treatment as expected, that is, the lesions shrunk and the fever dissipated. Treatment with antibiotics would naturally result in no ongoing signs of infection. Therefore, the defendants' argument that the lack of signs of infection allowed them to rule out brain abscess is unfounded. Further, the defendants knew that a lung infection could spread to the brain and cause an abscess.

[90] The defendants failed to properly rule out brain abscess, a serious condition which deserved priority. Their conduct implies that they either failed to consider the possibility of abscess or they considered it and then failed to take steps to rule it in or out. Either of these alternatives results in a failure to meet the required standard of care.

[91] The defendants focused on cancer to the exclusion of any other possibility and they cannot now say that they considered abscess at the material time. Their tunnel vision with respect to the diagnosis of cancer resulted in the pitfall that the proper application of a differential diagnosis is meant to avoid, namely, focusing on the most likely cause at the expense of failing to rule out the most serious potential cause first.

[92] Mr. Campbell's deficits were completely avoidable if treated with IV antibiotics. Instead he received treatments which were contraindicated such as whole brain radiation and steroids. The medical literature confirms that abscesses measuring less than three cm can be treated successfully with antibiotics alone. Without the proper treatment, Mr. Campbell's abscesses were left to grow and ultimately caused permanent neurologic deficits.

[93] The defendants' contention that Mr. Campbell would have suffered the same deficits even if the diagnosis had been made during the first admission is not tenable. In order to accept this theory means the court must accept that Mr. Campbell, once properly treated, would have suffered the neurological impairments in any event. That position simply lacks common sense.

[94] In terms of damages, the plaintiffs rely on the evidence of their experts with respect to Mr. Campbell's level of function and future needs. His permanent deficits mean he can no longer work and will require 24 hour care and supervision.

The Defendants

[95] The defendants quite properly concede that Mr. Campbell's presentation with multiple lesions in his lungs and brain was considered as primary lung cancer with metastases to the brain as the differential diagnosis throughout his admissions to Southlake. In hindsight it is now known that the lesions were brain abscesses, a more unusual condition than lung cancer. The question the court must consider in this case is whether the defendant physicians acted reasonably in their care of Mr. Campbell without the benefit of that hindsight.

[96] Mr. Campbell was a life-long smoker. The diagnostic imaging showed lung lesions with "spiculated" borders which are typical for cancer. Dr. Warde, an experienced radiation oncologist, agreed that the most likely diagnosis was cancer as did neurologist Dr. Sheldon Baryshnik. While both infection and cancer were investigated, neither condition was positively diagnosed while Mr. Campbell was at Southlake. The working diagnosis that Mr. Campbell had cancer remained most likely and was not dispelled by specialists as indicated above.

[97] While a small percentage of HHT patients can develop pulmonary AVMs, this will increase the risk of brain abscess only if the AVMs are clinically significant. Definitive diagnostic imaging has never shown that Mr. Campbell has clinically significant AVMs.

[98] Mr. Campbell presented with a very rare condition and symptoms that led the defendant physicians invariably towards a working diagnosis of lung cancer. They proceeded appropriately and with an overriding concern for Mr. Campbell's best interests. While the defendant physicians may have failed in their diagnosis of Mr. Campbell's condition, they met the standard of care required. Their care of Mr. Campbell cannot be judged using the hindsight now available. Further, even if the defendant physician's fell below the standard of care, their conduct did not contribute to Mr. Campbell's unfortunate outcome, which was inevitable due to the advanced stage of its process even before he was admitted to Southlake.

The Relevant Legal Principles

[99] Each physician must meet a certain standard of care when treating a patient. That standard is described as the physician exercising "the degree of care and skill which could reasonably be expected of a normal, prudent practitioner of the same experience and standing, and if he holds himself out as a specialist, a higher degree of skill is required of him than of one who does not profess to be so qualified by special training and ability."³

[100] In applying the requisite standard of care to the particular circumstances of this case, a number of important principles must be considered as follows:

- (a) A finding of negligence may still be made even where competent experts disagree about a physician's diagnosis or treatment;⁴
- (b) A higher standard of care is required when dealing with a potentially life threatening condition;⁵

³ *Crits v. Sylvester*, [1956] 1 D.L.R. (2d) 502, O.J. No. 526 (CA) at para. 13, aff'd [1956] S.C.R. 991.

⁴ *Crawford (Litigation Guardian of) v. Penney*, [2003] O.J. No. 89 (Sup. Ct.) (QL), at paras. 236-249.

- (c) An error in judgment is a valid defence only if it is made by a physician while exercising reasonable care;⁶
- (d) Where clinical judgment is exercised, it must be based on information that is as complete as reasonably available and possible in the circumstances, including tests or consultations that should have been carried out but were not;⁷
- (e) In properly diagnosing a patient, a physician must take a full history, perform proper examinations and testing, and consult or refer as necessary. When consultations are requested, the communication between the referring physician and the consultant should be meaningful and informative as to what steps were taken and what steps should be taken next;⁸
- (f) Using the process of differential diagnosis, a physician is required to identify the potential causes of a condition, and then through a systemic comparison and contrast of clinical findings eliminate the most serious potential causes first.⁹ A process of diagnosis that focuses on the most likely explanation is inconsistent with a proper differential diagnosis.¹⁰ Focusing on one diagnosis, to the exclusion of all others, without constant reassessment and reconsideration may be negligent.¹¹ However, a physician must not be held to a standard of practice in which he or she is always anticipating a worst case (but most unlikely) scenario;
- (g) Misadventure is not medical negligence and liability should not be imposed on doctors for everything that happens to go wrong;¹²
- (h) The standard of care should not be assessed with the benefit of hindsight, but in light of the medical knowledge a physician ought to have reasonably possessed at the relevant time;¹³
- (i) The law requires reasonable care and not infallibility. Reasonable physicians make mistakes.¹⁴ The honest and intelligent exercise of reasonable judgment by a

⁵ *Williams (Litigation Guardian of) v. Bowler*, [2005] O.J. No. 3323 (Sup. Ct.) (QL) at para. 250.

⁶ *Wilson v. Byrne*, [2004] O.J. No. 2360 (Sup. Ct.) (QL) at para. 28.

⁷ *Crawford*, *supra* note 4 at para. 229.

⁸ *Bergen Estate v. Sturgeon General Hospital District No. 100*, [1984], A.J. No. 2575, 52 A.R. 161 (Alta. Q.B.).

⁹ *Adair Estate v. Hamilton Health Sciences Corp.*, [2005] O.J. No. 2180 (Sup. Ct.) (QL) at para. 116.

¹⁰ *Ibid* at para. 153.

¹¹ *Williams*, *supra* note 5 at para. 243.

¹² *Crits*, *supra* note 3 at paras. 15-16.

¹³ *Ter Neuzen v. Korn* [1995] 3 S.C.R. 674, [2003] S.C.J. No 79 at para. 47.

¹⁴ *Felix v. Red Deer Regional Hospital Centre*, 2001 ABQB 545, [2001] A.J. No. 877 at para. 80.

physician, even if wrong, is not negligent.¹⁵ Further, an error in diagnosing a condition is not determinative of negligence;¹⁶ and

- (j) Given the normally lengthy time between the treatment at issue and the time of trial, evidence from a physician as to his or her invariable practice should carry great weight with respect to how the physician acted on the day in question.¹⁷

The Use of Expert Evidence on the Issue of Standard of Care

[101] There were six experts who testified for the plaintiffs and five for the defendants on the issue of negligence. The parties consented to having more than three experts testify for each of them given the number of defendants and the complex issues at stake.

[102] A few basic principles regarding the use of expert evidence should be set out at this point. First, given the technical nature of the evidence the court may not come to conclusions based on its own opinion but rather, conclusions must be drawn from the evidence of qualified witnesses.¹⁸ This includes an acceptance of the possibility that opinions of physicians may diverge. However, the court may not choose between competing opinions or accepted schools of thought. Rather, negligence occurs only where there the physician has violated universally accepted rules of medicine.

[103] The court must be careful not to make a conclusion about a breach of the standard of care without specific expert evidence which supports that conclusion.¹⁹ Further, the expert's opinion on standard of care must not be based on what that expert may have done in that set of circumstances, rather the opinion regarding the standard of care must relate to the standard of care required of the defendant(s) in particular circumstances.²⁰

The Evidence of the Defendant Doctors and the Related Expert Evidence

Defendant - Dr. Murray Weingarten

[104] Dr. Weingarten has been a specialist in internal medicine since 1983. He began working at Southlake (then known as York County hospital) thereafter and has been working there as an internal medicine consultant ever since. Dr. Weingarten was the Most Responsible Physician ("MRP") for Mr. Campbell during the first admission (March 17-26, 2009).

¹⁵ *Wilson v. Swanson*, [1956] S.C.R. 804.

¹⁶ Ellen I. Picard & Gerald B. Robertson, *Legal Liability of Doctors and Hospitals in Canada* 4th ed. (Toronto: Carswell, 2007) at 301.

¹⁷ *Turkington v. Lai*, [2007] O.J. No. 4418 (S.C.J.).

¹⁸ *Hajgato v. London Health Association* (1982), 36 O.R. (2d) 669 at para. 36 aff'd 44 O.R. (2d) 264 (C.A.).

¹⁹ *Campbell v. Hess* (unreported) (June 8, 1994), Toronto 51145/90 at p. 22 (Ont. Gen. Div.).

²⁰ *Ibid.*

[105] When Mr. Campbell presented at the ER on March 17, 2009, he complained of left peripheral vision changes. His HHT history was noted, as was his 40 year, pack-a-day, smoking history. Dr. Weingarten was consulted and ordered the St. Michael's Hospital HHT related records. He also ordered an ENT and a neurology consult and admitted Mr. Campbell. He did a physical examination which showed nothing abnormal. Mr. Campbell's lab results showed low iron but normal white blood count. Dr. Weingarten reviewed a chest x-ray done that day and another done three days earlier. Both showed pulmonary densities in the right side of the chest. Dr. Weingarten noted that the lung lesions could be metastases, HHT related or infectious disease. A CT scan from March 14 showed a head lesion in the right occipital lobe. He noted that this could be the result of infarction (stroke) or metastasis with edema (swelling).²¹

The March 17 Differential Diagnosis and the Standard of Care

[106] Dr. Weingarten recognized the importance of the process of a differential diagnosis, especially where a patient presents with complicated factors. This does not mean listing all possible causes but the most likely ones, and certainly those which are life-threatening and treatable should be considered first. He would order testing and investigation, based on the most likely diagnosis and then reconsider if the results of the investigations did not "pan out."²²

[107] Dr. Weingarten's note from March 17, 2009, indicates a differential diagnosis of malignant disease, inflammatory disease or vascular anomaly related to Mr. Campbell's HHT. He acknowledged that the patient had "multiple problems" including both head and lung lesions. He ordered follow-up by way of an MRI of the head, a consult with a neurologist and a CT scan of the chest.²³

[108] The plaintiffs called Dr. Akbar Panju as an expert in the field of internal medicine. Dr. Panju agreed that malignant disease, inflammatory disease and vascular anomaly were the three primary possibilities that should have been considered by Dr. Weingarten at this point.²⁴

[109] The plaintiffs also called Dr. Patrick Gerard Cox as an expert respirologist to comment on the standard of care regarding Dr. Roberts and Dr. Weingarten. In course of his evidence, Dr. Cox indicated that Dr. Weingarten's initial differential diagnosis of lung cancer, infection or AVM on March 17 were "very plausible possibilities."²⁵

[110] The defendant physicians called Dr. Elizabeth Powell. She was qualified as an expert in internal medicine and respirology to opine on whether Drs. Weingarten and Roberts met the standard of care. Her evidence was that Dr. Weingarten's initial consideration of infection,

²¹ Consultation Report of Dr. Weingarten dated March 17, 2009, Exhibit 2A, Tab 2A, p. 15.

²² Dr. Weingarten (Cross), May 29, 2013, p. 2, LL 5-19.

²³ Consultation Report of Dr. Weingarten dated March 17, 2009, Exhibit 2A, Tab 2A, p. 14.

²⁴ Dr. Panju (Cross), May 27, 2013, p. 8, LL 9-12.

²⁵ Dr. Cox (Direct), May 15, 2013, p. 23, LL 10-11.

tumour or AVMs as his differential diagnosis (before the March 17 CT results) was appropriate.²⁶

[111] Given all of the above I do not find that Dr. Weingarten's initial differential diagnosis on March 17 (and before the CT scan done on March 24 by Dr. Doran) with respect to Mr. Campbell fell below the standard of care.

[112] After the initial differential diagnosis on March 17, Dr. Weingarten received the results of the CT scan which showed spiculated lung lesions. The plaintiffs allege that it was not appropriate for Dr. Weingarten to consider lung cancer and metastases as the working diagnosis after the CT scan results. According to Dr. Panju, Dr. Weingarten formed a type of tunnel vision, focused only on cancer and failed to rule in/rule out other treatable possibilities. In fact, according to Dr. Panju, Dr. Weingarten simply ignored the brain lesions and failed to consider the possibility of brain abscess. He requested the consultation of a neurologist but did not follow-up on it in the nine days Mr. Campbell was under his care. He proceeded to consult with the defendant Dr. Roberts who had no expertise in brain lesions. Further, given that infectious cause had not been ruled out at that point, Dr. Weingarten should have consulted with an infectious disease specialist.

[113] According to Dr. Panju, another factor which Dr. Weingarten did not appropriately consider was the effect of the patient's HHT history. He should have consulted with Dr. Faughnan with respect to the effect of HHT on the possibility of infectious versus metastatic lesions.

[114] Dr. Powell disagreed. Her view was that the bubble study (June 2008) and subsequent CT angiogram (November 2008) done on Mr. Campbell did not show any clinically significant AVMs. She relied on literature and an Irish study of HHT patients, which showed that only patients with clinically significant AVMs ever developed brain abscesses.²⁷ As such, she did not assess Mr. Campbell as having any greater risk of developing a brain abscess than a non-HHT patient.

[115] Drs. Cox, Panju and Powell agreed that the most probable diagnosis when multiple lesions are found in multiple organs is cancer. They also agreed that the patient's smoking history (a pack a day for 45 years) would exponentially increase the risk of lung cancer as compared to a non-smoker. Mr. Campbell's age at the time was also a risk factor.

[116] Most importantly, Drs. Cox, Panju and Powell agreed that spiculated lung lesions are more typical for cancer than for pneumonia. Dr. Weingarten reviewed and relied on the report of Dr. Lisa Thain dated March 17, 2009, which showed brain lesions which she reported as likely being metastases.

[117] Dr. Weingarten also relied on the March 20 MRI report by Dr. Silmberg, which showed ring enhancing brain lesions that likely represented metastases with a differential diagnosis. Dr.

²⁶ Dr. Powell (Direct), June 17, 2013.

²⁷ Dr. Powell (Cross), June 18, 2013, p. 39, line 9.

Weingarten was not qualified to interpret the head MRI himself. He relied on Dr. Silmberg's expertise.

[118] Dr. Weingarten was criticized for the wording of his requisition to Dr. Silmberg. The requisition simply said "Lesion, right occipital lobe." His evidence was that he did not feel it was his position to lead the radiologist to a particular conclusion. He did not speak to Dr. Silmberg either before or after the MRI was done; however, he relied on its results. His view was that at this point he had a patient with multiple lesions that were described in multiple reports as possibly malignant. While he had not excluded abscess or infection entirely from his differential diagnosis, I infer from his evidence that the preponderance of information he had available favoured a diagnosis of tumour and he wanted to work towards excluding that first.²⁸

[119] I find that the change in Dr. Weingarten's differential diagnosis of preferring cancer over infection up until Dr. Doran's deferred needle lung biopsy on March 24 did not fall below the standard of care for the following reasons:

- (a) It was clear, and the experts agree, that Mr. Campbell's smoking habit and age increased his risk of cancer;
- (b) The lung imaging available to that point showed spiculated lung lesions which were often, although not always, typical of malignancy;
- (c) The brain imaging available to that point showed ring-enhancing lesions which were interpreted as possible metastases. I find that Dr. Weingarten was entitled to rely on this imaging, as he was not qualified to interpret such results on his own;
- (d) I do not find his requisition wording to be unreasonable or below the standard of care for two reasons:
 - (i) I accept that he wanted an objective opinion from the radiologist without wanting to "lead the radiologist to one diagnosis or another";²⁹ and
 - (ii) I accept Dr. Powell's evidence that the requisition for the March 20 MRI provided more than what is normally provided by a referring physician and did not fall below the standard of care. Dr. Panju did not testify that the wording of the requisition fell below the standard of care;
- (e) Based on a physician's tendency to make a unified diagnosis, a pathology found in the lungs is likely connected one found in the brain. This was a reasonable suspicion according to the plaintiffs' expert, Dr. Panju;³⁰ and

²⁸ Dr. Weingarten (Cross) May 30, 2013, p. 60, LL 22-30.

²⁹ Dr. Weingarten (Cross), May 30, 2013, p. 53, LL 28-29.

³⁰ Dr. Panju (Cross), May 27, 2013, p. 29, LL 28-32 and p. 30, LL 1-4.

- (f) The enhanced CT which Mr. Campbell underwent in November 2008 did not reveal any clinically significant AVMs. Dr. Powell testified, and I accept, that an enhanced CT scan is the “gold standard” for such testing. Therefore, I find that at this point Mr. Campbell’s HHT was not the risk factor it may have been in patients with clinically significant AVMs.

[120] Accordingly, I find that Dr. Weingarten applied reasonable care, skill and judgment to this point, and reasonably concluded that a diagnosis of lung cancer with brain metastases should take priority in the differential diagnosis.

Meeting with Mr. Campbell on March 18, 2009, and the Standard of Care

[121] It was alleged by the plaintiffs that Dr. Weingarten fell below the standard of care with respect to his interview with Mr. Campbell on March 18. The plaintiffs allege that in the course of an interview with Mr. Campbell in his hospital room on March 18, Dr. Weingarten told him that he had inoperable brain tumours and his condition was terminal. The plaintiffs allege these statements were made based only on the CT scans and without the benefit of the March 20 MRI.

[122] Dr. Weingarten denies making any such statement. Moreover, his evidence was that he has never told any patient they were “terminal.” It is not a term he ever uses. Based on his usual practice and his note from March 18, Dr. Weingarten’s evidence was that he would have told Mr. Campbell that the lesions in his lung were most likely cancerous, but a tissue diagnosis was needed for certainty. To be clear, Dr. Weingarten had no independent recollection of this conversation. He relied solely on his notes and his general practice in delivering such information to patients.

[123] The plaintiffs called Nurse Norris who gave evidence that it was her understanding that Mr. Campbell would be receiving information that day that he had a terminal condition. She was not in the room at the time the conversation took place. Mr. Campbell testified that after receiving this information he wrote an email that he had inoperable brain tumours. He then requested permission from Nurse Norris to leave the hospital so that he could visit a family friend and arrange for flowers to be delivered to his wife in order to cushion the blow of the bad news. Mr. Campbell’s evidence was that Dr. Weingarten delivered this devastating news to him in a curt and formal manner and then left the room.

[124] I do not accept that Dr. Weingarten’s actions this meeting with Mr. Campbell fell below the standard of care for the following reasons:

- (a) Nurse Norris’ notes from March 18 are clear that Dr. Weingarten spent as much as 45 minutes with Mr. Campbell. This contradicts the curt formality described by Mr. Campbell;
- (b) I accept that Dr. Weingarten does not use the word “terminal” with his patients. His evidence was that the word is meaningless in a medical context and I agree. I infer from this comment that Dr. Weingarten concedes that no physician can ever know if a patient is truly “terminal”;

- (c) Dr. Weingarten was frank that he had to rely on his notes as he had no independent recollection of the conversation. The March 18 notes are clear that no definitive diagnosis was available at that time. Dr. Weingarten specifically notes that the CT studies “so far suggest metastatic disease to brain and lung.” Further, the notes clearly indicate that he is awaiting the results of the head MRI and the lung biopsy. It could not be clearer from these notes that the meeting was informational. I cannot see how Dr. Weingarten can be faulted for wanting to keep Mr. Campbell abreast of his preliminary views, while suggesting caution in the face of pending results;
- (d) Mr. Campbell may have come to his own conclusions about this meeting which were inconsistent with those of Dr. Weingarten. However, Mr. Campbell has no notes of the meeting nor was anyone else in the room. In the absence of any further confirmatory evidence that Dr. Weingarten was unduly cruel or curt in his delivery of this information, I cannot find that the circumstances of this meeting would lead to a finding of negligence against the doctor; and
- (e) I note in passing that even if Dr. Weingarten had fallen below the standard of care with respect to this patient meeting, I am uncertain how that negligence would be related to causation or compensable in damages with respect to the actual deficits suffered by this plaintiff.

Dr. Doran’s CT Scan of March 24 and the Standard of Care

[125] Following the MRI on March 20, Mr. Campbell had been booked for a percutaneous lung biopsy. The biopsy had been deferred due to Mr. Campbell’s unacceptable INR levels. Otherwise, Mr. Campbell was doing better and Dr. Weingarten noted that “he could come and go as he pleases on passes over the weekend”, as long as it did not interfere with lab work testing.³¹ Dr. Weingarten noted on March 23 that Mr. Campbell was stable and that his visual field deficit had improved. He attributed this to the steroids which Mr. Campbell was taking. Outwardly, Dr. Weingarten agreed that by March 23 Mr. Campbell looked pretty healthy. The course of antibiotics had come to an end and Dr. Weingarten did not prescribe another course. His evidence was that the Levaquin (antibiotic) had been started for a lung infection, which had been treated.³²

[126] On March 24 Mr. Campbell was to have a lung biopsy. Dr. Doran was to perform that procedure. Prior to doing so he did a CT scan of Mr. Campbell’s chest. He noted that his lung lesions had shrunk by 50 per cent. Dr. Doran deferred the biopsy because the lesions were too small and performing a biopsy on them would have been difficult. At this point, Dr. Weingarten questioned his differential diagnosis. His note from March 24 reflects his reconsideration: “Very puzzling. Are these inflammatory rather than malignant lesions? Are they steroid

³¹ Exhibit 2A, Tab 2A, p. 20.

³² Dr. Weingarten (Cross), May 30, 2013, p. 74, LL 25-26.

sensitive tumours (e.g. lymphoma).”³³ Dr. Weingarten had a discussion with Dr. Doran who thought the lung lesions may have been arteriovenous malformations (AVMs). Dr. Weingarten did not hold that opinion as he knew that such formations did not shrink with steroid treatment.

[127] The plaintiffs allege that at this point Dr. Weingarten should have revisited his differential diagnosis. Cancer was no longer the most likely diagnosis. The fact that Dr. Weingarten insisted on retaining cancer as the most likely diagnosis was wrong given the new information available and, by continuing to test for cancer, he fell below the standard of care.

[128] Dr. Weingarten agreed that at this point the possibility of infection increased while that of cancer decreased, and that the pathology in the brain and lung were most likely the same.³⁴ Dr. Weingarten’s view was that more testing was necessary but drastic measures (such as brain surgery/biopsy) were not because, while cancer was less likely than prior to March 24, it still remained as the most likely diagnosis in his view. This was because the shrinkage of the lesions was still consistent with cancer or other complications and Mr. Campbell’s overall circumstances weighed heavily in favour of cancer.

[129] However, there are concerns about the approach taken by Dr. Weingarten at this point. They are as follows:

- (a) While AVMs do not shrink with steroid treatment, microscopic AVMs can potentially cause the hematogenous spread of infection from lung to brain. Dr. Powell confirmed this;³⁵
- (b) While Dr. Weingarten may have acted reasonably in considering other possibilities on March 24, such as inflammation, he did not act on them in a manner that reflected the possible serious effects of inflammation in the brain lesions. Rather, he focused on the lungs by ordering a bronchoscopy and referring the matter ultimately to Dr. Roberts, a respirologist;
- (c) Dr. Weingarten explained his focus on the lungs as being part of a unified diagnosis; that is, whatever the pathology was in the lungs was likely the same in the brain. This explanation may have been satisfactory after the March 20 MRI, but I do not find it sufficed after the March 24 CT scan information. That is, I find there was more reason at this point for Dr. Weingarten to have followed up on the neurology consult, which he did not do. Dr. Powell testified that she assumed Dr. Roberts would follow-up on the consult, but she conceded that was an assumption on her part;
- (d) Dr. Weingarten agreed that Mr. Campbell had presented with low-grade fever, headache and neurologic deficit in the ER. He agreed that these symptoms were classic for brain abscess. While they could also occur with any mass lesion, he

³³ Exhibit 2A, Tab 2A, p. 24.

³⁴ Dr. Weingarten (Cross), May 30, 2013, p. 93, LL 4-11.

³⁵ Dr. Powell (Cross), p. 39, LL 9-11.

was aware that Mr. Campbell had all of these symptoms before treatment with antibiotics and steroids;

- (e) The fact that infectious process was moving up the list on his differential diagnosis demanded that the brain lesions be investigated. This is based on Dr. Weingarten's own admission that the pathologies between lung and brain were likely similar. By deduction, this would mean that the risk of infection in the brain was increased significantly as of March 24;
- (f) While admitting that brain abscesses are rare and he had only seen 3 in 30 years, he agreed that this could have been the fourth and that he should have considered a brain abscess at this point in his differential diagnosis;³⁶
- (g) Dr. Weingarten did not quarrel with the proposition from the Mandell text that when an infection starts in the lung it can often end up as multiple lesions in the brain;
- (h) Dr. Weingarten had knowledge of brain abscesses and had had training in them as a resident. He knew that they are treatable by IV antibiotics and surgery, and that swelling caused by untreated abscesses can cause neurologic impairment or even death;
- (i) Dr. Weingarten admitted that he knew from the MRI that his top two differential diagnoses were infection and tumour. He also knew that infection was a treatable medical emergency for which the treatment (IV antibiotics) poses little risk to a patient. However, he never ruled infection in or out, did not treat it, or obtain an infectious disease consult. While he did order a neurologic consult, it never occurred and he did not follow-up on it. He admitted in cross-examination that by March 24 he needed to rule brain infection in or out as soon as possible;³⁷
- (j) Dr. Weingarten was aware of Dr. Doran's view in his report dated March 24, 2009, about the possibility of PAVMs (as Dr. Doran was aware of Mr. Campbell's HHT at that point). Dr. Weingarten did not recall having any discussion with Dr. Doran about this, but he did not change his focus on cancer notwithstanding Dr. Doran's "wondering" in his report about a vascular component (i.e. something other than cancer); and
- (k) At the time of Mr. Campbell's discharge on March 26, 2009, Dr. Weingarten knew that the possibility of abscess was far stronger than before, yet he released him from hospital without antibiotics. He conceded that it would have been a

³⁶ Dr. Weingarten (Cross), p. 34, LL 3-6.

³⁷ Dr. Weingarten (Cross), p. 97, LL 5-14.

“good idea” to leave Mr. Campbell on antibiotics on spec until the bronchoscopy results came back.³⁸

The Plaintiffs’ Expert – Dr. Akbar Panju

[130] The plaintiffs called Dr. Akbar Panju who was qualified as an expert in internal medicine. He was called to address the standard of care issues related to Dr. Weingarten. Dr. Panju completed his residency in internal medicine in 1981 and has been a professor of medicine since 2002. He has been a member of the panel who examines candidates for the competency for their fellowship in internal medicine and sets questions for the written portion of the exam and since 2009 he has been a panel member dealing with complaints and enquiries regarding the standard of care of physicians. He is currently a general internist practicing medicine at Hamilton Health Sciences and is the Deputy Chair of Clinical Services at McMaster University.

[131] While the defendants were critical of the fact that Dr. Panju worked in a teaching hospital as opposed to a community hospital (like Southlake), I accept his evidence that the only real difference between the two is the presence of residents. His view is that the difference in resources between teaching hospitals and community hospitals is becoming less and less marked.

[132] I accept Dr. Panju’s evidence that there would be little difference between his experience and that of Dr. Weingarten in terms of available resources. Dr. Weingarten did not testify that he lacked resources with respect to assessing Mr. Campbell

[133] In commenting on Dr. Weingarten’s standard of care in relation to Mr. Campbell, Dr. Panju’s view was that from the outset the differential diagnosis was narrow and focused on cancer without a tissue diagnosis. While Dr. Panju conceded that spiculated lesions can be a classic symptom of cancer, he was critical of the fact that ringed lesions in the brain were ignored. He testified that pursuing a neurologic consult was the right approach, but one which Dr. Weingarten never followed up on.

[134] According to Dr. Panju, it was not reasonable for Dr. Weingarten to focus on the lungs as he did. This was illustrated by his requesting help from Dr. Roberts, a respirologist, rather than someone with knowledge of the brain or an infectious disease specialist. So, while Dr. Weingarten made passing mention of an abscess he never took any steps to actually rule it out. Further, Mr. Campbell should not have been discharged given what was known about his brain. While infection was considered and even noted by Dr. Weingarten, his actions, testing and focus were all related to cancer.

[135] I accept Dr. Panju’s conclusion that Dr. Weingarten’s care of Mr. Campbell fell below the standard of care. While it is true that Dr. Panju did not make mention of Mr. Campbell’s smoking history in his reports he did not deny its importance as it related to the consideration of lung cancer as a possible diagnosis. However, the analysis should not end there. One cannot

³⁸ Dr. Weingarten (Cross), p. 109, LL 2-5.

invariably conclude that a smoker with an abnormal chest x-ray has lung cancer. There are other possibilities and those were not adequately considered by Dr. Weingarten according to Dr. Panju.

[136] Dr. Panju agreed that Dr. Weingarten's working diagnosis of a possible inflammatory rather than malignant process was appropriate as was his view that the process in the lung was related to that in the brain. However, the differential diagnosis for a lung lesion and a brain lesion do not necessarily mirror. Further, while the ordering of a bronchoscopy and the consultation with Dr. Roberts was not an inappropriate course, Dr. Weingarten was focusing on the lung lesions while the brain lesions received little attention other than the possibility of infection.

[137] The issue here is not the failure to consider a brain abscess by Dr. Weingarten, but as per Dr. Panju's evidence, the fact that it was considered but not acted upon which falls below the standard of care. By "acted upon" I mean that it was not ruled in or out, there was no follow-up with a neurologist, or an infectious disease expert. Further, the subsequent consultation with Dr. Roberts focused again on the lungs and not the brain. Finally, Dr. Weingarten did not take any substantial steps to consult with Dr. Faughnan. While this alone would not have been enough to fall below the standard of care (because Dr. Weingarten had reviewed the St. Michael's Hospital records from Dr. Faughnan and sent her a copy of his report), it forms part of the "tunnel vision" described by Dr. Panju with respect to focusing on cancer without having properly ruled out infection.

The Defendants' Expert – Dr. Elizabeth Powell

[138] The defendants called Dr. Elizabeth Powell to comment on the care provided by both Dr. Weingarten and Dr. Roberts. Dr. Powell is a specialist in both respirology and internal medicine. She is the Site Chief of the William Osler Health Centre in Brampton, Ontario.

[139] Dr. Powell opined that Dr. Weingarten met the standard of care required for his treatment of Mr. Campbell. In her opinion, lung cancer was an appropriate working diagnosis given Mr. Campbell's clinical history and the radiology results. Her opinion was that it was appropriate for Dr. Weingarten to transfer care when he began to question his working diagnosis.

[140] Dr. Powell agreed that a brain abscess is potentially life threatening. If no medical treatment is given, the abscess can grow. This is problematic because once abscesses reach a certain size they can no longer be treated by high dosage intravenous antibiotics; they must be excised. Therefore a prompt diagnosis of a brain abscess is important.

[141] She agreed that if she suspected a brain abscess she would want to involve an infectious disease expert so that antibiotics could be started. She also agreed that if lesions were shrinking on imaging with no signs of fever or infection she would not order a bronchoscopy. Her evidence was she "hoped" the appropriate investigations would be done.

[142] Dr. Powell was asked about how Mr. Campbell's HHT would factor into considerations in making a differential diagnosis. Her view was that since no clinically significant AVMs were detected on the CT scan Mr. Campbell had in 2008 brain abscess would be placed much lower on the differential list. However, she did not disagree with the conclusions of Dr. Dyer in his

article “Cerebral Abscess in Hereditary Haemorrhagic Telangiectasia: Report of Two Cases in a Family”³⁹ that “early diagnosis of cerebral abscesses is essential and should always be excluded by appropriate investigations.”

[143] However, while Dr. Powell’s concerns about detectable AVMs and their significance on a differential diagnosis were well placed, I find that she minimized the importance of the possibility of microscopic PAVMs and their effect on the risk of increased abscess formation, and therefore the requirement that abscess be ruled in or out.

[144] In summary, I agree with the plaintiffs’ submission that Dr. Powell’s opinion that brain abscess would be “low on the list” in terms of a differential diagnosis is in contrast with the evidence of Dr. Weingarten who agreed that by March 24 he needed to rule brain abscess out “as quickly as possible”, as brain infection was a “higher” risk than before.

Conclusions on Standard of Care – Dr. Weingarten

[145] In conclusion, I find that Dr. Weingarten’s care of Mr. Campbell fell below the standard of care as of March 24, 2009 for the following reasons:

- (a) He failed to adequately address what he conceded was his number one priority as of March 24, 2009 – brain abscess. He did not rule it in or out nor did he follow-up on a neurology consult;
- (b) I accept Dr. Panju’s evidence with respect to Dr. Weingarten focusing on cancer despite his concession that by March 24 infection was moving up on the list of potential diagnoses;
- (c) I reject Dr. Powell’s evidence that Dr. Weingarten was proceeding to rule abscess in/out in a “step-wise” fashion. I am unaware of any “steps” having been taken by Dr. Weingarten to rule abscess in or out. Further, there is no reference in Dr. Powell’s reports with respect to Dr. Weingarten proceeding in a “step-wise” fashion;
- (d) Dr. Weingarten failed to advert appropriately to the “differential diagnosis” mentioned in Dr. Silmberg’s March 24 report given what he knew at that point. Dr. Roberts, a co-defendant and also an internal medicine specialist testified that ring enhancing lesions carry an “acute inflammatory connotation”;
- (e) The underlying factor of Mr. Campbell’s HHT cannot be ignored. The fact that Mr. Campbell did not have any clinically significant AVMs on CT did not rule out the risk of microscopic ones. As per the Mandell text, HHT is a predisposing factor for brain abscess, such that the risk of brain abscess is 1,000 times greater

³⁹N.H. Dyer, “Cerebral Abscess in Hereditary Haemorrhagic Telangiectasia: Report of Two Cases in a Family” (1967) 30 J. Neurol. Neurosurg. Psychiat. 563.

in HHT patients than in the general population.⁴⁰ Dr. Weingarten was aware of Mr. Campbell's HHT condition, yet focused on the fact that he had no clinically significant AVMs on CT rather than a possible increased risk of brain abscess. Once Dr. Weingarten was at the point of finding that Mr. Campbell's diagnosis was "puzzling", I agree with Dr. Panju's report that it was incumbent on Dr. Weingarten to take positive steps to consult with Mr. Campbell's HHT specialists at St. Michael's Hospital; and

- (f) Dr. Weingarten focused on the mostly likely or probable cause (cancer) rather than dealing with the most serious cause. This offends the universally accepted rule of medicine as set out in *Adair Estate*⁴¹, "after the physician has identified the likely perils, these possible perils must be listed from the most serious to the least serious. The physician eliminates the peril that has the most severe consequences first." The experts agreed that a brain abscess is a serious and life threatening condition. As of March 24, 2009, it therefore should have been listed first in Dr. Weingarten's differential and ruled in or out as per the principle in *Adair Estate*.

Defendant – Dr. Erik Silmberg

[146] Dr. Silmberg did a five year residency in diagnostic imaging at McMaster. He explained that diagnostic imaging includes the study of x-rays but also magnetic resonance images (MRIs) and ultrasounds which are not x-rays. Dr. Silmberg has worked as a radiologist at Southlake since 2004 and since July 2012 has been the Chief of Diagnostic Imaging at Southlake and reports directly to the Chief of Staff there. In his day-to-day work he spends one to two days a week doing MRIs. He would review about 40 to 60 MRIs a day, and do about 5,000 brain MRIs a year.

[147] Dr. Silmberg became involved in this case because he was the MRI radiologist when Dr. Weingarten's requisition was received. Dr. Silmberg knew from this requisition that Mr. Campbell had had a CT scan on March 17. However, a CT scan does not give the same refined information than an MRI does. Although Dr. Silmberg has no specific recollection of this case his normal practice would be to review the previous CT scan where he would have seen two brain lesions and three chest lesions. The ring enhanced brain lesions and the spiculated lung lesions found in Mr. Campbell's case were, according to Dr. Silmberg, a classical radiological finding for metastatic disease.

[148] After completing the head MRI on Mr. Campbell, Dr. Silmberg gave a one line opinion in his report dated March 20, 2009, that "ring enhancing nodules in the brain as described likely representing metastatic lesions." His colleagues tell him that long differentials are useless to them. They want a one line conclusion. Putting something like abscess on this differential diagnosis would lead to wild goose chases and patients undergoing testing they didn't need.

⁴⁰ Mandell, *supra* note 1 at 1267.

⁴¹ *Supra* note 9 at para. 116.

While abscess would have been on the differential diagnosis it would have been very far down, and therefore not listed in his report for the reasons given above.

[149] In radiology one reports on the likelihood of findings and not what is or is not treatable. He did not agree that there were only two main diagnoses here. His view was that there was one diagnosis (metastases) that was very likely and a list of unlikely diagnoses, and abscess was at the top of that list. While he did not disagree with Dr. Cooper's finding that the most likely diagnosis is between metastases and abscess he differs with Dr. Cooper in terms of how the findings should be weighed.

[150] Dr. Silmberg had only seen one brain abscess in his practice up to March 2009, but he has seen MRIs showing multiple brain abscesses while at educational conferences. He agreed that metastases and abscesses can appear similar on conventional MRI. He agreed that while an abscess is more likely to have restricted diffusion, a considerable number of metastatic lesions will have restricted diffusion and metastatic disease is far more common than brain abscess. Because of the fact that an abscess is more likely to have restricted diffusion, Dr. Silmberg pointed out a differential diagnosis to the clinician in his report. Specifically, Dr. Silmberg said that indicating in his report that there was a differential diagnosis for ring enhancing lesions was his quiet way of mentioning a possible infection. He left the differential diagnosis "up for discussion" if the clinical finding did not match.

[151] Dr. Silmberg did not specifically mention infection in his report because based on the clinical history of the patient, the spiculated lung nodules (which are almost always malignant) and the ring enhancing lesions seen on the MRI, his view was that it was likely metastatic disease. His opinion would not have changed even taking out the clinical history. It should be noted that Dr. Silmberg was not made aware that Mr. Campbell has presented on March 14 with a fever, nor that he had been treated with antibiotics and was still on them at the time Dr. Silmberg reviewed the MRI findings. Had he known of the fever, Dr. Silmberg was candid that it would have affected the clinical history and the manner in which he evaluated the MRI images.

[152] Dr. Silmberg was not made aware that Mr. Campbell had HHT. This would not, however, have affected his findings because he was not aware that HHT patients were at a higher risk with respect to the spreading of infection.

The Plaintiffs' Experts

Dr. David Mikulis

[153] Dr. Mikulis is a staff neuroradiologist at Toronto Western Hospital. Neuroradiology is the study of the central and peripheral nervous system using imaging. He was called as an expert by the plaintiffs and qualified as an expert in radiology and neuroradiology to provide an opinion on the appropriate interpretation of the MRI imaging, the interpretation of the imaging and changes on imaging between November 2008 and March 20, 2009, and to provide an opinion on evidence with respect to the likely effect of those changes on imaging, if any, on human function.

[154] Dr. Mikulis has been instructing since 1985 and is currently a professor at the University of Toronto medical school and director of the Radiology Resident and Fellow Research Program there. As Dr. Mikulis is involved in training all radiology students who go through the radiology program at the University of Toronto he is, according to the plaintiffs, well qualified to testify about the standard one should expect of any resident seeking to qualify as a radiologist.

[155] Dr. Mikulis testified that Dr. Silmberg fell below the standard of care required in his reporting, review and interpretation of the MRI of Mr. Campbell's brain. He did a "blind review" of the March 20 MRI images and noted the two ring enhancing lesions. The two leading pathologies for such lesions are either infection or cancer. Using diffusion weighted imaging in the utilization of an MRI can assist in telling apart an abscess from a tumour. In reviewing the diffusion weighted imaging in the subject MRI he noted that there was "severe restricted water movement" for both lesions which was uncommon in cancer. Given this, he would not have removed brain abscess from the differential diagnosis as the severe restriction was a form of red flag which should have been reported to the clinician. Further, given that abscess is an urgent medical condition, it needs to be reported; otherwise the clinical importance of an MRI is minimized. As the imaging features on the MRI, in particular the restricted diffusion, strongly pointed toward cerebral abscess, Dr. Silmberg's failure to note this as a diagnostic likelihood fell below the standard of care.

[156] However, it should be fairly noted that in cross-examination Dr. Mikulis agreed that the presence of restricted diffusion does not unequivocally diagnose an abscess, that it should be viewed in the context of the case and that a brain tumour or metastases can also have restricted diffusion. He also agreed that it was clear from the requisition that the referring physician was leaning towards a likelihood of metastases, that Mr. Campbell's smoking history would have been relevant information as would have been the presence of spiculated lesions in the lung. He agreed that it is appropriate for a radiologist to narrow a differential diagnosis in order to narrow the possibility of multiple unnecessary tests on a patient.

[157] Dr. Mikulis did not find that Dr. Thain's CT scan report of March 17, 2009, fell below the standard of care, notwithstanding that she did not mention any other differential diagnosis than brain metastases, did not refer to any potential for abscess nor did she recommend an MRI.

[158] Dr. Mikulis conceded that he works in an academic setting and that he has not practiced as a general radiologist in 22 years. He also conceded that it would be difficult for him to put himself in Dr. Silmberg's shoes and that he would refuse a request to cover a shift for Dr. Silmberg.

Dr. Lori Moore

[159] Dr. Moore is a staff radiologist at Lakeridge Health in Oshawa, Ontario. She interprets MRIs about once every two weeks. The plaintiffs submitted that her familiarity with MRI imaging and her approach put her on all fours with Dr. Silmberg in terms of training and experience.

[160] Dr. Moore has served as an examiner for the Royal College and ensures candidates for a specialty in radiology met the standard of practice required to practice in the community. She has also worked as a peer reviewer for the College of Physicians and Surgeons of Ontario to review the conduct of other radiologists to determine if they have met the standard of care.

[161] Dr. Moore performed a blind review of the March 20 MRI similar to that of Dr. Mikulis. She noted two ring-enhancing lesions with surrounding edema and restricted diffusion. The lesions were described as thin walled, smooth and irregular. She gave a differential diagnosis of abscess and metastatic disease and less likely demyelination. In her second report she replaced demyelination with infarction. Her evidence was that this differential diagnosis was based on her opinion that a ring-enhancing lesion with restricted diffusion is more common with infection than metastases.

[162] Dr. Moore agreed that a radiologist is influenced by the diagnostic question in the requisition and the clinical context that the radiologist knows about. As well, the role of a radiologist involves much professional judgment in terms of analyzing and weighing findings, the drafting of reports and what to include in them. She acknowledged that the presence of individual judgment means that radiologists have different styles and variability in their reporting. She fairly conceded that although she was doing a blind review she knew that she was looking for “something” in the context of this lawsuit and she had the luxury of time and the ability to consult academic literature in the course of conducting her review.

[163] The defendants were critical of Dr. Moore for not having reviewed Dr. Silmberg’s discovery transcript in advance of preparing her report. In not doing so, Dr. Moore would not have known whether or not Dr. Silmberg reviewed Dr. Thain’s report of the CT scan on March 17, what knowledge he had of diffusion weighted imaging, or why he ruled certain things in and out of his report. Further, the defendants submitted that Dr. Moore was combative in her manner even on uncontroversial propositions, such as whether it would be reasonable to list every possibility suggested by an image in a differential diagnosis.

[164] The defendants were also critical of Dr. Moore for her position with respect to Dr. Thain’s report, claiming she did not apply the same standard to Dr. Thain as she did to Dr. Silmberg, and that her position with respect to Dr. Thain was influenced *ex post facto* by Dr. Thain’s release by the plaintiffs as a defendant.

[165] Dr. Moore’s view was that Dr. Silmberg’s report fell below the standard of care of a community radiologist in 2009 because the lesions were highly suggestive of abscess and this was not mentioned in this report. While conceding that abscess is rare, her view was that this did not excuse the radiologist from making such a finding or highlighting its importance. She went on to testify that Dr. Silmberg’s failure to mention abscess in his report was an egregious error and would have resulted in him failing to pass Royal College exams to qualify as a specialist in radiology.

The Defendants’ Expert – Dr. Evan Roberts

[166] Dr. E. Roberts completed his specialty certification in radiology in 1997. Since then, he has practiced as a community radiologist in Sudbury at the Health Sciences North hospital. He reviews MRIs one week out of six which is about 250-300 MRIs per week. Ten to fifteen per cent of these MRIs are of the brain. He was qualified as an expert in radiology and opined that Dr. Silmberg met or exceeded the standard of care required.

[167] Dr. E. Roberts conducted a blind review of the March 20 MRI. He also reviewed the March 20 MRI requisition, the March 17 CT scan and the reports by Drs. Mikulis and Moore. His evidence was that although the lesions showed restricted diffusion, which is common in abscesses, it can also be present in about 20 per cent of cancerous lesions. He stated some concern with the “imprecise nature” of diffusion restriction as well. He used the previous workweek as an example in which he saw three cancerous lesions, one of which had restricted diffusion. However, since cancer is much more prevalent than lesions, it would be a more likely finding than abscess in these circumstances. His view was that, in a patient who had spiculated lung masses and multiple lesions in the head, there was an overwhelming likelihood of brain cancer. He came to this conclusion using the same information as Dr. Silmberg.

[168] Dr. E. Roberts reminded the court that radiology is a tool in the diagnostic process but that radiologists do not diagnose patients. His evidence was that he sees about one brain metastases per week and one brain abscess every two to three years. He has seen only one brain abscess caused by hematogenous spread. He also gave evidence that, in his experience, brain abscess usually presents as a single lesion and not multiple ones. He described his experience as similar to Dr. Silmberg with respect to being a radiologist working in a community setting.

[169] Dr. E. Roberts was not aware that patients with HHT have a higher risk of infection including brain infection. However, as he did not see any AVMs in Mr. Campbell’s imaging, his opinion would not have changed even if he had known about the HHT factor.

[170] With respect to Dr. Silmberg’s report, Dr. E. Roberts’ view was that it contained clinically relevant and accurate information. Further, radiologists do not report on every differential diagnosis as that would not be helpful to a clinician.

[171] Dr. E. Roberts cautioned the court about a hindsight review in this case. The experts all knew immediately that there were masses in Mr. Campbell’s head with a presumptive diagnosis of metastases. Therefore the reviewer would obviously looking for an alternative; what Dr. E. Roberts referred to as a “confounding variable.”⁴²

Conclusions on Standard of Care – Dr. Silmberg

[172] I do not find that Dr. Silmberg fell below the standard of care expected of a community radiologist of his experience in reviewing and reporting on MRI images based on the following:

- (a) I do not find that Dr. Silmberg was required to specifically include abscess as part of the differential diagnosis in his report;

⁴² Dr. Evan Roberts (Cross), June 21, 2013, line 28.

- (b) I accept his evidence that metastases were the most likely diagnosis. While abscess was at the top of the list of unlikely diagnoses, it was not far enough up to be mentioned;
- (c) I accept Dr. Robert's opinion in this regard with respect to Dr. Silmberg attempting to be helpful to clinicians, as opposed to listing all the unlikely diagnoses. Dr. E. Robert affirms Dr. Silmberg's view that this would potentially subject patients to a "plethora" of un-useful tests.⁴³ Most importantly, being helpful to clinicians does not include listing severe but **improbable** diagnoses, according to Dr. E. Roberts.

[173] It is important to note Dr. Moore's evidence in this regard in which she emphasized that radiologists are called upon to exercise significant professional judgment when writing their reports. This involves weighing findings and determining what should be put into a report such that it will be helpful to a clinician. I find in this case that Dr. Silmberg's exercise of professional judgment did not fall below the standard of care. After reviewing Dr. Thain's CT scan, the requisition and the MRI images, Dr. Silmberg determined that the ring enhanced brain lesions and the spiculated lung lesions were a classic finding for cancer. While being aware of the possibility of abscess on restricted diffusion, he weighed his findings and determined that metastatic disease was most likely given all the information before him.

[174] I accept Dr. E. Roberts' evidence that, as recently as the week before he gave his evidence, he encountered three cancer patients, one of whom showed restricted diffusion. Dr. Silmberg was aware of the possibility of abscess and noted a differential diagnosis in his report. Dr. E. Roberts gave evidence that, and I accept, in his experience at least 20 per cent of brain metastases will show restricted diffusion. Dr. Silmberg using his experience, the information he had before him and exercising his professional judgment, chose not to enter into the arena of improbable but possible findings. I do not find in doing so he fell below the standard of care.

[175] The plaintiffs allege that Dr. Silmberg fell below the standard of care by reporting metastases when brain abscess was more likely. I do not agree. Dr. Silmberg exercised his judgment and the opinion in his report was well founded because:

- (a) Brain abscesses are often solitary as opposed to presenting as multiple lesions (corroborated by Dr. E. Roberts);
- (b) The majority of ring enhancing lesions in the brain are cancerous and not infectious (corroborated by Dr. Mikulis);
- (c) Restricted diffusion does not inevitably lead to a conclusion of abscess as a tumour or metastases may also (but not always) present with restricted diffusion;
- (d) The presence of spiculated lung lesions is a classic presentation for primary lung cancer (as per the evidence of Dr. Mikulis); and

⁴³ Dr. Erik Silmberg (Cross), p. 2, LL 19-23 and affirmed by Dr. Evan Roberts (Direct) June 21, 2013.

- (e) In the clinical information available to Dr. Silmberg there was no indication of infection. His evidence was that the history available to him would also not have supported a finding of abscess because there was no history of fever or low white blood count.

[176] Care must be taken in reviewing the evidence of the plaintiffs' experts with respect to any blind review. They were fair in their admission that they knew of the lawsuit and knew they were looking for something. Since brain metastases were obvious to them, they knew they were looking for something that was less common. While I accept that the plaintiffs' experts gave their evidence fairly, it would not be surprising if it was somewhat coloured by a degree of hindsight. As Dr. E. Roberts' put it; without hindsight, most radiologists would not have made this "weird one in a million curveball."

[177] The court must also ensure that Dr. Silmberg's actions are not judged by hindsight. As per *Lapointe v. Hopital Le Gardeur*⁴⁴ "in order to evaluate a particular exercise of judgment fairly, the doctor's limited ability to foresee future events when determining a course of conduct must be borne in mind. Otherwise...the doctor will be held accountable for mistakes that are apparent only after the fact." Dr. Silmberg made a judgment call based on the information available at that time as well as his personal experience. As we now know, his judgment was incorrect but that does not mean it fell below the standard of care.

[178] I reject the position of the plaintiffs with respect to Dr. Silmberg justifying the contents of his report on the basis that listing possibilities would mean clinicians would be faced with ruling all of them in or out. I find that based on the evidence of all of the experts, a radiology report is not meant to be a diagnosis but a likely finding or findings to help with a diagnosis. Dr. Silmberg was following what was clearly accepted practice of a community radiologist (corroborated by Dr. E. Roberts) with respect to listing the most likely finding. In doing so he had regard to all of the factors listed in paragraph 175 above and the opinion in his report therefore did not fall below the standard of care.

Defendant – Dr. David Kenneth Roberts

[179] Dr. Roberts has been a specialist in internal medicine and respirology since 1985. He has worked as a consultant staff physician at Southlake since he obtained his qualifications. Dr. Roberts is a consultant to the Canadian Medical Protective Association (CMPA) and prepares case statements for them for litigation purposes. He usually works at Southlake six days a week doing a combination of ICU work and his own cases.

[180] Dr. Roberts fairly conceded that he had no recollection of this case and his evidence was largely focused on the medical records from the relevant time period. He knows Dr. Weingarten

⁴⁴ [1992] 1 S.C.R. 351, S.C.J. No. 11 at para. 28.

very well as they both began work at Southlake in the same week. When he received this referral from Dr. Weingarten it would have been his practice to review the CT scan of March 17th and all reports and MRIs. He always reviews MRI reports because an MRI is more precise than a CT scan and better for viewing brain disturbances. His evidence was the MRI report showed metastases as the most likely possibility, however ring-enhancing lesions carry what he described as an acute inflammatory connotation.

[181] Dr. Roberts explained that a bronchoscopy was booked for Mr. Campbell with the purpose of obtaining a rich sample close to an inflammatory site. He was looking for abnormal cells. Dr. Roberts was aware of the fact that Mr. Campbell had HHT from his review of the file, however he knew that a pulmonary AVMs would not shrink in response to medication. After reviewing the file and speaking to Mr. Campbell's wife about the proposed bronchoscopy, he prescribed Decadron, an anti-inflammatory intended to reduce any swelling in the brain.

[182] He was asked about the shrinkage of the lung lesions. His view was that lesions, even lung lesions, can shrink and disappear depending on the circumstances and sometimes in response to Decadron.

[183] The bronchoscopy was booked for March 26. As of the evening of March 24 Dr. Roberts described Mr. Roberts as gravely ill with recent lesions. He wanted to proceed with the gentlest form of testing first. His evidence was that a negative result from the bronchoscopy would not mean the end to testing; other pursuits would have to be considered.

[184] Dr. Roberts met with Mr. Campbell for the first time on the evening of March 25. He examined him and noted his smoking history and prior residence in Mississippi. The bronchoscopy took place on March 26 without complications. As the results would take two days to process, Dr. Roberts allowed Mr. Campbell to leave the hospital (at Mr. Campbell's request) and kept him on Decadron because of the concern about inflammatory lesions. He gave Mr. Campbell his pager number (which he only does about twice a year) to ensure ease of contact and would have reviewed with him what to look for in terms of problems that might require contacting him or re-admission. In cross-examination Dr. Roberts agreed that at this point brain infection was on his differential diagnosis but he had not ruled it in or out nor mentioned it in his discharge note.⁴⁵

[185] Dr. Roberts asked Mr. Campbell to come back to the hospital on March 31 to review the bronchoscopy results. He prepared a discharge note and copied it to Dr. Faughnan. His conclusion in the discharge note of likely lung malignancy was because of Mr. Campbell's age, smoking history and the radiology reports. As of March 26 Dr. Roberts noted that Mr. Campbell looked stable, cognitively normal and did not live alone. Mr. Campbell knew how to reach Dr. Roberts and no test results would be available during the upcoming weekend, so Dr. Roberts was comfortable with allowing Mr. Campbell to leave.

⁴⁵ Dr. Roberts (Cross), p. 174, LL 2-9.

[186] Mr. Campbell was brought back to Southlake on March 30 due to clouded cognition and different behaviour. Dr. Roberts saw him on March 31 and noted “disorganized thinking.” Dr. Roberts was candid that he was alarmed about Mr. Campbell’s neurologic deterioration. Mr. Campbell was also suffering from delirium so Dr. Roberts ordered that he be put in restraints. He prescribed Dilantin, an anti-seizure medication, to be administered by IV. Mr. Campbell’s temperature was also elevated to 38.5 degrees.

[187] When admitted to Southlake on March 30, the emergency physician prescribed an antibiotic, Ceftriaxone for possible bronchitis or pneumonia. As of April 1 the antibiotic course was continued but Mr. Campbell was still cognitively impaired. The bronchoscopy results were negative, so Dr. Roberts wanted to move to the next stage which was a needle lung biopsy. Dr. Roberts wanted to perform another independent test to disclose possible cancers. Dr. Roberts was asked why he did not revisit his differential diagnosis at this time given that Mr. Campbell had returned to the hospital on March 30 in a different condition than expected. He agreed that given Mr. Campbell’s fever and delirium he would need to be re-evaluated regarding the possibility of infection.

[188] Dr. Roberts was aware that on re-admission, Dr. Zahirieh prescribed Ceftriaxone via IV. Dr. Roberts was under the impression that this antibiotic would be continued for seven days. He agreed however that charting indicated it was stopped as of April 2. Dr. Roberts was the Most Responsible Physician for Mr. Campbell at that time.

[189] Dr. Roberts was asked if he had arranged a consult with neurosurgery. He did not because he believed that Mr. Campbell had multiple manifestations emanating from one disorder. He felt that neurosurgery would have wanted him to exhaust the least invasive testing first in any event. However, Dr. Roberts did refer Dr. Roberts to a neurologist, Dr. Baryshnik, on April 2 hoping for a different perspective on the range of diagnostic considerations. Dr. Baryshnik provided three possibilities for the change in Mr. Campbell’s mental state; chemical disturbance from cancer, local swelling at the site of the two lesions and toxic effect from the steroid. He suggested an EEF but did not suggest surgery or suggest any reordering of priorities with respect to the diagnostic pursuit. He tacitly acknowledged metastases but was waiting for a tissue diagnosis.

[190] The needle biopsy took place on April 3. While awaiting the results, Mr. Campbell remained agitated and confused. Dr. Roberts thought this might be related to the continued use of steroids on the patient. Dr. Roberts then proposed a consultation with radiation oncology on April 6. Over the weekend (April 4-5), Mr. Campbell remained restless and confused. He had suffered a pneumothorax (leak in the lung) in the course of the needle biopsy and had been uncomfortable as a result.

[191] Dr. Warde attended at Southlake on April 6 at Dr. Roberts’ request. Dr. Roberts was looking to Dr. Warde for a review of the case as Mr. Campbell’s health had deteriorated since the previous week. Dr. Roberts’ evidence was that he thought Mr. Campbell was dying and did not want to lose any options for treatment. Dr. Roberts testified that he would have confidence in Dr. Warde’s choice of treatment and knew that Dr. Warde had the ability to refer him

elsewhere as well. He also knew that Dr. Warde had clinical experience with brain lesions, not all of which were cancerous.

[192] By April 9 Dr. Roberts had the results from the needle lung biopsy which were negative. He cannot recall if he discussed these results with Dr. Warde but he was confident that Dr. Warde would be aware of the limits of accuracy of any biopsy procedure. Dr. Roberts also could not recall if he told Dr. Warde about the shrinkage of Mr. Campbell's lesions while on antibiotics, his history of infection, his initial fever and the fever on March 30 or that he had been on Ceftriaxone. He agreed that all of this was relevant information with respect to a cancer diagnosis.

[193] Dr. Roberts decided to proceed with the next most invasive procedure with Mr. Campbell. This was an open lung biopsy; often called a wedge resection. Dr. Lee did this on April 11. In his note of April 10 Dr. Lee indicates that a tissue diagnosis was needed because Mr. Campbell had radiation scheduled for Sunday April 12. Dr. Lee's April 11 note did not indicate that he found any pus or signs of infection. The actual results of the wedge resection were not available until April 16. They showed abscess formation with no evidence of tumour or malignancy.

[194] Dr. Roberts' notes indicate that Mr. Campbell's condition remained unchanged by April 14. On April 15 his notes show that he made a telephone order for an infectious disease consult. He wanted Dr. Lingley (the infectious disease specialist) to review the case. He also ordered updated blood work and a cranial CT scan. By 6:15 that evening he had the results of the cranial CT which showed the same lesions as before but with more swelling and some signs of intracranial pressure. The next day Dr. Roberts had the written results of the wedge resection which showed no evidence of malignancy.

[195] By the evening of April 15 Dr. Roberts realized he had to change course. He wanted to review the results of the cranial CT with neurosurgery the next morning. He spoke to Mrs. Campbell about a transfer to St. Mikes on the evening of the 15 and the morning of the 16. On April 16 he spoke to Dr. Lingley who recommended a brain biopsy and Ceftriaxone IV both before and after the biopsy. Dr. Roberts agreed that an infectious disease consultation was less invasive than brain radiation, a needle biopsy or a wedge resection.

[196] Dr. Roberts' discharge note of April 16 indicates that he is still unable to distinguish whether or not Mr. Campbell has metastatic lesions or an inflammatory disorder. He noted his concern about the edema shown on the CT cranial scan from two days before and was endeavouring to have Mr. Campbell transferred as soon as possible for a brain biopsy.

[197] Dr. Roberts has seen some brain abscesses in his career and knew they can be both serious and life threatening because of increased cranial pressure. He was aware that increased cranial pressure can lead to downward herniation and permanent impairment of brain function or death. While Dr. Roberts was aware that brain abscesses are treatable with IV antibiotics and draining, he did not agree with the proposition that empiric antibiotics should be started as a precaution. His concern was that doing so could lead to an incomplete response or resistance. It is better to first identify the organism and then treat it. While he was aware of authoritative texts

that recommended empiric therapy, his view was that targeted therapy was better than just guessing.

[198] Dr. Roberts agreed that where brain abscesses are concerned, size matters. An abscess of less than three centimetres is treatable with IV antibiotics because the drugs are more likely able to penetrate the abscess when they are this size. For this reason he agreed that prompt diagnosis is important for treatment and to ensure that a patient does not suffer any neurologic deficit.

[199] Dr. Roberts was aware that brain abscesses can spread from other infected sites such as the lungs. He was also aware that the classic presentation for brain abscess is fever, headache and focal-neuro deficit. He agreed that 50 per cent of patients with brain abscesses present without fever.

[200] Dr. Roberts did not disagree that steroid treatment could be counter-productive because of the potential to allow bacteria to multiply. But, as is often the case, potential value must be weighed against adverse-affect where there is no firm tissue diagnosis. He agreed that both a bronchoscopy and a lung biopsy are invasive tests but disagreed that they lacked utility while a patient was on antibiotics. His view was that an antibiotic may not be useful against all organisms or there may be several diseases present at the same time.

[201] Dr. Roberts has seen some cases of HHT and is aware of the correlation between HHT and brain abscesses with respect to HHT patients having an increased risk of the spread of infection. He knew that microscopic AVMs can allow infection to travel within the body.

[202] Dr. Roberts agreed that, in his experience, it is rare for tumours to shrink by 50 per cent within seven days while on Levaquin and was not surprised that Dr. Doran and Dr. Weingarten raised alternative possibilities when they saw this. In cross-examination he agreed that this would support an infectious process.⁴⁶ He further agreed that the lung and brain conditions were likely the same in this case. While he agreed that lung infection can propagate to the brain, this is very rare, and he sees brain abscesses only about once a year in his practice. Dr. Roberts explained that he does not usually write out exhaustive differential diagnoses. While the possibility of brain abscess would have been on his list on the differential diagnosis as of March 24 he did not actually refer to it until his April 16 discharge note because the case had changed dramatically by that point.

[203] Like Drs. Weingarten and Silmberg, Dr. Roberts had no specific recollection of treating Mr. Campbell. His evidence was based on a review of the medical records and what he understood from them and what he likely would have been thinking at the time.

The Expert Evidence

The Plaintiffs' Expert – Dr. Gerard Cox

⁴⁶ Dr. Roberts (Cross), p. 150, line 24 to p. 151, line 2.

[204] Dr. Cox was tendered and accepted as an expert in respirology. He was certified in respiratory medicine in 1988. He has been a full professor at McMaster since 2006 where he continues at present. He also continues to practice as a clinician and has treated patients with lung cancer, which is common in his practice. He has frequently taught on the subject of lung cancer and infections. Dr. Cox also acts as an independent assessor for the Royal College and, when requested, does reviews on their behalf to ensure that individual respirologists are upholding practice standards.

[205] Dr. Cox was retained by the plaintiffs regarding the standard of care with respect to Dr. Roberts. Dr. Cox has given expert legal opinions in the past but this was his first time testifying in court.

[206] Dr. Cox was familiar with HHT. He testified that it is an inherited condition which results in malformed blood vessels. An AVM in the lung may not get properly filtered as a result. This puts patients at risk of particles getting through to the systemic circulation in the body. Dr. Cox also testified that he knew about cerebral abscesses and how serious they can be, but he would not treat them. He would refer such cases to Infectious Disease, Neurology or Neurosurgery. In a differential diagnosis, cerebral abscess would be at the top of his list. However, he did not disagree that the literature supported that HHT on its own without evidence of pulmonary AVMs would not increase the possibility of brain abscesses. He was aware that the CT scan done on Mr. Campbell in 2008 did not show any clinically significant PAVMs.

[207] Dr. Cox was aware of the bubble study done on Mr. Campbell in 2008. In his view AV malformations in Mr. Campbell were possible given that saline bubbles did get through on the ECG and the fact that Mr. Campbell had a 15 year history of coughing up blood. Dr. Cox's view was that this rare genetic disorder "should have prompted earlier reflection on alternative diagnosis and greater urgency for the active pursuance of these."⁴⁷

[208] Dr. Cox testified that Dr. Roberts' thought process on March 24 fell below the standard of care. Mr. Campbell had been diagnosed with pneumonia the week before and treated with an antibiotic. So while Dr. Roberts thought of infection as part of his differential diagnosis as of March 24, he did nothing to demonstrate that this was actively entertained in future decisions. In fact, the contrary is indicated in that the antibiotic was terminated on March 23 or 24 and a steroid prescribed. While steroids can decrease an inflammatory response, they are immuno-suppressant and so can put a patient at risk of infection. Dr. Cox agreed, however, that it was likely the steroid that caused the lesions to shrink and explained the visual improvement. In Dr. Cox's view, Dr. Roberts failed to consider all of the relevant background information including headache, fever, respiratory problems and HHT. Fever is an important element, as both malignancy and infection can present with fever.

[209] Dr. Cox was directed to Dr. Roberts' discharge summary dated March 26. He testified that this report was excellent insofar as it dealt with the possibility of cancer. However, he could not accept that cancer was so far ahead of everything else no other considerations should have

⁴⁷ Report of Dr. Gerard Cox dated October 30, 2012, p. 3, para. 2.

been pursued. While he did not disagree that doing a bronchoscopy would have provided some tissue for analysis of cancer and culture, it would be of less value for obtaining cultures to rule out infection because it was done after Mr. Campbell had been on antibiotics for 10 days.

[210] Dr. Cox was also critical of Dr. Roberts' course of action when Mr. Campbell returned on March 30 because the diagnostic tests that were ordered focused on cancer. The main reason for ordering a lung biopsy is where lung cancer is suspected. Dr. Cox saw no evidence that Dr. Roberts took steps to consider infectious process or reconsider his differential diagnosis at that time.

[211] Further, Dr. Cox felt that Dr. Robert's referral to an oncologist (Dr. Warde) was putting all of one's eggs in one basket. This is supported by Dr. Roberts' completion of the referral to radiation oncology in which he indicated a diagnosis of metastatic brain cancer without a tissue diagnosis.

[212] Overall, Dr. Cox opined that Dr. Roberts fell below the standard of care by actively pursuing only one diagnosis, prescribing a potentially injurious treatment (steroids) and failing to consider the possibility of infection when it had been considered (but not acted upon) by Dr. Weingarten, Dr. Doran and the emergency physician who prescribed Levaquin. While the tests and consultations ordered by Dr. Roberts (bronchoscopy, lung biopsy, wedge resection, and neurology consult) were reasonable when taken in isolation they failed to appropriately address the alternate diagnosis of infection.

The Defendants' Expert – Dr. Elizabeth Powell

[213] Dr. Powell supported the care provided by Dr. Roberts. Her qualifications are set out above in relation to her expert evidence with respect to Dr. Weingarten.

[214] Dr. Powell testified that bronchoscopy done by Dr. Roberts was an appropriate course of action. It involves a fibre optic used to look at the lung in order to take cultures and make a diagnosis. Because a unitary diagnosis is preferred, it was reasonable for Dr. Roberts to proceed on the basis that the pathology in the lung was the same as in the brain. Doing testing on the lung is less invasive than on the brain, so it was reasonable to test there first.

[215] Dr. Powell's view was that it was appropriate to discharge Mr. Campbell on March 26. It was a temporary discharge, there were safety checks in place, Dr. Roberts had spoken to Mr. Campbell's wife about what to look for if there were problems, Mr. Campbell was stable and Mr. Campbell had Dr. Roberts' pager number. As there was no definitive diagnosis at that time and test information was not yet available, the discharge was reasonable. In any event, Mr. Campbell had frequently been out of hospital on day passes prior to this. Given these facts she would have sent Mr. Campbell home even if he did have a known cerebral abscess as he was quite stable. It was also quite proper for Dr. Roberts to send a copy of this discharge note to Dr. Faughnan.

[216] Dr. Powell's opinion was that the fine needle biopsy was an appropriate next step as was the consultation with Dr. Baryshnik. The fact that Dr. Baryshnik did not make a referral to neurologic service validates Dr. Roberts' working diagnosis of metastases. Dr. Powell was aware that Dr. Baryshnik was asked to comment on Mr. Campbell's confusion in the context of a

diagnosis of brain cancer. She disagreed with the proposition that Dr. Baryshnik would not be required to assess Mr. Campbell's lesions. In her view he would be obliged to do so and give an opinion as to what was happening in Mr. Campbell's brain.

[217] There was no reason for Dr. Roberts to continue the antibiotics after the fine needle biopsy because there was no evidence of fever, further infection or elevated white blood count. Dr. Powell disagreed that Dr. Roberts would be obliged to convey the results of the fine needle biopsy to Dr. Warde immediately. The test was inconclusive and did not establish a non-malignant disease and therefore would not assist Dr. Warde.

[218] When Mr. Campbell returned to the hospital on March 30 his neurological condition had changed. At this point, Dr. Powell would have revisited her differential diagnosis and consulted a neurologist first and then an infectious disease specialist. She agreed that by this point cancer was becoming less likely. She thought that, in a case such as this, the neurologist would want to meet with the radiologist where there was a "difficult film."

[219] In Dr. Powell's view, the lack of AVMs in Mr. Campbell's lungs played a significant part as to why cancer was pursued. Although she was aware that microscopic AVMs could be present and undetected on CT, her view was that these would be unlikely to increase the chances of a spread of infection. Her overall conclusion was that, in these circumstances, Mr. Campbell's risk of brain abscess was little different from that of the overall population.

[220] Dr. Powell was surprised to learn that Dr. Roberts was unaware that antibiotics had been stopped after the March 30 admission. As MRP, Dr. Powell would certainly have known if her patient was on or off antibiotics.

[221] Dr. Roberts' step to obtain a wedge resection when the needle biopsy came back negative was reasonable. He moved this forward quickly and arranged to have it done on the holiday Easter weekend. When those results were negative he arranged for a consultation with Dr. Lingley, an infectious disease specialist, which was a natural next step. It was reasonable for Dr. Roberts not to have requested a neurologic consult earlier than this given the lack of evidence of infection, fever and the platelet count.

[222] With respect to the requisition for Dr. Warde, she agreed that the wording of it clearly pointed to cancer. She also agreed that Dr. Warde would not be consulted with respect to the cause of a brain lesion; one would consult a neurologist. In Dr. Powell's view, it was important to ensure that Dr. Warde knew about Mr. Campbell presenting with fever on two occasions, that he had had two courses of antibiotics, that the bronchoscopy was negative and that the lung lesions had shrunk by 50 per cent. She did not know what information Dr. Roberts actually passed on to Dr. Warde, however she had no reason to believe that Dr. Warde had not reviewed the entire chart before administering radiation. Dr. Roberts' was therefore entitled to take comfort in Dr. Warde's confidence he was treating cancer.

[223] According to Dr. Powell, it was reasonable for Dr. Roberts to obtain the infectious disease consult when he did (April 15) because of his expanded differential diagnosis at that point. He had a preliminary verbal report that the open lung biopsy did not reveal any

malignancy and, given the lack of fever, white cell count and the validation of his working diagnosis by radiology, neurology and radiation oncology physicians, his decision to obtain the infectious disease consult at this point rather than earlier was reasonable.

[224] Dr. Powell disagreed that Dr. Roberts should have ensured that Mr. Campbell was transferred to St. Michael's Hospital on the evening of April 15. Her view was that this would have accomplished little as he was unable to speak to anyone in neurosurgery personally until the morning of April 16. In doing so, Dr. Roberts was approaching the transfer in the most efficient manner given the resources available.

Conclusions on Standard of Care – Dr. David Kenneth Roberts

[225] I find that Dr. Roberts care of Mr. Campbell fell below the standard of care for the following reasons:

- (a) As of the March 24 transfer of care Dr. Roberts was aware that infection had moved up the list on Dr. Weingarten's differential diagnosis. That was part of the reason for the transfer of care. Knowing this, however, Dr. Roberts continued with testing for cancer taking the view that he needed proof that Mr. Campbell did not have cancer before he would take steps to consider infection. While he testified that part of the reason he ordered a bronchoscopy was to help him further consider infection his discharge note of March 26, 2009, makes no mention of this;
- (b) Dr. Roberts was aware that Mr. Campbell had been diagnosed with pneumonia and treated with an antibiotic, however he did not continue with the course of antibiotics but prescribed a steroid. Although the steroid was likely responsible for the shrinkage of the lesions and the visual improvement, I agree with Dr. Cox that Mr. Campbell was put at increased risk of infection due to their being immuno-suppressant;
- (c) While the ordering of the bronchoscopy on its own is not below the standard of care, I find that Dr. Roberts' intention in ordering such a test was to prove cancer notwithstanding that he agreed that abscess was on his differential diagnosis. He was aware that Mr. Campbell had been on antibiotics such that any culture from the bronchoscopy would likely not be helpful with respect to any infection;
- (d) I do not disagree that at this point there were many reasons why the possibility of cancer would influence Dr. Roberts' decisions. He had the March 20 MRI results showing spiculated lung lesions, he had Mr. Campbell's smoking history and he knew that the testing in 2008 did not reveal any significant PAVMs. It was clear that finding the cancer was uppermost in Dr. Roberts' mind. That is what he told Mr. Campbell he would do when he spoke to him about the bronchoscopy. The steps he took thereafter clearly show that he was focused on this goal to the exclusion of ruling in or out abscess which he admitted was on his differential diagnosis at this point;

- (e) Dr. Roberts continued with cancer focused testing even in the face of Mr. Campbell's deteriorating mental condition after March 30 and the results of such testing being consistently inconclusive for cancer. Further, I agree with Dr. Cox that, while the various testing and neurology consult ordered by Dr. Roberts taken in isolation were reasonable, overall they failed to address the alternate diagnosis of infection;
- (f) When Mr. Campbell was re-admitted on March 30 in a state of confusion with delirium, Dr. Roberts did not take steps to revisit his differential diagnosis. Instead, he continued more invasive testing on Mr. Campbell's lungs (biopsy and resection). He did not make a referral to an infectious disease specialist for over two weeks after this presentation;
- (g) Dr. Powell's evidence was that the changes in Mr. Campbell's condition would prompt revisiting the differential diagnosis and include a neurology consult and then an infectious disease consult. Dr. Roberts consulted neurologist Dr. Baryshnik and had his report by April 2. Dr. Baryshnik noted "looks like 2 mets....enhancing ring lesions + edema – extensive." While Dr. Baryshnik's opinion favoured cancer, infection had still not been ruled in or out at this point. In fact, Dr. Roberts did not seek an infectious disease consult until April 16th. He waited until the results of the wedge resection were available before doing so. This is contrary to Dr. Powell's opinion with respect to steps to be taken after the change in Mr. Campbell's condition on March 30;
- (h) I find that Dr. Roberts had sufficient information after March 25 to actively rule abscess in or out. This includes:
 - (i) The initial ER note of March 17 listing a differential diagnosis for pulmonary lesion;
 - (ii) Dr. Weingarten's referral note questioning whether the lesions were inflammatory;
 - (iii) Two courses of previously prescribed antibiotics – Levaquin and Ceftriaxone;
 - (iv) The shrinking of the lung lesions in response to Decadron and Dr. Doran's wondering about a "vascular component";
 - (v) The pre-existing HHT. In my view this unusual background was something that should have prompted "earlier reflection or alternative diagnoses" as per Dr. Cox's report.

[226] I find that Dr. Roberts ignored one of the key features of differential diagnosis in that he did not eliminate the most serious possibility first, but rather he was bent on eliminating the most

probable. As per the text *Legal Liability of Doctors and Hospitals in Canada*⁴⁸, “[I]f doctors were to diagnose based on probability, rare and severe ailments would regularly be ignored....”

[227] Dr. Roberts’ conduct goes beyond an error in judgment as he failed to reconsider his initial diagnosis where Mr. Campbell was not improving or responding to treatment. Negligence may result where a physician falls prey to tunnel vision and adheres to an incorrect diagnosis particularly in the case of new symptoms consistent with a life threatening condition.⁴⁹ While I agree with Dr. Cox that Dr. Roberts’ failure to reconsider his diagnosis as of March 24th falls below the standard of care, if I am wrong in that conclusion, then certainly he fell below the standard of care on Mr. Campbell’s readmission on March 30th when as per *Williams* Dr. Roberts’ adhered to the diagnosis of cancer even in the face of new symptoms which concerned Dr. Roberts enough that he testified that he thought Mr. Campbell was going to die.

Defendant – Dr. Padraig Warde

[228] Dr. Warde qualified as a radiation oncologist in 1986. Since 1987 he has worked at Princess Margaret hospital in that capacity. He is the deputy head of the radiation medicine program and has been since the 1990s. Dr. Warde also maintains a busy clinical practice and has been teaching students, residents and fellows at the University of Toronto since 1987.

[229] At the time of Mr. Campbell’s illness Dr. Warde was spending one morning a week at Southlake, as it had no radiation oncologist at the time. On April 6 he received the consult by fax from Dr. Roberts. In the diagnosis box it was noted “metastases brain tumour.” He asked the nurse to ensure he received the result of the needle biopsy. Dr. Roberts told Dr. Warde he did not have a firm diagnosis. Dr. Warde indicated he would give radiation, but only if he was convinced that cancer was very, very likely.

[230] According to Dr. Warde he had no specific recollection of this case but his practice would have been to thoroughly review Mr. Campbell’s chart including the imaging in chronological order. Dr. Warde went to see Mr. Campbell that day and noted he was confused and mumbling. Dr. Warde took this to be a neurological deficit consistent with brain metastases. He also raised the possibility that Mr. Campbell’s delirium resulted from either steroids or swelling of the brain.

[231] Dr. Warde reviewed Dr. Thain’s March 17 CT scan and concluded there were two brain lesions and three lung lesions with spiculated borders. He also reviewed Dr. Silmberg’s March 20 MRI and took from it that there were two ring enhancing lesions in the brain with extensive edema around them. His review of the CT scan from March 24 showed that something had resolved in the lung. He also reviewed the x-ray from April 4. He did not recall having reviewed the results of the bronchoscopy, but if it were in the file he testified he would have looked at it.

⁴⁸ *Supra* note 16 at pp. 299-311.

⁴⁹ *Williams*, *supra* note 5 at para. 243.

[232] Dr. Warde agreed that the shrinkage of the lung lesions by 50 per cent as reported by Dr. Doran was unusual, but not an abnormal finding for cancer. He was not certain that he was aware that Dr. Doran thought it was somewhat less likely to be lung cancer and that Dr. Weingarten had also moved cancer down on his list in his differential diagnosis. He could not recall if he read Dr. Weingarten's note of March 24 to this effect.

[233] Dr. Warde could not recall if he knew that Mr. Campbell had been diagnosed with pneumonia and had a fever when admitted on both March 14 and 30. Dr. Warde agreed that fever, pneumonia and lesion shrinkage with antibiotics would be consistent with a lung infection as opposed to lung cancer. He agreed that without a biopsy and with the knowledge (if he had it at the time) that some doctors had expressed a degree of uncertainty about the case, this would increase the uncertainty of a cancer diagnosis. However, he felt he had a good overall picture of the case and that the appropriate investigations were done, and he made a judgment call that the use of radiation was in order. He also knew that Mr. Campbell was on high dose steroids. His experience was that patients on steroids who were radiated did not significantly deteriorate during radiation.

[234] Dr. Warde's view was that it was likely lung cancer with metastases to the brain. However, he was clear that radiologists do not diagnose cancer. A biopsy is usually required but here he did not wait for the biopsy result because he had a patient with a high likelihood of metastases who was deteriorating and so he had to make a considered decision in Mr. Campbell's best interest.

[235] Dr. Warde referred to his note of April 6, 2009, in which he recommended that whole brain radiation commence on April 7 even without a formal diagnosis of cancer. Dr. Warde had a conversation with Mrs. Campbell in which he explained that the purpose of the radiation was palliative and not curative. That is, it would prolong her husband's life and reduce his pain but it would not cure him. He would have discussed with her the risks which included hair loss, nausea, possible damage to brain tissue and possible deterioration of consciousness.

[236] Dr. Warde's evidence was that he was reasonably if not very certain of the diagnosis or he would not have proceeded with the radiation. While Mrs. Campbell and her daughter testified that Dr. Warde told them there was a 99.9 per cent probability that Mr. Campbell had cancer, Dr. Warde's evidence was that he does not use that type of language; he would probably have discussed it with them in terms of cancer being more likely or less likely.

[237] Dr. Warde did not disagree that it was unusual to proceed with radiation without a formal diagnosis. He estimated that this type of situation comes up every two to three years. In Mr. Campbell's case, Dr. Warde weighed the risks and benefits and felt that radiation would help reverse Mr. Campbell's deteriorating condition while waiting for the histology results. Mr. Campbell then received four fractions of whole brain radiation between April 7-10.

[238] Dr. Warde's note of April 13 indicates that he cancelled the final (and fifth) fraction of radiation because Mr. Campbell showed no improvement. The note indicates that he was still awaiting the results of the lung biopsy. Dr. Warde was uncertain whether he had the results of the needle lung biopsy by April 13 although he was aware that an open lung biopsy was

scheduled. In any event he did not request the results of the needle lung biopsy when they were available on April 9.

[239] Dr. Warde has never seen or treated a brain abscess. He knows it is a very rare diagnosis. There was nothing about Mr. Campbell that would have cued Dr. Warde that he had a rare condition. The doctor agreed that there was no part of Mr. Campbell's presentation that made a brain abscess unlikely other than the fact that it is an exceedingly rare diagnosis. Dr. Warde was unaware that Mr. Campbell had HHT. If he had known this he may have considered a brain abscess as less rare in Mr. Campbell's case.

[240] Dr. Warde was aware that swelling of the brain can be a short-term side effect of radiation. He was also aware that the swelling in Mr. Campbell's brain had been increasing over time despite being on a high dose of Decadron steroid. The point was to kill cancer cells which would thereby reduce the swelling and assist with his delirium and confusion. Dr. Warde agreed that the long-term side effects of radiation can include damage to brain tissue and cognitive issues. He agreed that neurosurgery and radiation were both options for Mr. Campbell at that time. Unfortunately, and in hindsight, Dr. Warde chose the wrong option. However, he did so with the best of therapeutic intentions.

The Plaintiff's Expert - Dr. Roy Ma

[241] Dr. Ma was asked to comment on the standard of care required by Dr. Warde in these circumstances. He is a radiation oncologist on staff at the Vancouver Cancer Centre. He confirmed that the examinations and requirements to become a radiation oncology specialist are the same across Canada.

[242] Dr. Ma has both a clinical practice and teaches at the University of British Columbia. He is a director of the residency training program to ensure that residents were fit to practice as radiation oncologists.

[243] Dr. Ma opined that Dr. Warde fell below the standard of care by prescribing whole brain radiation without a tissue diagnosis. While radiation is occasionally done on patients without a tissue diagnosis, those situations would be limited to ones in which neurosurgical options are too risky or there is a known history of cancer in another part of the body.

[244] In Dr. Ma's opinion, Mr. Campbell's case did not fit within his listed exceptions to proceeding to radiate without a tissue diagnosis because:

- (a) Mr. Campbell had never been diagnosed with cancer before;
- (b) He had HHT and therefore an increased risk of infection;
- (c) The needle biopsy suggested evidence of infection;
- (d) One of the brain abnormalities was close to the surface and therefore easy to get to by a neurosurgeon; and

(e) The brain MRI results were equally consistent with abscess.

[245] Dr. Ma expressed concern that Dr. Warde would proceed without the results of the fine needle biopsy. The results were actually available on April 9 but there is no note, nor can Dr. Warde recall if he pursued them. In fact, he continued with radiation on April 10. Dr. Ma agreed that the fine needle biopsy results actually indicated inflammatory process or inflammation caused by cancer and that a biopsy is only as good as the tissue retrieved.

[246] Dr. Ma's view was that the steroids being given to Mr. Campbell were not working as the swelling in his brain was increasing and he remained in a state of confusion. Giving whole brain radiation in such a setting is dangerous because of the possible side effect of more swelling. Dr. Ma testified that, by April 6, the only effective form of treatment to reduce the swelling would have been neurosurgery. Although Southlake did not have a neurosurgeon on staff, the arrangement could easily have been made as it was on April 16 when Mr. Campbell was transferred to St. Michael's Hospital.

[247] Dr. Ma testified that he was aware that the presumptive diagnosis for Mr. Campbell was cancer. He agreed that the lesions with spiculated borders in Mr. Campbell's lungs were typical for cancer.

[248] He agreed that whole brain radiation is not curative but palliative, and was meant to both buy time and relieve symptoms for Mr. Campbell.

The Defendants' Expert – Dr. James Wright

[249] Dr. Wright was qualified to opine on standard of care and causation issues in this action. He has been a radiation oncologist for 19 years. At the time of trial he was the Acting Head of the Juravinski Cancer Centre in Hamilton where he oversees the academic and clinical performance of 23 other radiation oncologists. He is also an Associate Professor at McMaster University, where he instructs students and residents in oncology.

[250] Dr. Wright opined that Dr. Warde did not fall below the standard of care required. Dr. Warde's review of the chart was thorough and Dr. Wright agreed that the low-grade fevers on March 4 and 30 and the slightly elevated white cell count were not enough for Dr. Warde to have considered infection as an alternative diagnosis.

[251] Dr. Wright's view was that the spiculated borders on the lung lesions in the March 17 CT scan were typical of cancer as were the ring enhanced brain lesions. Dr. Wright testified that Dr. Warde was correct in his view that the March 20 MRI was consistent with March 17 CT scan and that generally oncologists will defer to a radiologist's opinion regarding the most likely diagnosis.

[252] Dr. Wright minimized the importance of the March 24 CT scan and Dr. Doran's testimony that he did not think cancer was a likely possibility as of that date. Dr. Wright's evidence was that the March 24 shrinkage was unusual, but not completely uncommon in cancer.

[253] Dr. Wright opined that Dr. Warde considered the risks and benefits of taking action as opposed to inaction and made a judgment call. Dr. Wright gave evidence that, given the same information as Dr. Warde had, he would have recommended radiation as well. His view was that the risk of morbidity is greater for surgery than the risk of marginal swelling from radiation. He did not agree with the characterization of brain radiation as invasive therapy. It is surgical intervention that is the most invasive.

[254] While Dr. Wright felt that a neurosurgical consult may have been helpful, he was satisfied that the consultation with Dr. Baryshnik was sufficient and it was reasonable for Dr. Warde to rely on it to support lung cancer and metastases.

[255] Dr. Wright testified that the non-diagnostic result of the needle biopsy would not change the assumptive diagnosis of cancer because of the existing radiological findings which pointed to cancer.

[256] Dr. Wright disagreed with Dr. Ma and noted that he was not aware of any evidence that prophylactic brain radiation causes any impairment of neurocognitive function. In fact, to the contrary, his report dated October 10, 2012, cited a report from 2011 which reported “no significant differences in global cognitive function or quality of life after prophylactic brain radiation.”

[257] Most importantly, Dr. Wright opined that the progression of Mr. Campbell’s swelling would have continued notwithstanding the radiation. He explained that swelling of the brain is a side effect of brain cancer not radiation. Patients without diagnosed cancer often therefore undergo radiation without steroids.

Conclusions on the Standard of Care – Dr. Padraig Warde

[258] I find that Dr. Warde did not meet the standard of care with respect to Mr. Campbell for the following reasons:

- (a) The evidence is unclear as to whether Dr. Warde took into account Mr. Campbell’s full history or was aware of the concerns of Dr. Doran and Dr. Weingarten regarding the changing differential diagnosis of Mr. Campbell. The reason for this is that Dr. Warde had no specific recollection of treating Mr. Campbell and was only able to give evidence based on the charts and notes. He agreed that, if some doctors had expressed some uncertainty about the diagnosis, this would have affected his view of the case;
- (b) While there was certainly imaging that reflected a likely cancer diagnosis, Dr. Silmberg’s MRI indicated a differential diagnosis;
- (c) While the lung lesion shrinkage could have been related to cancer there were other potential explanations. Mr. Campbell’s fever and pneumonia were also symptoms which Dr. Warde testified he would have associated with an infection as opposed to cancer;

- (d) There was also the issue of Mr. Campbell's HHT and the potential for increased brain abscess;
- (e) Dr. Warde's evidence was that he had to be very certain of cancer as being likely when proceeding without a tissue diagnosis. As such, I find that he fell below the standard of care in not taking into account the questions that had been raised in Mr. Campbell's chart to that point. Dr. Warde was clear in his evidence that "I tried to look, does the whole picture fit?" In this case the scrutiny of Mr. Campbell's chart was paramount and I find that Dr. Warde, like Drs. Weingarten and Roberts, was focused on cancer to the exclusion of other diagnoses;
- (f) Both Dr. Warde and Dr. Wright agreed that radiation could create a potential for increased swelling in the brain. Dr. Wright agreed that it was clear that Mr. Campbell had swelling in his brain from the March 20 MRI and that the swelling had increased by March 30th as per the CT scan. There was no doubt that Mr. Campbell's mental state was altered. Dr. Wright agreed that, although neurosurgery was more invasive, it would have given a tissue diagnosis and would not have increased brain swelling. Further, Dr. Wright's evidence was that he would likely have steered the decision maker (in this case Mrs. Campbell) away from doing anything at all (i.e. no radiation and no neurosurgery) as he described doing nothing as "high on his list of options." Doing nothing was not high on the list of Dr. Warde's options;
- (g) Dr. Warde had to weigh the risks and benefits in this case. Both neurosurgery and radiation had some risk of harm according to Dr. Wright. However, according to Dr. Wright, neurosurgery had the immediate potential benefit of decompression whereas radiation did not have the same short-term benefit;
- (h) Dr. Warde made a judgment call but he did so in the face of what he testified to as the "uncertainty of the diagnosis." He knew that the results of the lung needle biopsy were pending. Those results were not conclusive, showing both the potential for inflammation from cancer or from infection. Dr. Warde did not follow-up on these results even though after the referral on April 6 he indicated he was awaiting them with interest. He could have waited until April 9, when the results were available, and reassessed at that point. Instead, he continued to give Mr. Campbell radiation up to April 10. While Dr. Wright testified that the non-diagnostic result of the needle biopsy would not have changed the landscape given the assumptive cancer diagnosis, I prefer the evidence of Dr. Ma that, by April 6, the only way to effectively reduce the swelling and obtain a proper tissue sample was by way of neurosurgery;
- (i) I accept that Dr. Warde gave his evidence fairly in conceding that he had options (radiation or neurosurgery) but in hindsight chose the wrong one while having the best of therapeutic intentions. While an error in judgment on its own may be insufficient to find that the standard of care has not been met, a series of errors in

judgment may cumulatively result in a finding of negligence.⁵⁰ In this case, those errors in judgment may be generally listed as follows:

- (i) Pursuing whole brain radiation, with a patient whose brain swelling was increasing, was potentially dangerous as both Dr. Wright and Dr. Warde conceded that the radiation could cause more swelling;
- (ii) Dr. Warde proceeded with the radiation being aware that the high dose steroid prescribed to Mr. Campbell (Decadron) was not controlling the swelling in his brain. Dr. Wright agreed that “doing something that only gives you the risk of increased pressure, I would agree doesn’t seem a reasonable thing to want to do”,⁵¹ and
- (iii) Dr. Wright agreed that neurosurgery represented the best and most immediate method of reducing swelling to the brain and obtaining an immediate diagnosis, whereas radiation could take up to six to eight weeks for results with potentially no amelioration to the swelling Dr. Wright’s evidence is that neurosurgery or no action at all should have been raised as options with the caregiver. There is no evidence that Dr. Warde did so with Mrs. Campbell;
- (j) Finally, there was a certain degree of inconsistency in Dr. Wright’s evidence with which I found it hard to grapple. On the one hand he was clear that in similar circumstances he would not have pursued either neurosurgery or radiation with Mr. Campbell at that point in the process. At the same time he was clear that he did not think Dr. Warde’s approach was unreasonable, notwithstanding his concession that radiation could increase the swelling in the brain of a patient who had continually increasing swelling (even with Decadron) and whose condition was declining; and
- (k) I should add at this point that, although I have found that Dr. Ward’s care of Mr. Campbell fell below the required standard, I am not satisfied that there is sufficient evidence to connect the brain radiation to Mr. Campbell’s cognitive deficits. This will be explored further under Causation issues set out below.

Causation – Applicable Legal Principles

[259] If one or any of the defendants in this case are found to have breached the standard of care, he will not be liable if the particular breach of the standard of care is not found to have caused the injury. This principle is often referred to as the “but for” test. That is, the plaintiff

⁵⁰ *Ibid* at para. 236.

⁵¹ Dr. Wright (Cross), p. 51, LL 8-10.

must show on a balance of probabilities that he would not have suffered the injury but for the negligent act(s) of the defendant(s).⁵²

[260] Causation cannot be inferred by, for example, a delay in treatment. Even the use of common sense must have a factual underpinning and the court is not entitled to bridge a gap in the evidence by way of inference.⁵³

[261] Causation can be supported by “common sense inferences” which allow the court to determine what would have happened if something which ought to have been done but was not done had been done. A court may be pragmatic in its approach to causation.⁵⁴

[262] The plaintiffs need not establish that the defendants’ negligence was the sole cause of the injury. As long as the defendants’ were part of the cause of the injury, they or he is liable.⁵⁵

[263] “Loss of chance” is also not compensable in medical negligence cases. That is, the actions of a physician which increased the risk of injury to a plaintiff or reduced his or her chance at a better outcome are insufficient to establish causation. The plaintiff must establish that the actual outcome would likely have been materially better.⁵⁶

The Evidence of Causation - General

[264] When Mr. Campbell came to Southlake on March 14 it is well documented that he had no detectable cognitive deficits. By March 17 he returned to hospital with a left side visual field deficit. By March 24 he was fully recovered from that deficit and had no health complaints upon his temporary discharge by Dr. Roberts on March 26.

[265] The first cognitive deficits appeared on March 30 and became increasingly worse until he was transferred to St. Michael’s Hospital on April 16, by which point he was significantly neurologically impaired.

[266] The position of the plaintiffs is that there is a common sense inference that the delay in treatment was causally related to Mr. Campbell’s injuries.

[267] The defendants’ view is that the date by which brain injury was inevitable was much earlier than March 17, 2009 and, in any event, the standard of care was not breached on March 17, 2009. Further, Mr. Campbell’s brain injury was not preventable.

The Position of the Plaintiffs on Causation

⁵² *Clements (Litigation Guardian of) v. Clements*, [2012] 2 S.C.R. 181, 2012 SCC 32 at para. 37.

⁵³ *Kozak v. Funk*, [1997] S.J. No. 743 at para. 22 (C.A.).

⁵⁴ *Snell v. Farrell*, [1990] 2 S.C.R. 311 at paras. 30-31.

⁵⁵ *Athey v. Leonati*, [1996] 3 S.C.R. 458 at paras. 13-17.

⁵⁶ *Laferriere v. Lawson*, [1991] 1 S.C.R. 541, [1991] S.C.J. No. 18 at paras. 154-163.

[268] Dr. Weingarten agreed that a brain injury would require urgent diagnosis and treatment. Although a treatable condition when the abscesses are small, a failure to treat promptly will worsen the patient's neurologic outcome, increase brain swelling and lead to permanent neurologic impairment.

[269] Dr. Roberts also agreed that an abscess is a serious medical condition that requires prompt treatment. Prompt treatment will affect outcome because the smaller the abscess the greater likelihood that it can be penetrated by antibiotic treatment. He agreed that, in the absence of treatment, abscesses can grow and cause brain swelling which will increase intracranial pressure and potentially lead to permanent impairment of brain function.

[270] The plaintiffs submit that the process identified by Drs. Weingarten and Roberts is the very one which caused Mr. Campbell's injuries. Those admissions and the evidence of non-treatment are sufficient to establish the mechanism of the injury and causation.

[271] Dr. Mikulis was also qualified as neuroscientist, which is the scientific study of the brain, spinal cord and nerves. Dr. Mikulis testified that the lesions in Mr. Campbell's brain increased in size significantly between the March 20 MRI and the March 30 CT scan. They increased in size again between March 30 and April 15 such that by April 15th they measured 4.6 centimetres and 4 centimetres, respectively. By the time Mr. Campbell was admitted to St. Michael's Hospital the lesions had grown to 5.2 centimetres and 4.3 centimetres. At that point Dr. Mikulis found a six millimetre mid-line shift of Mr. Campbell's brain structures which would explain his significant neurological defects. There was no mid-line shift detectable on the March 20 MRI.

[272] Dr. Mikulis reviewed a follow-up scan from September 3, 2010, in which he noted a global loss of brain substance and injury to the brain's white matter. He attributed this to the increase in intracranial pressure suffered over time and the toxic substances released during the inflammatory reaction to the abscesses. He testified that the administration of radiation likely added some degree to the mass effect.

[273] The plaintiffs note that the defence expert, Dr. Wright, agreed that the more swelling that is present in the brain and the amount of time such swelling is present will determine the amount and seriousness of the damage to the brain.

[274] The defendants' expert Dr. Zoutman agreed that untreated abscesses can cause increased brain swelling which over time will cut off blood supply to vessels in the brain and cause brain damage or death.

[275] The defendants' behavioural neurology expert, Dr. Sharon Cohen, explained it succinctly when she testified that:

It [the brain] is compromised by pressure effects, and that's the raised intracranial pressure, and when adjacent healthy brain is compromised, it loses its blood supply, and it loses its ability to function normally, and brain cells die. You'll have an area where some brain cells will be on threshold of death and maybe can

recover and other situations where the brain cells have died and cannot recover.⁵⁷

[276] The plaintiffs submit that the evidence of Dr. Mikulis, Dr. Weingarten, Dr. Roberts and Dr. Cohen confirmed the primary process by which abscesses can cause brain damage. As such, the only remaining question is whether earlier treatment would have prevented the growth of the abscesses and the corresponding increase in swelling.

The Plaintiffs' Expert – Dr. Christine Lee

[277] Dr. Lee is an Associate Professor at McMaster University and a specialist in Infectious Diseases and Medical Microbiology. She was qualified as an expert in Infectious Disease (ID) and provided an expert opinion on whether the defendants' negligence and the resulting delay in treatment caused or contributed to Mr. Campbell's brain injuries.

[278] During her career, Dr. Lee has had occasion to diagnose brain abscesses. Symptoms often include fever, headache and neurological deficit. Radiological imaging is needed to support the diagnosis and the treatment is IV antibiotics. If the organism is not known, there is little risk in initiating an antibiotic treatment and assessing the patient frequently in hospital. Dr. Lee has initiated IV antibiotic treatment for brain abscesses before. Treatment usually takes about six weeks plus follow-up.

[279] Dr. Lee's evidence was that it is often difficult to identify an organism due to its location, especially if it is in a location where a brain biopsy would be deleterious. In her view, patients who present with neurological and radiological symptoms of abscess should be started on empiric IV antibiotics and monitored through radiological imaging. This should occur as soon as the diagnosis is considered. In her view the risk of prescribing empiric antibiotics was negligible compared to the huge benefit of starting antibiotics in a timely manner. Drs. Powell and Zoutman did not agree with Dr. Lee on the issue of empiric antibiotics and testified concerning possible negative consequences of treating patients with the wrong choice of antibiotic.

[280] Abscesses larger than three centimetres may require neurosurgery and drainage. Generally patients with abscesses smaller than three centimetres will respond to antibiotic treatment. Generally an abscess must measure at least three centimetres before aspiration or excision will be considered, although that is not a rule and the decision to use surgical intervention will often depend on the clinical state of the patient. Dr. Lee accepted that the Mandell text *Principles of Infectious Disease*⁵⁸ is an authoritative text on ID. As per the text, she accepted that other physicians may view 2.5 centimetres as the cutoff for the excision or aspiration of an abscess, but she maintained her opinion that 3 centimetres is a better benchmark.

[281] Dr. Lee's view was that there was sufficient evidence to support a brain abscess diagnosis because Mr. Campbell presented with fever, headache, neuro deficit and ring enhancing lesions. Her view was that, had IV antibiotics been administered to Mr. Campbell at Southlake when his

⁵⁷ Dr. Cohen (Cross), p. 12, line 17 to p. 13, line 16.

⁵⁸ *Supra* note 1.

abscesses were small (i.e. as at the time of the March 20 MRI), he likely would have fully recovered without any neuro deficit and without the necessity of a craniotomy. Dr. Lee has treated two patients with abscesses similar to those of Mr. Campbell. The patients made a full recovery after IV therapy and were able to return to work. The largest abscess treated was 2.9 centimetres. Mr. Campbell's largest abscess as of March 20 was 2.6 centimetres.

[282] Regarding the radiation therapy, Dr. Lee's evidence was that radiation destroys white blood cells which fight infection. This can allow bacteria to multiply and can cause further inflammation and swelling. Dr. Lee's initial evidence was that Mr. Campbell received radiation between April 7-14. She was unaware that he had only received four fractions between April 7-10.

[283] Dr. Lee was asked about Mr. Campbell's temporary discharge on March 26 when he showed no deficit. Her opinion was that the Levofloxacin antibiotic he had been taking can often get to the brain and that may have assisted. She noted that, once the antibiotic was stopped, the patient's condition deteriorated.

[284] Dr. Lee opined that Mr. Campbell fell into a subset of patients who could be treated by antibiotic therapy alone because his lesions, as of March 20, 2009, were less than three centimetres, there were multiple abscesses, he had noted clinical improvement after oral antibiotics and Decadron and no noted change of consciousness through the first admission. The plaintiffs submit that this opinion substantiates that Mr. Campbell was fully treatable with antibiotics alone up to March 26.

[285] The defendants disagree and seek to limit Dr. Lee's opinion to the provision of antibiotics as of March 17. As none of the plaintiffs' experts offered any opinion that the standard of care required the defendants to diagnose infection or abscess as of March 17, 2009, Dr. Lee's causation opinion is irrelevant to this case. For reasons that are outlined below, I reject this argument.

The Defendants' Experts

Dr. Dick Zoutman

[286] Dr. Zoutman is a specialist in ID and medical microbiology. Dr. Zoutman has had a distinguished career and is currently Chief of Staff of the Quinte Health System, which includes four hospitals. Dr. Zoutman was qualified as an expert in internal medicine, medical microbiology and infectious disease in this case.

[287] Dr. Zoutman has cared for patients with brain abscesses and both treated them with antibiotics and referred them to surgery. He agreed that brain abscesses require treatment as soon as possible in order to avoid increased brain swelling. His opinion was that Mr. Campbell would have needed surgical intervention even if he had been properly diagnosed on March 17.

[288] Dr. Zoutman opined that the use of empiric antibiotics has risks, including septic shock and bacterial resistance. Dr. Zoutman did not necessarily agree with the Mandell text that Ceftriaxone is a recommended empirical approach to antimicrobial therapy for bacterial brain

abscesses. He said it was his practice to use that antibiotic but only after the neurosurgeon has done the craniotomy and not before.

[289] Mr. Campbell's brain abscess came to the brain by way of hematogenous spread from the lungs. Although the presence of HHT will increase the risk of developing brain abscesses, they are still rare in such people. He agreed that Dr. Faughnan's recommendation of antibiotic prophylaxis was because of the possibility of the spread of infection.

[290] Dr. Zoutman reviewed the March 20 MRI and opined that it revealed either tumour or abscess; it was impossible to tell from the imaging. His view was by March 17 the abscesses required surgery because the largest one was 2.6 centimetres. The centre of the abscess would contain dead brain tissue which would be hostile to antibiotics, so giving Mr. Campbell antibiotic therapy on March 17 would have been of no use even if the right antibiotic had been chosen. His view is that, once the lesion is excised and you have the tissue sample, you can then prescribe the appropriate antibiotic. It is different, for example, for a lung infection. In that case, Dr. Zoutman would start an empiric antibiotic because he is familiar with what organisms are most likely. He agreed that there would be a subset of patients whose abscesses are small enough that they could be treated with antibiotics alone, but he would still want to know the microbiology.

[291] The Mandell text⁵⁹ on Infectious Disease was put to Dr. Zoutman with respect to the statement that antimicrobial therapy alone can be used for abscesses under three centimetres. His evidence was the chapter had inconsistencies and that he was relying on a specific reference from a text in 1980 as well as his own experience and other literature.

[292] Dr. Zoutman was asked about Mr. Campbell's symptoms in late February 2009; the persistent headache and a painful cough. His view was that this implied a possible tumour or abscess. Mr. Campbell's reporting of "flashing lights" and trouble with his left visual field also confirmed that an abscess had been growing there for some time. His view was that by March 17 the abscess had already been growing for about 30 days. Mr. Campbell's reported symptoms of fatigue and forgetfulness in February were also indicative of global brain dysfunction. The defendants submit that this evidence remains unchallenged as Dr. Lee did not give an opinion on when the abscesses began to form. The defendants' expert Dr. Muller testified to the same effect.

[293] Dr. Zoutman testified that abscesses are rare but part of an ID doctor's education. They can be treated by IV antibiotics if they are small enough. Dr. Zoutman would be fine with sending a patient home with a suspected brain abscess if they were on steroids as the immunosuppressant aspect of steroids is a long-term effect. He was unsure if he would send that same patient home without antibiotics.

[294] Dr. Zoutman was aware that on March 17 Mr. Campbell presented with left hemianopsia and headache. His opinion was that earlier treatment would not have affected Mr. Campbell's outcome. Dr. Zoutman agreed that Mr. Campbell's mental status on March 26 was normal, but

⁵⁹ *Supra* note 1 at 1272-1275.

far from normal on April 17; however, his opinion was that Mr. Campbell's condition demanded intervention but it would not have changed his deficits. There was a point of no return and Mr. Campbell's cognitive defects would be the same even if he had been treated before March 26 because of the edema and because the scar tissue had already formed from the brain lesions. He did not agree that the radiation and mid-line shift would have made a difference in the outcome. The delay that occurred in treatment did not change the probability of him having a deficit. His view was that Mr. Campbell would have needed a craniotomy even if he had been treated on March 17.

[295] The plaintiffs submit that in order to accept Dr. Zoutman's opinion above, the court would have to find that even if Mr. Campbell's abscesses were treated appropriately by March 26 he would still have gone on to experience neurological deficits. The plaintiffs submit this is not a supportable proposition. The plaintiffs position is that the only real remaining question is whether he could have been treated with antibiotics alone as per Dr. Lee's opinion or whether he required a neurosurgical procedure as Dr. Zoutman opines.

[296] Dr. Zoutman agreed that a patient's neurologic status on presentation was an important predictor of outcome. If a patient is in an altered state as defined by the Glasgow Coma Scale, intervention must take place within hours or sooner to avoid the effects of increased swelling in the brain.

Dr. Paul Muller

[297] Dr. Muller is a senior neurosurgeon with a distinguished career at St. Michael's Hospital where he has served as both Surgeon in Chief and Chief of Neurosurgery. He was qualified to give expert evidence on behalf of the defendants in the specialty of neurosurgery on the issue of causation including the etiology, diagnosis, treatment and consequence of brain abscesses.

[298] In his four decades as a neurosurgeon Dr. Muller has had occasion to diagnose brain abscesses, although he testified that they were much rarer than brain tumours. The incidence of brain cancer is about 1,000 per 100,000 as compared to 1 per 100,000 in the general population. He has been involved in brain abscess related surgery from both diagnostic and therapeutic perspective. He estimated that he has seen some 70-80 patients for brain abscess. He performed craniotomies on a small number of these patients.

[299] Dr. Muller's opinion was that the time taken between mid-March and mid-April 2009 to arrive at the correct diagnosis for Mr. Campbell did not make any difference in terms of his ultimate outcome. His situation was not life threatening even at the point of being transferred to St. Michael's Hospital as the surgery was delayed for two days. Emergency surgery would have taken place (even with the nosebleed factored in) if Mr. Campbell's condition had been truly life threatening. Further, Dr. Muller opined that Mr. Campbell did not have any general atrophy in his brain. The modest change in ventricular size is attributable to the two neurosurgeries and the presence of abscesses prior to March 17.

[300] Dr. Muller opined that existence of PAVMs in Mr. Campbell's lung is immaterial; that is, infection does not need a PAVM to get to the brain from the lungs.

[301] Dr. Muller was asked to comment on Mr. Campbell's discovery evidence concerning his headache, which he was experiencing prior to March 14 and which was very painful when he coughed. Dr. Muller opined that it could be an indication of increased cranial pressure or an inflammatory process. By March 17 Mr. Campbell had developed a visual field change on the left side which Dr. Muller opined was reflective of a clinical change in the brain most likely in the back of the brain. His opinion at trial was that the embolization occurred while Mr. Campbell was on vacation in Louisiana in February 2009. Given that he knew that what the type of bacteria was strep milleri, his view was that it would have taken two to three weeks for the abscesses to form. His symptoms of forgetfulness and headache while he was on vacation would be consistent with this.

[302] Dr. Muller confirmed that the homonymous hemianopsia with which Mr. Campbell presented on March 17 was clearly a manifestation of local tissue pressure in the occipital lobe region of the brain. Dr. Muller did not disagree that, as of the date of Mr. Campbell's temporary discharge on March 26, his headache and neurologic symptoms had completely recovered after treatment with antibiotics, steroids and several blood transfusions. It was perfectly reasonable for Dr. Roberts to have temporarily discharged him at this point while awaiting the results of the bronchoscopy. He agreed that by the time of Mr. Campbell's readmission on March 30 he had neurologic issues which were causing swelling in the brain and that swelling in the brain continued until his transfer to St. Michael's Hospital.

[303] Dr. Muller did not disagree that brain abscess was on the differential diagnosis for Mr. Campbell during the first admission. He agreed it would be sub-optimum not to include it in the differential diagnosis.

[304] It was Dr. Muller's view, based on a review of the records and Mr. Campbell's discovery transcript, that he was employable, perhaps not at his previous occupation but at some occupation. He was surprised to learn at trial that the parties had conceded that Mr. Campbell was not employable in any capacity. Dr. Muller's opinion was that Mr. Campbell had suffered a visual field deficit as a result of a drop in blood pressure when he had a serious nosebleed on April 17 just prior to surgery at St. Michael's Hospital. This is contrary to the note of Dr. Bhardwaj, who reported that Mr. Campbell's blood pressure at the time of the nosebleed remained stable. Dr. Muller insisted that his review of the anaesthetic record showed otherwise. There was no reference to Dr. Bhardwaj's note in Dr. Muller's report.

[305] At the time of authoring his report, the visual deficit had resolved based on reports on Mr. Campbell's progress in 2012. Dr. Muller's view was that any visual field loss was unrelated to the conduct of the defendants in this case. Dr. Muller agreed that when Mr. Campbell returned to hospital on March 30 with visual agnosia that this was a visual processing issue (neurologic and due to mass effect).

[306] Dr. Muller's opinion was that Mr. Campbell's abscesses were well formed by the time he came to the emergency department at Southlake on March 14. Dr. Muller did not agree that abscesses the size of Mr. Campbell's could fully form within six days (i.e. between March 14 and the date of the MRI – March 20). The size of the abscesses by mid-March (2.4 or 2.6 centimetres, depending on which measurement is accepted) would not be amenable to antibiotic

treatment alone. Mr. Campbell required two surgeries to deal with the right occipital abscess even while on antibiotics, so antimicrobial treatment would be insufficient. This is because the volume of pus inside the abscess would make it impossible for an antibiotic to penetrate. The infection at that point can only be eradicated through draining supplemented by high dose antibiotics. Draining (or aspiration) is accomplished by placing a needle near the center of the abscess and sucking out its contents. In Dr. Muller's view, a tissue sample from the lesion was needed to determine the proper antibiotic post aspiration.

[307] Dr. Muller testified that the growth of the abscesses did not damage Mr. Campbell's brain because the brain compensates for this by pushing spinal fluid out of the brain. Only when the available spinal fluid is exhausted does brain damage occur.

[308] Dr. Muller went on to opine that, in his view, neither the radiation therapy or immune suppression had any effect on Mr. Campbell's outcome or on brain atrophy. The tissue that was being radiated was already dead, so there would be no difference. In cross-examination he conceded that there is a risk of swelling in the brain with any radiation treatment. He also agreed that there had been problems with the medical management of Mr. Campbell pre-radiation, but he did not include this in his report and told the court in his evidence in-chief that the medical management of Mr. Campbell's case had been reasonable. He agreed he saw problems with Mr. Campbell's medical management but did not tell anyone this because he was retained to give an opinion on causation. For example, he testified that the MRI was "incomplete" as there was no assessment of the lesions with respect to the diffusion-weighted imaging that might have tilted the diagnosis.⁶⁰ It should be noted that no standard of care expert opined that further brain imaging should have been ordered, and Dr. Muller was not qualified on standard of care issues. In any event, he was clear that even with improved imaging studies, a definitive diagnosis would only be available from tissue.

[309] Dr. Muller was cross-examined on the draft reports he generated in relation to his October 17, 2012, report. His evidence was that he had received no new documents between the date of the October 10, 2012, and the final report of October 17. The final report contains a reference to Mr. Campbell's haemoglobin levels on April 17, 2009, which were not included in the October 10, 2012, draft. Dr. Muller was not sure if he had obtained the information about the haemoglobin levels from the material provided to him by counsel, from the St. Michael's Hospital electronic records system or whether this information was something outside of the materials provided to him. On re-examination, his evidence was that he was of the view that all of the relevant records from St. Michael's hospital had been produced to him in paper form. It was simply easier to review them electronically.

[310] He was certain that the visual field deficit had been caused by a hypotensive event and he wanted to be able to corroborate that. He agreed that the drop in haemoglobin was not 23 per cent as he indicated in his report. After lab reports from the relevant times were put to him, he agreed that the drop was only from 91 to 86 which would not be as he described it a "seminal

⁶⁰ Dr. Muller (Cross), Vol. 2, p. 84, LL 23-27.

event.’⁶¹ He therefore agreed that his conclusion about the hypotensive event causing the visual field deficit was wrong and that there was no evidence to substantiate that any hypotensive event occurred during surgery. Later in re-examination Dr. Muller confirmed that the anesthetic report showed a significant drop in blood pressure and increase in heart rate in the early interval of his monitoring. Dr. Muller testified that this would create a hypotensive effect and returned to his view that this hypotensive event would have caused an altitudinal field deficit regardless of the haemoglobin issues because there would have been a stretching of Mr. Campbell’s optic nerve combined with a lack of blood resulting in effectively a stroke of the optic nerve.

[311] Dr. Muller was cross-examined about a preliminary draft of the October 17, 2012, report dated September 5, 2012. In that draft there is a note indicating a differential diagnosis for the March 20 MRI of tumour or abscess. The conclusion, in those exact words, did not appear in the October 17, 2012, report; however, Dr. Muller did not disagree that that is what he felt the differential diagnosis was.

[312] Dr. Muller was asked about a draft of his January 30, 2013, report dated December 12, 2012. He agreed that the third line of the draft read “The lesions which indeed were abscesses, did increase in size and did cause a tissue distortion that led to Mr. Campbell’s neurological decline.” This was changed from “neurological decline” to “transient neurological symptoms” in the final report. He did this after meeting with one of the defendants’ counsel but cannot remember the content of that meeting. He denied making any substantive changes to his report based on any outside influence. There was no evidence that Dr. Muller allowed counsel to make changes to his report or that counsel was involved in the writing of his report.

[313] Dr. Muller conceded that, as part of his corroboration of opinions, in his final report he accessed the private records of four patients via the hospital database and did not excise their personal information from those pathology results. Although ultimately those results did not make their way into any appendix, plaintiffs’ counsel saw this information when reviewing the file. Dr. Muller agreed that accessing those private records was an inappropriate exercise of his position, although the failure to redact was entirely inadvertent on his part.

The Position of the Defendants on Causation

[314] According to the defendants, the determination of causation is a three-step process in this case. First, the court must determine based on expert evidence what the latest date was that Mr. Campbell’s injury was preventable.

[315] The defendants submit that Dr. Lee’s evidence must be restricted to whether Mr. Campbell’s injuries could have been avoided by antibiotic treatment as of March 17. Therefore, any opinion from Dr. Lee that relates to causation issues after March 17 is arguably inadmissible.

[316] The defendants submit, however, that the brain injury was inevitable much earlier than March 17, because Mr. Campbell had both an infection and neurological symptoms for several

⁶¹ Dr. Muller (Cross), Vol. 2, p. 60, line 5.

weeks prior to March 17. By March 17 his abscesses had fully formed and encapsulated, and one of his brain abscesses measured 2.6 centimetres, according to Dr. Mikulus' testimony, which made surgery inevitable. His neurological symptoms prior to March 17, 2009, were consistent with what Dr. Muller described as global brain dysfunction. Dr. Zoutman also noted that he had one abscess measuring 2.5 or 2.6 centimetres on March 17, which meant his infection had been present for almost a month. Dr. Lee gave no opinion on this issue and, as such, the evidence of Drs. Muller and Zoutman is unchallenged in this regard.

[317] The second step in determining causation, according to the defendants, is whether the standard of care was breached on March 17. No standard of care expert testified that antibiotics ought to have been started on March 17. The physician caring for Mr. Campbell at the time was Dr. Weingarten. The plaintiffs' only allegation against him at that point was that his differential diagnosis was insufficiently broad. A breach of the standard of care after March 17 is irrelevant as the brain injury could not have been prevented.

[318] As per Dr. Muller's evidence, the swelling in Mr. Campbell's brain had no permanent impact on him. The damage done to Mr. Campbell's brain had already happened. Dr. Muller's view was that, based on radiographic images from both before and after the events of March/April 2009, there was only some scarring and a small increase of ventricular size which were the consequences of his initial infection surgery and not from swelling. The plaintiffs did not call a neurosurgeon to counter this point and the evidence is therefore uncontested.

[319] There is no evidence about exactly what the delay in treatment actually caused and therefore causation cannot be inferred without expert evidence.

[320] With respect to the third step; i.e. that Mr. Campbell's brain injury was not preventable – the defendants argue that no expert evidence was lead that the standard of care required the defendants to diagnose the abscess as of March 17, 2009.

The Causation Analysis

[321] The defendants are effectively asking the court to find that even if Mr. Campbell's lesions had been aspirated on March 17 his cognitive impairments would have been exactly the same as in the agreed statement of facts. On the issue of causation and based on my review of the evidence presented and the factual record, I find that I am unable to make such a leap.

[322] Further, I am unable to find, based on the factual record and on a balance of probabilities, that the delay in treatment caused by the negligence of Drs. Weingarten and Roberts had no effect on Mr. Campbell's outcome. With respect to Dr. Warde, the analysis is slightly different and I am unable to conclude that the whole brain radiation increased the brain swelling or had any effect on outcome. My detailed reasons are set out below.

[323] With respect to the evidence of Dr. Muller, he was very confident in his view that any swelling in Mr. Campbell's brain would have had only minor long-term effects. He did not explain how it was that Mr. Campbell's serious cognitive impairments therefore arose. His evidence contradicts that of the defendants' expert Dr. Cohen, who testified that Mr. Campbell's brain injury was the result of raised intracranial pressure related to cerebral abscesses. It also

contradicts that of Dr. Mikulis, who testified that Mr. Campbell suffered a global loss of brain substance.

[324] Dr. Muller conceded that Mr. Campbell has a visual processing deficit which is a neurological issue. At the same time he insisted that his visual deficit was caused by a form of stroke which occurred when lowered blood pressure from a severe nosebleed on April 17 caused a stretching of Mr. Campbell's optic nerve. He did not include in his report the note of resident Dr. Bhardwaj who reported that Mr. Campbell's blood pressure remained stable throughout. I find that Dr. Muller's evidence on the issue of Mr. Campbell's visual deficit was inconsistent and at times confused. The court is left with a doubt about whether the visual deficit was caused in the manner suggested by Dr. Muller. As a result, it is preferable to rely on Dr. Muller's concession that the visual deficit is a neurological issue and therefore related to events which occurred prior to April 17.

[325] I do not question Dr. Muller's competence or professionalism, but there was a certain flavour to his evidence that favoured the defendants. For example, his view was that Mr. Campbell was employable. He was surprised to learn that the parties had agreed, along with experts in the area, that Mr. Campbell was not employable. Dr. Muller never personally assessed Mr. Campbell before making this conclusion. Further, Dr. Muller admitted that he had concerns about Mr. Campbell's medical management, but never included this concern in any of his reports.

[326] While I do not accept that changes to his draft reports were influenced or written by any of the defendants' counsel (who were professional, prepared and fair throughout); the changes he made were subtly favourable to the defendants' position. For example, the change from the December 12, 2012, draft of "neurological decline" to "transient neurological symptoms" in the final report of January 30, 2013, tempers his conclusion. Other examples raised by the plaintiffs' counsel in cross-examination were similar.

[327] I do not make any finding with respect to Dr. Muller's accessing St. Michael's Hospital electronic records or the non-redaction of personal patient information in his litigation file. Nothing malicious was intended. Better communication between the doctor and counsel as to exactly what he should have reviewed is the only thing to be taken from this.

[328] I take issue with Dr. Muller's view that Mr. Campbell's situation was not life threatening at the time of transfer to St. Michael's Hospital. Dr. Mikulis' evidence was that, by this point, Mr. Campbell had suffered a six millimetre mid-line shift of the brain and uncus herniation. One cannot ignore, of course, the evidence of Dr. Roberts who candidly conceded that he thought Mr. Campbell was going to die.

[329] Drs. Weingarten, Roberts, Lee and Zoutman all agreed that an untreated brain abscess can cause increased swelling and permanent neurological impairment. Dr. Muller's view was that Mr. Campbell was already at the point of no return by March 17. The court cannot reconcile Dr. Muller's opinion with that of Drs. Lee and Zoutman on this issue. Even if it is true that not all swelling causes brain damage, what is the level of swelling required to do so? The abscesses in Mr. Campbell's brain had increased almost twice in size from March 17 to his admission to St.

Michael's Hospital. Mr. Campbell's cognitive impairment also continued to decline. All of this occurred during his treatment by Drs. Weingarten and Roberts. Ignoring these facts is simply not possible on a causation analysis.

[330] The medical experts agreed that most abscesses can be successfully treated by either antibiotics or surgery if they are small enough and if the patient is not already compromised. In Mr. Campbell's case, he was temporarily discharged on March 26 in stable condition having been treated with antibiotics and steroids. He no longer had any neurological deficits. It is therefore impossible for this court to say that had he been treated at this point, the outcome would have been the same. No evidence was led by the defendants' to show that timely diagnosis and treatment would have had led to those effects.

[331] The defendants will no doubt raise a concern about the fact that no finding of a breach of the standard was made as of March 17. However, in my view not much turns on this for the causation analysis. The brain lesions measured 1.6 and 2.2 centimetres on March 17 according to Dr. Thain's CT scan. They were 1.5 cm and 2.4 centimetres as of the March 20 MRI according to Dr. Silmberg. Dr. Mikulis' evidence was that on his blind review of the MRI the lesions had not changed much. His measurement of the largest lesion was 2.6 centimetres at that point. The defendants want me to accept Dr. Mikulis' evidence for the proposition that the larger lesion already measured over 2.5 centimetres when Mr. Campbell was admitted on March 17, 2009. However, I accept the measurements of Drs. Thain and Silmberg on the size of these lesions at these points in time, as they were the attending radiologists, and no evidence has been raised which would cause me to question their initial measurements. By the time of Dr. Doran's CT scan, Mr. Campbell was stable and his symptoms from March 17 had improved while on antibiotics and steroids.

[332] The defendants further want me to limit Dr. Lee's evidence to whether Mr. Campbell's injuries could have been avoided by antibiotic treatment as of March 17. They argue, therefore, that any opinion from Dr. Lee that relates to causation issues after March 17 is inadmissible. I reject this argument by the defendants and maintain that Dr. Lee's evidence stands for the proposition that antibiotic treatment can alone be used to treat brain abscesses as long as the abscesses are smaller than three centimetres. Therefore, if I accept Dr. Lee's evidence in this regard, and if I accept that the abscesses measured 1.5 centimetres and 2.4 centimetres as of the March 20 MRI, then Dr. Lee's evidence substantiates the proposition that the abscesses were likely treatable by IV antibiotics alone up until the March 26 discharge.

[333] The evidence of Dr. Lee cannot be ignored with respect to the treatment of brain abscess by antibiotics. In her own practice she has successfully treated a patient with a brain abscess measuring 2.9 cm using IV antibiotics. Her evidence was that Mr. Campbell's neurological improvement was attributable to the oral antibiotic Levaquin, which Mr. Campbell received between March 14 and 23. He continued to improve to the point where Dr. Roberts temporarily discharged him on March 26 while awaiting the bronchoscopy results.

[334] There was a different of medical opinion as between Dr. Lee and Dr. Muller concerning the size of abscess which can be successfully treated by antibiotics. Dr. Muller was adamant that antibiotics would have had no effect on Mr. Campbell's lesions because the amount of pus inside

the lesions would have made penetration by antibiotics impossible. However, I prefer the evidence of Dr. Lee on this point because:

- (a) The lesions were less than three centimetres in size and therefore treatable by antibiotics as per the Mandell text;
- (b) Mr. Campbell responded to oral antibiotics very positively;
- (c) Dr. Lee is an infectious disease specialist and has treated abscess patients with antibiotics whereas Dr. Muller has not. Dr. Muller is obviously more comfortable with the surgery route as he is a brain surgeon and that is his experience;
- (d) Dr. Weingarten agreed if that he thought Mr. Campbell had a brain abscess he would have started him on Ceftriaxone as an empirical antibiotic approach; and
- (e) Dr. Zoutman agreed that it was his practice to use Ceftriaxone as an initial therapeutic regimen, as per the Mandell text, and he would continue to watch the patient in consultation with a neurosurgeon.

[335] With respect to Dr. Warde, it is not possible to find a causal link between the whole brain radiation and Mr. Campbell's outcome. More specifically, the factual record is clear that Mr. Campbell's abscesses continued to grow throughout the time he was being radiated. There is no evidence that the swelling in Mr. Campbell's brain at that time was caused or increased by the radiation as opposed to the swelling which the experts agreed accompanies abscess growth. Further, while Dr. Wright agreed that whole brain radiation can cause swelling of the brain, it does not always do so. In my view, there is therefore a strong possibility that the whole brain radiation had no effect whatsoever on Mr. Campbell's outcome. While I have already found that the decision to radiate Mr. Campbell's brain fell below the standard of care, the result of that decision cannot be causally linked to any cognitive impairment ultimately suffered by Mr. Campbell.

[336] As per *Clements v. Clements*⁶² a robust and pragmatic approach to determining whether a plaintiff has established that a defendant or defendants' negligence caused the plaintiff's loss. This includes common sense inferences about what ought to have been done but was not. Applying that approach to this case results in the simple proposition, that if Drs. Weingarten and Roberts had properly expanded their differential diagnosis to include a brain infection, and taken steps in that regard, there is evidence to support that Mr. Campbell's outcome would have been different, on a balance of probabilities.

[337] Further, it is not necessary for the plaintiffs to show that Drs. Weingarten and Roberts were the sole cause of Mr. Campbell's injuries. As long as each of them is part of the cause of the injury then they are liable.⁶³ Therefore, while Dr. Weingarten's contribution to the injury

⁶² *Supra* note 52 at paras. 38 & 46.

⁶³ *Athey v. Leonati*, *supra* note 55 at) paras. 13-17.

may be considered to have been less than that of Dr. Roberts, given the timing, that is of no consequence with respect to causation.

[338] A causal link may be presumed where a potential danger materializes.⁶⁴ In this case, the “medical mismanagement” of the case as described by Dr. Muller and the tunnel vision of Drs. Weingarten and Roberts with respect to a cancer diagnosis created a danger. That danger was the lack of prompt diagnosis and treatment of brain abscesses which all of the experts agreed would be life threatening if left untreated. I find that the danger materialized and caused Mr. Campbell’s injuries on a balance of probabilities.

[339] The plaintiffs’ theory need not be proven to a standard of scientific accuracy. The law does not require such a standard. In *Snell v. Farrell*⁶⁵ the court relied on *Dunlop Holding Ltd.’s Application*⁶⁶ with respect to the fact that little affirmative evidence of causation is likely to be within the knowledge of the plaintiff. Quoting *Dunlop* the court said, “This does not mean that the peculiar means of knowledge of one of the parties relieves the other of the burden of adducing some evidence with respect to the facts in question, although very slight evidence will often suffice.” While the burden of proof is not shifted, little affirmative evidence on the part of the plaintiff will justify the drawing of an inference of causation.

Contributory Negligence

[340] The defendants allege that Mr. Campbell contributed to his own injury through failing to seek medical advice earlier when his symptoms first arose in mid-February. The defendants further allege he did not act reasonably by failing to report to his doctors that these symptoms first arose in mid-February.

[341] In order to prove that the plaintiff was contributorily negligent the defendant’s bear the onus of showing both that:

- (a) The plaintiff failed to act reasonably; and
- (b) That any such failure to act reasonably contributed to the damages at issue.⁶⁷

The Reasonableness of Mr. Campbell’s Conduct

[342] The evidence is that Mr. Campbell first began to feel ill in mid-February before he went on his trip to Louisiana with his wife. Mr. Campbell began to experience headaches, cough and fatigue before his trip and these symptoms worsened during the trip. He complained of these symptoms to his son and daughter, which they said was unusual on his part. However, these symptoms were not too severe as his wife failed to notice them, and only noted that he hadn’t

⁶⁴ *Lafferriere v. Lawson*, *supra* note 56 at para. 162.

⁶⁵ *Supra* note 54 at 31.

⁶⁶ [1979] R.P.C. 523 (C.A.).

⁶⁷ Allen M. Linden and Bruce Feldthusen, *Canadian Tort Law*, 8th ed. (Markham: Lexis Nexis Canada Inc., 2006) at pp. 490-491.

played golf on their trip. She further observed that he returned to work when they returned home from the trip. The first that she heard of the symptoms was when he returned from the emergency room on March 14, 2009. His evidence was that these symptoms did not really start getting really bad until he got back from his trip.

[343] It was at this point that Mr. Campbell went to the Emergency Room at Southlake on March 14, 2009, and there reported symptoms of cough, head and body aches and low grade fever. These symptoms were again reported when he went to the emergency room on March 17, 2009. According to the note of Dr. Weingarten on March 17, 2009, Mr. Campbell had returned because of ongoing headache, visual disturbance and severe fatigue. From his note, we also know that Dr. Weingarten was aware that “[o]ver the last couple of weeks he has been feeling generally tired and achy,” that he had developed visual scintillations and that his headaches would “worsen considerably if he had a coughing spell.”⁶⁸

[344] I conclude that Mr. Campbell acted reasonably by seeking medical assistance after he returned home from his trip on March 14, 2009, and later by returning to the emergency room on March 17, 2009. While it appears that he was not feeling well during his trip, his symptoms were not at the level that would have mandated that he attend to a clinic or the emergency room. The most his wife can say about this trip is that he didn’t play golf. Further, when he returned from his trip, he was well enough to go to work. When Mr. Campbell realized that his symptoms were getting worse upon his return, he promptly took a trip to the emergency room at Southlake. It was there that he was given a likely diagnosis of community acquired pneumonia, and prescribed antibiotics. There were some questions over some findings over the CT scan of his head, and Dr. Kouchakan wished to admit Mr. Campbell to do a CT scan of the chest to rule out lung malignancy. However, Mr. Campbell opted to follow-up with his family doctor. While staying at the hospital may have been the most cautious course of action, given the likely diagnosis of pneumonia at this point, I do not find that Mr. Campbell acted unreasonably by choosing to follow-up with his family doctor.

[345] When Mr. Campbell lost his left field of vision while driving to work on March 16, 2009, he again acted reasonably through driving directly to the office of his family doctor. While he did not immediately follow his doctor’s advice by going to Southlake right way, he did go the very next day. While it would have been preferable to go straight to Southlake, I do not find that anything would turn on this, as no damage to him was suffered, nor was there any likely delay in treatment, in the brief intervening period between March 16 and March 17.

[346] The defendants state that Mr. Campbell contributed to his own injuries by failing to communicate to his doctors the fact that his symptoms had started in mid-February. I cannot find that, by failing to communicate the precise time period he began feeling unwell, Mr. Campbell was somehow acting unreasonably. The evidence demonstrated that Mr. Campbell communicated a great deal to his doctors. Most importantly, in his note on March 17, 2009, Dr. Weingarten demonstrates that he knew:

⁶⁸ Dr. Weingarten, Consultation Report, Tab 2A, pg. 14-15.

- (a) Mr. Campbell had returned because of ongoing headache, visual disturbance and severe fatigue;
- (b) That Mr. Campbell had been feeling generally tired and achy over the last couple of weeks;
- (c) That he had developed visual scintillations a few days before; and
- (d) That his headaches would “worsen considerably if he had a coughing spell.”

[347] This does not give the impression that Mr. Campbell was withholding information. Likely, Mr. Campbell only wished to emphasize that period when his symptoms were becoming noticeably worse. If any discomfort or fatigue he had been feeling up to a month before was relevant, it was incumbent on Mr. Campbell’s doctors to elicit this information.

Contribution to His Injuries

[348] Even if I were to find that Mr. Campbell failed to act reasonably by failing to attend to the hospital earlier and by failing to report his earlier symptoms to his doctors, I cannot find that these actions in any way contributed to his injuries in the case at bar.

[349] While Mr. Campbell had been generally not feeling well in the weeks prior to his admission to Southlake on March 14 and 17, 2009, he suffered no real cognitive impairment until his loss of left field vision on March 16, and no significant cognitive change until March 30, 2009, when he returned to Southlake by ambulance. As per my findings above with regards to negligence, Mr. Campbell’s condition was treatable and curable upon his admission to Southlake on March 17, 2009, and up to a point before the serious changes on March 30, 2009.

[350] This case is somewhat analogous to that of *Chan (Litigation Guardian of) v. Tang*⁶⁹, where Mulligan J. highlighted the importance of logically connecting the injury suffered with the negligence of the plaintiff. In that case, the plaintiff failed to attend for blood testing for a period of time preceding his eventual stroke, but then subsequently completed three blood tests which were available to the treating physician. It was the courts view that the “preceding period of non-testing” did not in any way contribute to the plaintiff’s stroke. In the same way, the preceding brief period of failing to seek treatment did not contribute to Mr. Campbell’s injuries. It was the failure of Mr. Campbell’s treating doctors to properly consider abscess in the differential diagnosis and treat it accordingly that led to his injuries.

[351] The defendants have further failed to demonstrate that Mr. Campbell’s alleged failure to disclose information on when his symptoms first began in any way contributed to his injury. As stated above the treating doctors knew that he presented ongoing headache, visual disturbance and severe fatigue, that he had developed visual scintillations a few days before, that his headaches would worsen considerably if he had a coughing spell, and that he been feeling generally tired and achy over the *last couple of weeks*. Quite frankly, they knew, or ought to

⁶⁹ 2012 ONSC 2050, [2012] O.J. No. 1436 (QL) at para. 164 (Sup. Ct.).

have known based on the information collected, that he had not been feeling well for some time. While they did not precisely know that he began feeling unwell in mid-February (which was only a couple of weeks before when they knew his symptoms had begun) they have presented no evidence whatsoever that their diagnosis and treatment options would have differed had they known this.

[352] Further, given the “tunnel-vision” they demonstrated in focusing on cancer to the neglect of ruling out the serious possibility of brain abscess, I am not persuaded that their treatment of Mr. Campbell would have changed had they known his symptoms had begun earlier.

[353] In conclusion, I do not find that Mr. Campbell acted unreasonably in failing to seek medical advice when he did and in failing to notify his doctors of the precise date when he began not feeling well. In the event that I am wrong, I find that, even if Mr. Campbell could be found to have acted unreasonably, this did not contribute in any way to his eventual injury.

DAMAGES

Summary and Principles

[354] The plaintiffs seek damages totaling \$4,484,851, including the *Family Law Act* claims and OHIP’s subrogated interest. In this case, Mr. Campbell is entitled to damages which would return him to the position he was in had the injury not occurred due to the negligence of the defendants. Mitigation of damages is not a consideration in personal injury claims. He seeks general damages of \$300,000.

[355] Mr. Campbell seeks damages for future losses on the basis that it is a substantial possibility that such losses will occur. He seeks damages for future care costs (\$2,758,378) and future loss of income (\$512,000) on the basis that but for the injuries, Mr. Campbell would have been able to care for himself and continue to work. The plaintiffs submit that there is no legal basis to place the burden of Mr. Campbell’s future care on his family members.

[356] Mr. Campbell also seeks damages for past loss of income (\$346,008) and for management fees (\$195,819). The other plaintiffs seek damages related to the loss of care, guidance and companionship of their father/spouse (\$160,000) and Mrs. Campbell seeks damages for past loss of services (\$75,000.).

[357] The defendants position is that Mr. Campbell is entitled to \$289,709 for past income loss, \$220,863 for future income loss, \$130,000 in general damages, \$556,260 for future cares costs and a total of \$55,000 to be distributed to Mr. Campbell’s family under the FLA. They submit that there is no evidence to maintain a claim in damages for past care costs or for an OHIP subrogated claim. Further, they maintain that there is no basis to include a management fee in the damages.

[358] It is an agreed fact in this case that Mr. Campbell suffered permanent cognitive defects which affect his memory, concentration, language, executive function, informational processing and vision.

The Nature and Extent of Mr. Campbell's Injuries

The Plaintiffs' Evidence

Dr. William Fulton

[359] Dr. Fulton is a registered clinical neuropsychologist with a Ph.D. in Clinical Psychology. He has assessed over 4,000 patients with various head injuries and other neuropsychological disorders.

[360] Dr. Fulton assessed Mr. Campbell's neuropsychological function over a period of two days on February 15 and March 5, 2011. This included a clinical interview, personal history and a review of data gathered from 32 neuropsychological tests.

[361] Dr. Fulton diagnosed Mr. Campbell with adjustment disorder, anxiety, depressed mood and cognitive disorder NOS of moderate to severe impairment for right hemisphere tasks. Validity tests were also applied which revealed that Mr. Campbell gave his full and best effort to the testing.

[362] In Dr. Fulton's view, Mr. Campbell cannot return to work and needs supervision. He assessed that Mr. Campbell's visual issues are related to processing caused by brain damage, and not any actual difficulties with the eye itself. Mr. Campbell also has trouble with organizing and multitasking, which leave him vulnerable to mental fatigue, especially in busy or noisy environments. He has difficulty following instructions and will lose track of time and location. This poses a possible safety risk for him. These deficits are all caused by his brain injury according to Dr. Fulton.

[363] Mr. Campbell's brain injury has also resulted in him taking longer and needing more energy to complete tasks. He therefore becomes prone to mental fatigue and is unpredictable, which could also become a safety risk.

[364] Other observations made by Dr. Fulton were that Mr. Campbell limits himself from doing certain things due to a fear of injuring himself. This leads to despondency and discouragement. He also tended to deny his symptoms and did not acknowledge the psychological changes associated with his injuries.

[365] All of the above changes and injuries render Mr. Campbell unemployable and permanently and chronically impaired. He needs constant supervision.

[366] Dr. Fulton did not disagree with Dr. Cohen's conclusions regarding Mr. Campbell's neurological deficits.

Ms. Sumeet Shergill

[367] Ms. Shergill is a Behavioural Consultant who works at the York Region Brain Injury Association and has been working one on one with Mr. Campbell once a week since June 2011. She helps him identify his functional difficulties and provides coping and memory strategies.

[368] She testified that Mr. Campbell feels lonely and uncertain about his future. He also feels anger, sadness, depression and worries that he can never compensate his wife for what has happened. Mr. Campbell's biggest concern is his vision and it has been very hard for him not to drive. It has taken some time for Ms. Shergill to learn Mr. Campbell's real deficits, as he does not want to show people that he cannot do certain things.

[369] Mr. Campbell becomes angry when he cannot complete tasks he could do pre-injury and has difficulty solving problems. He worries about how his injuries will affect his relationship with his family and his interaction with his grandchildren. He worries about how his family will be taken care of when he is gone.

[370] Ms. Shergill accompanied Mr. Campbell to a day program, which he attended only once. Mr. Campbell kept to himself and did not socialize. He did not want to be told what to do and did not like the structured group setting. Ms. Shergill felt the programs were geared towards those with more severe injuries than Mr. Campbell and that it was not a good fit for him. He has not attended any further day programs although they have been offered to him.

[371] She testified that he has resumed playing chess and when they play together he always beats her.

The Defendants' Expert

Dr. Sharon Cohen

[372] Dr. Cohen is a behavioural neurologist specializing in disorders of memory and cognition. She was qualified to give expert evidence on assessing neurological function and recommending future care needs. Dr. Cohen's evidence was that almost all of her patients have acquired brain injuries and she frequently assesses and makes recommendations for their future care needs.

[373] Dr. Cohen reviewed five important factors that influence one's ability to rehabilitate functional abilities after a brain injury:

1. The extent of the injury;
2. The person's pre-morbid psychological and intellectual abilities;
3. The frequency and timing of proper rehabilitation;
4. The amount of elapsed time since the brain injury occurred; and
5. The person's family and social support network.

[374] Reviewing the five factors as they relate to Mr. Campbell's case she found that: he was an intelligent person before the injury; his brain injury is static and not degenerative; he underwent rehabilitation and is seen weekly by Ms. Shergill, and he has a loving and supportive family.

[375] Dr. Cohen did a functional assessment on Mr. Campbell on February 7, 2013. During the course of the three hour assessment she asked Mr. Campbell about his history, his abilities and his self-maintenance. There was also a general cognitive exam and a paper and pencil test.

[376] Dr. Cohen's conclusion was that Mr. Campbell's impairment is of moderate severity. He also has a moderate visual impairment. The cognition and vision impairments impact Mr. Campbell's day to day abilities. It is not disputed in this case that Mr. Campbell cannot work, needs assistance with certain activities and cannot complete some of the leisure activities in which he formerly engaged.

[377] Dr. Cohen made conclusions after the assessment relating to memory, concentration, language, executive functioning, speed of information processing and visual impairment. She concluded that Mr. Campbell has moderate to severe memory impairment. He is forgetful and requires repetition to encode information. Dr. Cohen relied on some of Mr. Campbell's self-reporting in coming to this conclusion, including the fact that he now cannot remember the rules of chess. This is contrary to the evidence of Ms. Shergill who testified that she had been playing chess with Mr. Campbell since 2012.

[378] Dr. Cohen concluded that Mr. Campbell has moderate impairment to his concentration. However, he had developed certain adaptive strategies to assist in this regard, such as counting steps. More could be implemented (i.e. automatic stove shut off) that would assist him.

[379] Dr. Cohen found that Mr. Campbell has moderate impairment to his language skills. This conclusion is mostly based on Mr. Campbell's self-reporting that he forgets words, cannot complete sentences and goes off on tangents. The defendants submit this is inconsistent with the manner in which Mr. Campbell gave his testimony at trial, when none of these alleged language difficulties were exhibited.

[380] Dr. Cohen testified that the results of her assessment showed that Mr. Campbell had moderate to severe impairment to his cognitive domain of executive functioning and moderate impairment to his speed of information processing. Much of the information relating to Mr. Campbell's speed of information processing came from self-reporting, which the defendants' submit was not corroborated by other members of Mr. Campbell's family.

[381] As for Mr. Campbell's visual issues, she was somewhat uncertain as to whether they were related to his brain injury and deferred that question to a neuro-ophthalmologist.

Non-Pecuniary Damages

General Damages

[382] Mr. Campbell seeks \$300,000 in general damages. The defendants' view is that \$130,000 is a reasonable assessment of his damages.

[383] There are certain incontrovertible facts about the effects of Mr. Campbell's brain injury. The first is that it will shorten his life. Dr. Segel opines that it will do so by two years and Dr. Armstrong by almost the same amount.

[384] The second important fact is that the cognitive impairments from the brain injury prevent Mr. Campbell from working. Mr. Campbell was described by his wife as a "workaholic." He told the court that he "loved" his job at Magna. He wanted the biggest problems that work could throw at him and he reveled in solving them. When Mr. Campbell spoke of his work, he impressed me as a man who was passionate about this work and thought of it as much more than a way to support his family, but as a form of stimulation and a foundation for his self-worth. That has now been taken away from him. He spends a significant amount of time alone now. He is understandably bored much of the time. He feels guilty that he cannot support his family. He is sensitive about admitting the extent of his limitations or talking about them with his wife or children. His impairments are extensive but not so much so that he cannot understand what he has lost, thus demonstrating the depth of his frustration and guilt.

[385] The third important fact relates to Mr. Campbell's loss of independence. Evidence was heard about his attempts to requalify for his driver's license. He was unable to do so. He must now rely on others to take him where he wants to go or take public transit. Mr. Campbell loved being a consultant in part, I find, because he liked the aspect of being his own boss. He was an independent self-starter and a problem solver both at home and at work before his brain injury. His now forced reliance on others has been very difficult for him and a blow to his pride.

[386] The fourth fact relates to Mr. Campbell's actual suffering while in the hospital. He thought he was going to die because he was told he had brain cancer. He suffered numerous invasive tests including a bronchoscopy, a needle biopsy, and a wedge resection. He then underwent whole brain radiation. During the course of these tests, the doctors caring for him did not know he had a brain abscess which continued to grow. After he was properly diagnosed he had a craniotomy and was in Intensive Care for two weeks following that procedure. He underwent extensive rehabilitation once released from hospital. According to Dr. Cohen he is past the point of ideal rehabilitation, which I take to mean that Mr. Campbell's cognitive impairment will not make any further significant improvement.

[387] In *Gordon v. Greig*⁷⁰ one of the plaintiffs suffered a brain injury which left him unemployable, lacking control over bodily functions and having difficulty with interpersonal relationships, but with enough mental function to realize his shortfalls and that he could not do anything about them. The court awarded general damages in that case at the then maximum of \$310,000.

⁷⁰ [2007] O.J. No. 225 at paras. 75 & 76.

[388] On the positive side, Mr. Campbell is able to function at a level much higher than many others with acquired brain injuries. He is able to dress himself, make basic meals, do some light housekeeping and gardening and walk or take public transit on his own.

[389] The defendants submit that Mr. Campbell's general damages should not be in the range suggested by the plaintiffs. They cited several cases in which the plaintiffs' brain injuries resulted in more serious impairments than those of Mr. Campbell. In each case the plaintiff received between \$100,000 and \$125,000 before adjustment for inflation.

[390] In *Van Dyke v. Grey Bruce Regional Health Centre*⁷¹ the plaintiff suffered brain damage from toxicity as a result of the administration of an inappropriate antibiotic. The plaintiff suffered from dizziness, headaches, blurred vision and had difficulty walking on even ground. He was no longer employable nor could he drive. He could not participate in hockey, baseball or soccer as he had before the accident. The court awarded damages of \$100,000⁷² (\$119,167 adjusted for inflation) for what the defendants submit is plaintiff in circumstances similar to Mr. Campbell.

Conclusions Regarding General Damages

[391] The pain and suffering endured by Mr. Campbell in hospital and the changes in his life he has endured since then are of significance. His work and personal life are forever changed. But for the negligence as found in this judgment Mr. Campbell would have continued in his beloved work as a consultant and he and his wife would have been able to pursue the adventures in their "wish jar".

[392] However, I accept the defendants' argument that the factual circumstances in this case would not put Mr. Campbell at the top end of the general damages range. He is not in the unfortunate situation of the plaintiff in *Gordon, supra*, where he has no control over bodily functions and difficulty with relationships. Mr. Campbell enjoys a loving relationship with his wife and children. Indeed, Mr. Campbell has a certain level of functionality which allows him a degree of independence not available to the plaintiffs in the cases cited by the defendants.

[393] In all of the circumstances, I award Mr. Campbell **\$200,000** in general damages.

Family Law Act Damages

[394] The *Family Law Act* in Ontario provides in section 61(2) the right of a person to claim damages for the loss of "guidance, care and companionship that the claimant might reasonably have expected to receive from the person if the injury or death had not occurred." Section 61(2)(d) also provides compensation for "a reasonable allowance for loss of income or the value of the services" where the claimant provides "nursing, housekeeping or other services" for the person who has been injured.

⁷¹ [2003] O.J. No. 3047, [2003] O.T.C. 692 rev'd in [2005] O.J. No. 2219, 197 O.A.C. 336, 255 (Ont. C.A.) on the issue of negligence and causation.

⁷² *Ibid* at para. 71.

[395] Mrs. Campbell's evidence was that her husband's injuries have resulted in a complete reversal of their previous roles and a dramatic change in their relationship. Whereas Mr. Campbell often used to cook, did chores and attended to many things around the home, Mrs. Campbell has had to take on the majority of responsibilities. As the sole income earner for the family she must work outside the home. She has had to take on debt to manage things financially.

[396] Mr. Campbell worries constantly about leaving Mr. Campbell alone. She is afraid he may not be able to contain the spontaneous nosebleeds he gets from his HHT or that he will injure himself while trying to cook something. She is fearful of talking to him about much of this as she knows her husband is sensitive about his loss of independence and feels guilty about how much she has had to take on. She knows his self-esteem has been affected by the fact he is no longer financially contributing to the family.

[397] The plaintiffs claim *Family Law Act* damages of \$100,000 for loss of care, guidance and companionship on behalf of Mrs. Campbell.

[398] Mrs. Campbell has had to take on a caregiving role which has left her tired and with little time for herself. This role has continued since July 2009 when Mr. Campbell returned from the hospital. While she has not been able to provide full time care for her husband she has provided a substantial amount of care for more than five years. She seeks an award of \$75,000 for that care.

[399] Mr. Campbell's son, Noel, is 34 and has resided in the family home with his parents since 2010. Prior to his father's injuries they would spend leisure time together golfing, watching sports on television and playing horseshoes. They are no longer able to engage in physical activities together.

[400] Noel has observed that his father is frustrated with his vision and memory and the fact that he can no longer use his hobby machinery and had to sell it.

[401] Mr. Campbell's daughter, Riki, is 36 and lives with her son, Harley, and her husband in the Campbell home. Riki teaches high school in Newmarket.

[402] Riki's evidence is that, prior to her father's brain injury, she and her father would spend time together golfing, having dinner and going on outings. Sometimes he would come over and fix things for her when she lived in an apartment. Since the injury she has observed that her father becomes frustrated easily and even angry because he can no longer do the activities he could do before the injury. He is more dependent on her mother because she has to drive him everywhere and do things for him.

[403] Riki testified that her relationship with her father has changed since the brain injury. It is now strained because they avoid topics of conversation that she knows her father finds difficult. She told the court that her father is a different man since the injury.

[404] Riki and Noel each seek \$30,000 to compensate them for the fundamental change in their relationship with their father.

Conclusions Regarding Family Law Act Damages

[405] I accept Mrs. Campbell's evidence that her husband's brain injury has resulted in a significant change in their roles. She has had to take on more responsibilities, including being the sole breadwinner, and she has the added stress of worrying about Mr. Campbell's well-being when she is at work and he is unattended. However, most fortunately, it appears that their marriage has weathered these changes and they continue to have a close relationship. Mrs. Campbell testified that Mr. Campbell still makes the bed the way she likes it, does little things for her and still plans surprises for her. It does not appear that Mrs. Campbell has lost the "companionship" of her husband in the same way that less fortunate spouses have. For example, in *Bannon v. Hagerman Estate*⁷³ the court awarded *Family Law Act* damages in the amount of \$35,000 (\$49,752 adjusted for inflation) to a spouse where the injured spouse was unable to dress, bath or even get up from a chair. The uninjured spouse essentially became a full time caregiver.

[406] On these facts, however, I find that Mrs. Campbell has lost the care and guidance of her spouse given his cognitive impairments and the significant change in their roles in the marriage.

[407] In the circumstances I award *Family Law Act* damages in the amount of **\$50,000** to Mrs. Campbell.

[408] Turning to Mrs. Campbell's claim for damages for past care costs, claimed under section 61(2)(d), it is clear that Mr. Campbell requires some level of supervision. The defendants' expert, Dr. Cohen, testified that Mr. Campbell should not be left alone for more than two hours. It is Mrs. Campbell who has provided almost all of the care for Mr. Campbell since he was discharged from the hospital. This has made her tired and prevented her from doing other things she may have wanted to do during this time. She has provided this care for four and a half years.

[409] The defendants argue that such past care costs cannot be inferred without being quantified or without proper supporting evidence. Neither Mrs. Campbell nor Dr. Katz gave such evidence.

[410] I find that the amount requested by the plaintiffs for past care damages is more than reasonable. They originally claimed \$250,000, but the reduction of the request to \$75,000 reflects, in my view, an acknowledgment that such damages are difficult to prove and quantify. I therefore award Mrs. Campbell the sum of **\$75,000** representing the cost of the care she has provided her husband for the past four and one-half years.

[411] With respect to the damages for Mr. Campbell's children it is clear that they have both lost some degree of companionship with their father because they are no longer able to engage in the same activities with him as they did before the injury. Further, their relationship with their father appears to be less relaxed and open due to their fear of upsetting him or making him feel guilty. I agree with the defendants, however, that as they are both living with their father, their

⁷³ [1995] O.J. No. 539, 22 O.R. (3d) 396 at paras. 100-101 (Gen. Div.), quantum aff'd on appeal: [1998] O.J. No. 1673.

claim in relation to a loss of “companionship” is somewhat diminished because they have contact with him every day. There was no evidence that their father’s injury had impacted either Noel or Riki psychologically or with respect to their employment. There was also no evidence that either child had had to sacrifice any time or energy caring for their father.

[412] Therefore, while their relationship with their father has been changed, the change is not one that is so compelling or significant as to attract an award in the upper range as suggested by the plaintiffs. I award each of Riki and Noel Campbell the sum of **\$7,500**.

Future Care Costs

Mr. Campbell’s Future Care Needs

The Plaintiffs’ Expert – Ms. Yvonne Pollard

[413] Ms. Pollard is a rehabilitation consultant. She was qualified as an expert in vocational rehabilitation, Life Care Plans (“LCPs”) and in assessing and developing medical and vocational rehab plans for individuals. She testified that the goal of life planning is to bridge the gap between a person’s current difficulties and their pre-injury function and quality of life.

[414] In determining the needs of an individual, a life care planner will review the medical information and interview the individual and their family and friends. A geographic boundary area is determined for a costing of appropriate services and the plan is sent to a practitioner for review.

[415] A life care planner is taught to ignore a person’s spouse as capable of taking over the injured person’s care, as that person could pass away, nor is it fair to “co-opt” a spouse permanently into the care program. However, Ms. Pollard could not point to any evidence suggesting the existence of this view as a best practice.

[416] Ms. Pollard assessed Mr. Campbell’s future care needs in May 2012 and again in January 2013 and wrote three reports. In her May 22, 2012, and November 22, 2012, reports she recommended two hours of attendant care per day for Mr. Campbell. However, after reviewing Dr. Mikulis’ opinion and meeting with Dr. Fulton (who had not met with Mr. Campbell since 2011) she determined in her January 2013 report that Mr. Campbell needed 24 hour care and could not be left alone. Ms. Pollard was unable to medically justify the requirement that Mr. Campbell have attendant care throughout the night. She stated that she had deferred to the opinion of Dr. Fulton on 24 hour care, and was surprised to learn that, in his testimony, Dr. Fulton had in fact deferred to her opinion on the 24/7 care. She also agreed that she had never met with Dr. Mikulis nor did she request any other reports (such as that of Dr. Muller) which offered different opinions from that of Dr. Mikulis.

[417] According to Ms. Pollard, the care should be provided by a professional agency with bonded and reliable workers. Full-time care would allow Mr. Campbell to stay in his own home and would meet his various needs.

[418] Ms. Pollard opined that the January 2013 Summary of Costs (Exhibit 27) were reasonably necessary for Mr. Campbell's quality of life and to maintain his independence. Dr. Fulton agreed that Ms. Pollard's recommendations are supported from a neuropsychological perspective.

[419] Ms. Pollard testified that she was concerned about Mrs. Campbell's view that Mr. Campbell had too much time on his hands. She suggested he join a club for structure and guidance.

[420] Ms. Pollard recommended 25 sessions of visual therapy for Mr. Campbell despite this never having been recommended by Mr. Campbell's neuro-ophthalmologist, Dr. Thompson.

[421] Ms. Pollard also recommended a rehabilitation therapist to establish the "routine and rapport" that Mr. Campbell needs based on the strategies created by other therapists. In addition, she recommended extensive speech therapy, occupational therapy, physiotherapy, an exercise therapist, a volunteer specialist and 30 hours of psychological counseling.

[422] Ms. Pollard recommended a grab bar, a programmable phone, a voice recorder, an iPad and a white board by way of assistive devices.

[423] Ms. Pollard's evidence was that she would defer to a neurologist with respect to her recommendations. The only neurologist who testified with respect to future care costs was Dr. Cohen. Dr. Cohen opined that many of Ms. Pollard's recommendations were medically unjustified.

The Defendants' Experts

Dr. Sharon Cohen

[424] Dr. Cohen's qualifications and scope of expertise are outlined above. Dr. Cohen opined that, in terms of future care needs, Mr. Campbell does not need 24/7 care. He can be left alone for periods of up to two hours because he is insightful, aware of his own limitations and is neither erratic nor irrational.⁷⁴ Her evidence was that Mr. Campbell was not the type of patient that would, for example, leave the home with his slippers on in the middle of winter.

[425] Dr. Cohen recommended that Mr. Campbell attend a day program for individuals with neurological problems. Dr. Cohen is very familiar with such programs and felt it would be beneficial for Mr. Campbell to be in the community with individuals with similar issues and with whom he can socialize and exercise. Her experience is that individuals with brain injuries can often no longer engage in the same activities as before the injury. Therefore, they become bored and frustrated at home. A day program provides a safe environment with structured activities. She agreed that, at first, many patients resist such an idea, but over time they settle in and benefit

⁷⁴ Dr. Cohen (Direct), March 7, p. 31, LL 15-22.

from the program and in most cases do better than when cared for by a family member.⁷⁵ It was her initial view that Mr. Campbell could attend such a program for up to six hours a day.

[426] In cross-examination she agreed that, if Mr. Campbell tried the day program and could not tolerate it, he would need the services of a PSW or the support of his family to assist him if alone for more than two hours. She agreed that, prior to the injuries, Mr. Campbell had been very independent and self-sufficient and that there is a possibility that cognitively he may not be able to tolerate a day program for six hours a day.

[427] Dr. Cohen recommended a personal support worker or companion for Mr. Campbell when he is not at the day program and his family is not present for more than two hours. Dr. Cohen agreed that there was no magic in the two hour number she had chosen. The personal support worker or companion would not require specialized training and could be hired privately as opposed to going through an agency, although she agreed that hiring through an agency had the advantage of bonded and licensed personnel. Hiring privately would mean that Mr. Campbell would be able to interview the proposed companion himself, which Dr. Cohen felt was important.

[428] Dr. Cohen's view was that Mr. Campbell was past the ideal period for rehabilitation and at this point needs rehabilitation focused on developing adaptive strategies. She therefore recommends a speech pathologist twice a week for two hours for a period of six weeks, and an occupational therapist twice a week for two hours per session until age 65, after which time occupational therapy is not medically indicated. Dr. Cohen's view was that these recommendations were very generous. She disagreed that any rehabilitation therapy was required because the occupational therapist can develop a plan to be communicated to the personal support worker for implementation. She also rejected the recommendation of physiotherapy and kinesiology because his injuries are not related to strength or coordination. Again, the occupational therapist can be relied upon for adaptive strategies.

[429] Rather than a neurobehavioural psychologist, Dr. Cohen recommended a social worker or clinical psychologist for Mr. Campbell and his wife to attend psychosocial counseling if needed. She recommended one hour per week for five to nine weeks, and then every six weeks for one year.

[430] Dr. Cohen disagreed with Ms. Pollard's recommendation of a programmable phone, iPad, white board, markers and hand held recorder. She felt there was too much overlap with these devices. She agreed that he required a grab bar and that one of the devices above should be provided, but on the recommendation of the occupational therapist.

[431] Dr. Cohen did not disagree that Mr. Campbell needs assistance with transportation because of his visual issues. However, she did not see the need for such transportation for leisure assessment or volunteer work. As Mr. Campbell cannot work, Dr. Cohen opined it would not be appropriate for him to have the pressure of a volunteer position. Further, transportation

⁷⁵ Dr. Cohen (Cross), March 7, 2014, p. 79, LL 10-32, p. 80, LL 8-15.

for leisure activities can be provided either by the day program, the personal support worker or his family.

[432] Finally, Dr. Cohen agreed that Mr. Campbell does not have the capacity to clear snow, or do heavy maintenance at his home. He will need support for these tasks.

Ms. Angie Blazkowski

[433] Ms. Blazkowski was qualified as an expert in LCPs and as a Functional Assessor. Approximately 80 per cent of her patients have brain injuries. Interestingly, prior to this case, Ms. Blazkowski has only given evidence on behalf of plaintiffs.

[434] In order to prepare her report, Ms. Blazkowski reviewed medical records and assessments. In July 2013 she conducted a one hour structured interview with Mr. Campbell, his counsel and Mrs. Campbell and a two hour test to evaluate Mr. Campbell's functional ability (again with his counsel present.) This was all done in Mr. Campbell's home. Ms. Pollard did not evaluate functional ability. She conducted pen and paper tests with Mr. Campbell in her office.

[435] Ms. Blazkowski noted that Mr. Campbell walks to a local coffee shop every morning by himself. He has good balance and can walk down his steep basement steps. He can read a newspaper, use a telephone and manage his nosebleeds without assistance. Mr. Campbell uses a white board to help remember things. He can also go on public transit and do basic kitchen preparation.

[436] Ms. Blazkowski's recommendations in her reports, which were dated September 27, 2012, and April 2, 2013, were consistent. She did change some of the recommendations in her April 2, 2013, report to defer to specific recommendations related to future care needs made by Dr. Cohen, as she always defers to the medical specialist's recommendations.

[437] Ms. Blazkowski recommends that Mr. Campbell have a programmable phone with large buttons, a grab bar and an iPad. She also recommends a transportation and parking allowance, heavy housekeeping assistance for two hours bi-weekly, two hours twice annually for spring and fall cleaning, outdoor assistance for snow removal, some gardening and lawn maintenance, and the cost of a handyman for two hours a month.

[438] With respect to rehabilitation, Ms. Blazkowski recommended 12 sessions of speech and language therapy, 2 sessions a week of occupational therapy for 26 weeks, and 15 sessions of psychological counselling.

[439] While Ms. Blazkowski agreed that Mr. Campbell needed assistance with transportation, she only included \$218 per year to cover transportation to medical appointments.

[440] Ms. Blazkowski supported Dr. Cohen's recommendation of an adult day program for six hours a day for Mr. Campbell. She felt it would more stimulating for him than staying at home all day. He would have supervision but still be active and could pursue skills suited to his functioning. Mr. Campbell could choose from a number of different day programs that were

offered. She agreed that she had never asked Mr. Campbell about day programs and was not aware that he had not enjoyed the program he previously attended. She agreed that if he did not attend a day program he would have insufficient means for the required attendant care.

[441] Ms. Blazkowski also supported Dr. Cohen's recommendation that Mr. Campbell should have attendant care when he would be alone for more than two hours. She did not support Ms. Pollard's view that best practices dictated that she essentially ignore that Mr. Campbell had a spouse. As such, she recommended 14 hours of attendant care each weekend and 8 hours a day for week days. She also recommended that attendant care be available when the day program did not run (i.e. statutory holidays) and for other times when it may be required unexpectedly. Ms. Blazkowski's view was that attendant care could be provided by a support worker whose median range of pay would be \$15.23 an hour. Care necessary for weekends and overnight when the support worker would not be present would be provided by Mrs. Campbell. Ms. Blazkowski agreed in cross-examination that Mrs. Campbell should be compensated for this; however, this compensation is not accounted for in her calculations.

[442] The rates for the caregiver were obtained by Ms. Blazkowski from a website called www.payscale.com. She did not interview any caregivers or confirm that this was a current rate being charged by actual caregivers.

Conclusions Regarding Future Care Costs

[443] I reject the conclusion reached in Ms. Pollard's report on future care costs for the following reasons:

- (a) Her change from a two hour a day attendant care to 24/7 attendant care seems to be based solely on Dr. Mikulis' opinion (Dr. Fulton having deferred to her view). Dr. Mikulis never assessed Mr. Campbell. Ms. Pollard never met with Dr. Mikulis. Her somewhat drastic change of opinion is without proper foundation in my view;
- (b) Her underlying reasoning for recommending that Mr. Campbell join a club for structure and guidance is remarkably similar to Dr. Cohen's recommendation of a day program;
- (c) I agree with the defendants that Ms. Pollard's recommendations with respect to rehabilitation intervention are excessive and repetitive. I prefer the approach of Dr. Cohen which allows for some follow-up therapy but is reasonable given Mr. Campbell's age, improvements and impairments;
- (d) With respect to the assistive devices I prefer Dr. Cohen's approach. A grab bar and one of either a voice recorder or an iPad. Mr. Campbell already has a white board that he uses at home; and
- (e) She was not qualified to do a functional assessment of Mr. Campbell (as Ms. Blazkowski was).

[444] I prefer Dr. Cohen's evidence over that of Dr. Fulton where they differ for the following reasons:

- (a) Dr. Fulton's report is stale dated going back to February and March 2011. He seemed surprised to learn that Mr. Campbell was winning chess games against Ms. Shergill and agreed that this would be indicative of an improvement in Mr. Campbell's condition;
- (b) Dr. Fulton's testing has only general application and he agreed that two people with the same test scores may still be functionally different. It depends on their environment. By all accounts Mr. Campbell lives in a loving and supportive environment;
- (c) Dr. Fulton's evidence was contradictory on the point of Mr. Campbell underreporting his symptoms. In his evidence he indicated that he suspected Mr. Campbell was minimizing some of his symptoms while in his report he indicated that Mr. Campbell had no tendency to minimize his concerns;
- (d) Dr. Fulton adopted Ms. Pollard's recommendations with respect to attendant care. However, in first report he made no such recommendation. In his report of October 22, 2012, he recommended two hours of attendant care (similar to Ms. Pollard's recommendation at that point in time). He only recommended 24/7 care after meeting with Ms. Pollard in January 2013. Ms. Pollard had changed her opinion to 24/7 attendant care after reviewing Dr. Mikulis' opinion. However, Dr. Mikulis never met or assessed Mr. Campbell nor does he have any expertise in assessing patients for future care needs.

[445] I prefer Dr. Cohen's evidence generally because she did a full neurological assessment of Mr. Campbell in her office in February 2013. This was a much more recent assessment than that conducted by Dr. Fulton in 2011. Dr. Cohen carefully justified each of her conclusions. I agree with the defendants that the focus of the court should not be what Mr. Campbell needs in relation to his brain injury but with respect to his various functional abilities.

[446] Mr. Campbell has, thankfully, benefited from rehabilitation to the point where he can play chess, bathe and dress himself, make the bed, make coffee, use a keyboard, wash dishes, go on walks by himself, water the garden, maintain the garden pond and shovel snow (apart from heavy storm accumulation). He has the support of a loving family, all of whom live with him. He attended almost every day of the trial by himself (although transportation was provided). Mr. Campbell is at ease with the use of adaptive devices such as a pocket planner, white board and a telephone with large numbers.

[447] I accept that Mr. Campbell is bored and frustrated at home. That is not at all surprising given his active and independent life prior to the brain injury. However, Mr. Campbell did not testify about any day program or his preferences in that regard. Ms. Shergill testified that he tried a day program and it did not work out. I accept Dr. Cohen's evidence that many people are resistant to day programs initially but enjoy them over time. Further, there are many options for

day programs and Mr. Campbell will not be **required**, as a result of this judgment, to go to a day program. It is recommended, and provides the basis for the future care cost calculation, but ultimately the decision over whether to attend is up to him.

[448] I do not find that Mr. Campbell requires 24/7 attendant care. This is not medically justified based on the opinion of Dr. Cohen and the report of Ms. Blazkowski. I have already set out my reasons above for preferring the evidence of the defendants' experts in this regard over those of the plaintiffs.

[449] I also agree that there is no basis to suggest that Mr. Campbell's family support network should not be considered in determining future care costs. There was no evidence to support the proposition suggested by Ms. Pollard that the care of a spouse should essentially be ignored when preparing a LCP. Indeed, Mr. Campbell is in the favoured position of living with his wife, son, daughter, son-in-law and grandson. There is no evidence to contradict that he has a supportive family network. Taking the position that this should be completely ignored or discounted in assessing future care costs is not tenable in my view.

[450] I agree with the defendants' and adopt Dr. Cohen's recommendations as the "generous benchmark" suggested by them. I find that her recommendations are fair, non-duplicative, based on medical evidence and reflective of Mr. Campbell's reality, which is that he is past the stage of ideal rehabilitation, his conditions are mostly static and he is now in his sixties.

[451] Turning to a breakdown of the actual costs, I make the following findings:

- (a) Assistive Aids and Devices – I accept Ms. Blazkowski's recommendation of a grab bar, programmable phone and iPad. Mr. Campbell already has and uses a white board and a pocket diary, so more than Ms. Blazkowski's recommendations would be excessive. The amount awarded is therefore **\$1,810.87** based on Table 3 of Professor Hyatt's report;
- (b) Transportation – Mr. Campbell requires transportation to certain medical appointments throughout the year to Southlake and St. Michael's Hospital. Family members must take him and parking paid. The amount awarded is therefore **\$2,399** based on Dr. Katz' Future Care Cost report;
- (c) Rehabilitation Intervention – As indicated above, I prefer Dr. Cohen's approach to this. I do not agree that Mr. Campbell needs a rehabilitation therapist, visual therapy, physiotherapy, kinesiology, a neurobehavioural psychologist or a gym membership for the reasons given by Dr. Cohen in her report and in her evidence. The amount awarded is therefore **\$24,157.50** as per Dr. Hyatt's report;
- (d) Leisure/Avocational Supports – Mr. Campbell is not a good candidate for volunteer work. He has not shown interest in it and Dr. Cohen felt it would put undue pressure on him. A day program is preferable for the reasons articulated by Dr. Cohen. If Mr. Campbell is bored at home, as his wife says, then this is an opportunity for him to socialize, pursue activities and meet people. The amount awarded is therefore **\$79,225.44** as per Professor Hyatt's report;

- (e) Home Maintenance and Activities of Daily Living – As I have accepted the recommendation of Dr. Cohen that a day program is reasonable, this will mean that Mr. Campbell will not be at home as much as he was. Upon his return from the day program he may well be tired or disinclined to do the things he is currently doing because he is at home all day. As such, the amount recommended by Ms. Pollard for this category is more appropriate. The amount awarded is therefore **\$78,687** as per Dr. Katz’ report;
- (f) Attendant Care – While I agree with Dr. Cohen’s recommendations with respect to attendant care, the higher end of Professor Hyatt’s range must be awarded to allow for times when family are simply not available to care for Mr. Campbell, even if it is a weekend or evening. I do not agree that Mr. Campbell needs full time attendant care. That is simply not borne out by Dr. Cohen’s evidence, the evidence of lay witnesses who see Mr. Campbell day to day and his presentation in court throughout the trial. The amount awarded is therefore **\$547,886.55** based on Dr. Hyatt’s report.

[452] Total damages for future care costs are therefore **\$734,166.36** which I round up to **\$750,000** for contingencies, such as a higher than anticipated rate of inflation and the potential under-estimation of his life expectancy.

Life Expectancy

Dr. Lawrence Segel and Dr. Andrew Armstrong

[453] The plaintiffs called Dr. Segel and the defendants called Dr. Andrew Armstrong to opine on Mr. Campbell’s life expectancy. A determination of life expectancy is needed in order calculate the duration of Mr. Campbell’s need for future care.

[454] Dr. Segel’s evidence was that Mr. Campbell had a life expectancy of 10.4 to 11.6 years as of the date he gave his evidence (March 11, 2014). Originally Dr. Armstrong opined that Mr. Campbell would live another 9 years, but he revised his opinion in cross-examination after reviewing newer census data and agreed that Mr. Campbell should live another 9.63 years.

[455] Both life expectancy experts used the current census data, addressed the impact of Mr. Campbell’s brain injury, his HHT, his diabetes and took into account his smoking history.

[456] While both experts agreed on the “mortality debits” to be assigned to smoking and HHT, they varied somewhat in their views on the debits for HHT and the brain injury. In summary, however, the experts were not far apart in their assessment of life expectancy and agreed that it would not be unreasonable for the court to accept one doctor’s opinion on some issues and the other doctor’s opinion on others. In the circumstances it is reasonable, therefore, to assess Mr. Campbell’s life expectancy at 10.4 years.

Section 116 of the Courts of Justice Act

[457] As Mr. Campbell's future care costs exceed \$250,000, a structured settlement is mandated by section 116. The issue to be determined is what structured settlement product will adequately protect Mr. Campbell's lump sum future care cost award from inflation.

[458] The plaintiffs seek a structured settlement indexed at three per cent annually as per Exhibit 66 in order to minimize the risk of future under compensation. The defendants argue that Mr. Campbell's needs can be met by providing a stream of periodic payments for a fixed number of years. The plaintiffs did not agree with this proposal, their concern being that, if Mr. Campbell outlived the set number of years, he would not be able to afford the required care.

[459] The defendants called Mr. Ralph Fenik, the President of McKellar Structured Settlements, to give evidence regarding the indexation of structured settlements. In cross-examination, Mr. Fenik conceded that, if inflation increases beyond the percentage at which the structure was indexed, Mr. Campbell would not be able to afford the future care as awarded by the court and this could jeopardize his safety.

[460] The defendants argue that the legislation is meant to give flexibility to such settlements and there is therefore no requirement that they be indexed to the Consumer Price Index. Mr. Fenik testified that such annuities are not a mainstream offering and those that exist are significantly more expensive. By way of example, Mr. Fenik noted that, in the past five years, his company has sold more than 4,000 structured annuities and only 2 of them were CPI linked. Based on Mr. Fenik's 25 years of experience, fixed rate annuities are the product of choice to protect an individual from inflation. Further, CPI linked annuities are more appropriate where the payments are meant for a longer term (such as the case of an injured young person).

Conclusions on Structured Settlement Issues

[461] I accept Mr. Fenik's evidence that a fixed term annuity is appropriate in this case for the following reasons:

- (a) Inflation has averaged 1.85 per cent since 1992 (as per the defendants' submissions) and, according to Mr. Fenik, it is unlikely that inflation will skyrocket over the next decade;
- (b) The Bank of Canada monetary policy is aimed at keeping inflation at two per cent;
- (c) CPI inflation was 1.5 per cent in 2013;
- (d) A fixed term annuity is appropriate given the expert opinions on Mr. Campbell's life expectancy; and
- (e) Mr. Fenik does not recommend a CPI indexed annuity based on the facts of this case.

[462] Given all of the above, I order that the structured settlement related to Mr. Campbell's future care costs be set at a 2 per cent fixed rate annuity for a period of 10.4 years.

Past and Future Income Loss

Mr. Campbell's Employment History

[463] Mr. Campbell obtained his certification as a tool and die maker at George Brown College in 1974. In 1976 he began working with Magna where he worked in various capacities, including management and strategic positions where he developed innovative manufacturing strategies.

[464] In 2006 Mr. Campbell began working as a consultant with Magna and was charging \$50 per hour plus mileage. He often worked up to 80 hours per week and prided himself on taking on the toughest jobs the plants could throw at him.

[465] Mr. Campbell was on the verge of negotiating an increase in his hourly rate when he returned home from Louisiana and then became ill. His wife described her husband as a workaholic and both Mr. and Mrs. Campbell testified that he intended to continue working into his 70s, but taking on fewer projects as he got older.

[466] The defendants' view is that Mr. Campbell could not have kept up a pace of 80 hours a week into his 70s and there is no evidence that he would continue working full-time into his 70s.

The Evidence of John Pearson and Darrell Hill

[467] Mr. Pearson was Mr. Campbell's superior at Magna. He testified that Mr. Campbell was very good at his job and could always be counted upon. He told the court about a door panel design problem which Mr. Campbell solved and which saved the company \$6 million.

[468] Mr. Pearson would not hesitate to re-hire Mr. Campbell if he was capable of resuming his former duties. He has not been able to replace Mr. Campbell's talents. He currently pays consultants in Mr. Campbell's former position \$87 an hour and would pay Mr. Campbell that amount if he were working for him now.

[469] Mr. Darrell Hill is an Assistant General Manager with Magna International. He worked with Mr. Campbell on the Dodge Challenger launch program. Mr. Campbell always asked for the biggest problem to solve and he solved them. He was able to coordinate the movement of an entire plant on a tight timeline without any loss of production.

[470] At the time when Mr. Campbell became ill, Mr. Hill was discussing the possibility of hiring Mr. Campbell as an Assistant General Manager at Ontegra for a salary in the range of \$300,000 to \$400,000 per year. This never came to fruition due to Mr. Campbell's illness.

[471] The defendants take the view that Mr. Campbell is not entitled to rely on Mr. Pearson or Mr. Hill's evidence about what he may have earned. Such evidence can only come from a vocational expert (whom the plaintiffs did not call), especially where the plaintiffs are claiming a past loss of more than what he historically earned.

The Evidence of Dr. Eli Katz

[472] Dr. Katz has a Ph.D. in economics and was retained by the plaintiffs to provide an estimate of Mr. Campbell's past and future income losses.

[473] Dr. Katz determined Mr. Campbell's past loss of income as of May 13, 2013, could be conservatively estimated at \$346,008. Dr. Katz assumed that Mr. Campbell would have returned to work in May 2009 but for the injuries. His evidence was that workers of Mr. Campbell's age and experience would earn over \$120,000 per year and over \$135,000 after age 65. In cross-examination Dr. Katz agreed that he was not able to predict accurately when Mr. Campbell would have returned to work and that, for example, there may have been complications or a need for more rehabilitation.

[474] Dr. Katz then factored in participation rates and reduced Mr. Campbell's assumed past income on the basis that, between the relevant periods of May 2009 to May 2013, Mr. Campbell may have been unable or unwilling to work in any given year. Dr. Katz's end point income by 2013 was, therefore, \$118,358 per year.

[475] Dr. Katz's view was that the present value of Mr. Campbell's future income losses was \$512,485. This number was based on participation rates, a review of the incomes of individuals with similar education and experience, an assumption that Mr. Campbell would work to age 70 and then reducing the income to take into account Mr. Campbell's reduced life expectancy on a year over year basis. Dr. Katz testified that it is most often the case that self-employed people work longer than employees and, as such, stopping his calculation at age 70 is more than reasonable.

[476] Dr. Katz also felt this was a conservative number because Dr. Segel expected that Mr. Campbell would live another 11 years and because he did not increase Mr. Campbell's estimated average income (\$118,358 per year) over time, despite the fact that the income for workers in Mr. Campbell's field generally increases over time.

[477] Dr. Katz acknowledged that, when trying to determine Mr. Campbell's income profile, he could not find an exact match with respect to Mr. Campbell's position. He further agreed that past income is a reasonable predictor for future income and that Mr. Campbell invoiced \$50 per hour prior to 2009. He considered Mr. Hill's opinion that Mr. Campbell had the skill set to earn \$150,000 per year, but acknowledged that neither he nor Mr. Hill are vocational experts. He was surprised when told that Mr. Campbell rendered invoices in 2010 (post-injury) for consulting at the rate of \$50 per hour.

[478] Dr. Katz calculated Mr. Campbell's future lost gross income as between \$255,086 and \$591,908.

The Evidence of Professor Douglas Hyatt

[479] Professor Hyatt has an M.A. in Economics and a PhD in Industrial Relations from the University of Toronto. He was qualified by the defendant's as an expert in assessing loss of income and future care costs.

[480] Professor Hyatt determined that Mr. Campbell's past loss of earnings between June 1, 2009, and May 29, 2013, are between \$289,709 and \$345,261.

[481] The calculation for the lower past loss of earnings number was based on an average of Mr. Campbell's earning in 2007 and 2008, when he was a consultant and, updating those figures for inflation. The yearly earnings number was therefore \$75,374.

[482] The calculation for the higher number was based on an analysis of all income data available from 2003-2008 when Mr. Campbell was working as an employee and then adjusting the average wage number of \$89,827 for inflation.

[483] Professor Hyatt calculated Mr. Campbell's future loss of earnings to be between \$133,356 and \$371,305. The range presented is dependent upon whether the court accepts the income numbers based on Mr. Campbell working as an employee or as a consultant and when he would have retired from full-time work.

[484] Professor Hyatt agreed that he did not have any discussions with past employers or co-workers of Mr. Campbell or research the statistics on income for persons in similar jobs. Further, he agreed that his calculations did not take into account that Mr. Campbell's earnings may have increased over time, nor did he consider the evidence of Mr. Hill and Mr. Pearson in his calculations.

Conclusions Regarding Past and Future Income Loss

[485] While I accept that Mr. Campbell was good at his job, well respected in his industry, and on the verge of growing his consulting business, none of that evidence is sufficient for me to conclude that Mr. Campbell would not have continued to earn income in the same range as he earned at the time he became ill. Mr. Campbell's own evidence was that he enjoyed being a consultant as he liked to solve problems and he liked the independence. I do not accept the evidence of Mr. Hill and Mr. Pearson as evidence upon which I can rely as a satisfactory predictor of future income. I therefore accept Professor Hyatt's position that future income should be calculated based on Mr. Campbell's annual earnings as a consultant, or \$75,734 annually.

[486] I do not, however, agree with Professor Hyatt regarding the date of retirement. Rather, I accept Dr. Katz' evidence that self-employed people tend to work longer. I also accept Mr. Campbell's evidence that he wanted to continue to work for as long as he could. In all, it is reasonable to infer that Mr. Campbell would have continued to work until age 70. Therefore, using the table prepared by Professor Hyatt related to the present value of Mr. Campbell's earnings, with retirement between age 65 and 70 (and allowing for the same contingencies as Dr. Katz), Mr. Campbell's future income loss would be **\$311,562**.

[487] For past income loss, I follow the same reasoning as above in concluding that Mr. Campbell's average income as a consultant in 2007 and 2008 (\$75,734) would be the most reliable evidence of past income, given that Mr. Campbell was actually working as a consultant, and to calculate his income as if he were an employee would be unrealistic. Therefore, in 2013

dollars (and relying on Exhibit 61), Mr. Campbell's past loss of income to May 29, 2013, was **\$289,709**.

OHIP Subrogated Claim

[488] Pursuant to section 30 of the *Ontario Health Insurance Act*, the Ministry of Health has a subrogated right to recover monies spent on Mr. Campbell's health care as a result of the defendants' negligence. In this case the total subrogated claim is \$184,669.23.

[489] The plaintiffs agree that care provided to Mr. Campbell on the first Southlake admission and after July 16, 2009, should be deducted, leaving an amount of \$151,332.

[490] Based on Dr. Lee's evidence, if Mr. Campbell had received IV antibiotics as proper treatment, he would have needed 14 days of hospitalization and some follow up homecare. At \$977 per day at Southlake, this would further reduce the subrogated claim to \$137,654.

[491] The defendants' view is that the plaintiffs have led no evidence distinguishing health care costs that would have been incurred in the normal course from those that would not have been incurred but for the negligence of the defendants.

Conclusions Regarding the OHIP Subrogated Claim

[492] In my view, the plaintiffs have been fair in deducting the costs of the first admission (up to March 17) and treatment after July 16, 2009. The sum of \$151,332 is therefore accepted.

Management Fee

[493] The defendants argue that the advantage of a structured settlement is that it obviates the need for a management fee. That seems logical. There appears to be a general agreement in the courts that a management fee is not payable on a structured settlement as compared to a lump sum payment.⁷⁶ In *Wilson v. Martinello* Finlayson J.A., speaking for the Ontario Court of Appeal, accepted the following as the explanation for why a management fee is not applicable:

[494] Since the payments are predetermined and paid periodically by a life insurer (usually over the lifetime of the recipient) there is no need to provide for a management fee to assist in investment as with the income generated by the investment of a lump sum award.⁷⁷

[495] The question remains as to whether a management fee should be applied to that lump sum part of the award that is not part of the structured settlement. Again, in *Wilson v. Martinello* the Court of Appeal remarked that a management fee would only be appropriate "where the recipient of a lump sum award needs professional help in order to preserve the award and invest

⁷⁶ *Melvin (Litigation Guardian of) v. Ontario (Minister of Correctional Services)*, 2013 ONSC 5432, 232 A.C.W.S. (3d) 353 at para. 39; *Sovani v. Jin*, 2005 BCSC 1285, 47 B.C.L.R. (4th) 97 at para. 78; and *Wilson v. Martinello* (1995), 125 D.L.R. (4th) 240, 23 O.R. (3d) 417 at para. 10.

⁷⁷ (1995), 125 D.L.R. (4th) 240, 23 O.R. (3d) 417 at para. 10.

it so as to provide for periodic payments to meet future care costs or other anticipated future needs.”⁷⁸ Given that the future care costs and other needs are provided for through the structured settlement (which does not attract a management fee), I can find no rationale to award a management fee on top of the lump sum portion of the award.

[496] Therefore, I do not find that a management fee is payable in this case where the award for costs of future care will be paid out through a structured settlement.

Total Damages Awarded

[497] In summary, I award damages of \$799,271, exclusive of costs and interests, for general damages, past loss of income and future loss of income, which is to be paid out in a lump sum payment to Mr. Campbell. I also award damages of \$750,000 to Mr. Campbell for his cost of future care, which is to be paid out in a structured settlement through a two per cent fixed rate annuity for a period of 10.4 years.

[498] Further, I award a total of \$140,000 in damages to Riki, Noel and Mrs. Campbell (see the breakdown below) for claims under section 61(2) of the *Family Law Act*.

[499] Finally, I award \$151,332 for the OHIP subrogated interests claim.

[500]

HEADS OF DAMAGES	QUANTUM OF DAMAGES
Richard Campbell	
1. General Damages	\$ 200,000.00
2. Past Loss of Income	\$ 289,709.00
3. Future Loss of Income	\$ 311,562.00
4. Future Care Costs	\$ 750,000.00
FLA Claims	
1. Carole Campbell	
(a) Care and Guidance	\$ 50,000.00
(b) Past Loss of Services	\$ 75,000.00
2. Riki Campbell	\$ 7,500.00
3. Noel Campbell	\$ 7,500.00

⁷⁸ *Ibid* at para.33.

OHIP Subrogated Claim	\$ 151,332.00
TOTAL DAMAGES EXCLUSIVE OF INTEREST AND COSTS	\$1,842,603.00

Costs

[501] If the parties are unable to come to an agreement on costs, I will accept written submissions by both parties on the following schedule: from the plaintiffs by October 31, 2014; followed by responding submissions from the defendants by November 28, 2014; then reply submissions, if any, by December 12, 2014. Cost submissions shall be no more than 10 pages in length, exclusive of any bill of costs or offers to settle. All costs submissions shall be delivered via email through my assistant at Alissa.Livesey@ontario.ca.



Justice C.A. Gilmore

Released: October 7, 2014